Sequential Redistribution in Networked Exchange Economies*

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Abstract

This paper studies decentralised trade in networks through a sequence of Walrasian exchanges, each within a completely connected group. Agents have quasilinear HARA preferences with common curvature, and the groups that keep trading connect all agents. I show that each agent ends with the same amounts of non-numeraire goods as in the centralised market but, since each group trades at its own price, with a different amount of numeraire. Hence, the final allocation is Pareto efficient, and the sequence affects only the speed of convergence and the distribution of the numeraire. I obtain a closed-form approximation of this redistribution for repeated sequences and for random sequences. Finally, I show that larger groups need not accelerate convergence or reduce redistribution.

Keywords: Trading networks · Sequential exchange · Path-dependent redistribution · Speed of convergence
JEL: D51 · D85 · C62 · D47

1 Introduction

In standard general equilibrium theory, all agents trade simultaneously in a centralised market. In practice, exchange is decentralised: agents trade sequentially with subsets of counterparties determined by a network of relationships. For

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example, the spot foreign exchange market is one of the largest financial markets in the world, and it has no central exchange on which all participants trade at once. Dealers can trade with their customers and pass the resulting imbalances among themselves before the market absorbs them, so a transaction reaches the rest of the market through a chain of relationships rather than through a single venue (Chaboud et al., 2023). More generally, over-the-counter[1] financial markets, bilateral commodity agreements, and social media marketplaces all share this structure: transactions are bilateral or multilateral, prices are determined locally, and trading relationships are persistent. This paper asks what such a structure determines and what it leaves undetermined: whether the allocation of goods is the one a centralised market would have reached, and whether the numeraire is distributed as that market would have done.

Consider three agents: Alice, Bob and Charlie. Bob knows Alice and Charlie, but Alice and Charlie do not know each other. Therefore, in this economy, Alice and Charlie can trade only with Bob, while Bob can trade with both of them. For example, Bob can be a dealer, and Alice and Charlie two of Bob's clients. I take this network of relationships as given. Suppose first that Alice and Bob meet, and then Bob and Charlie. After the first meeting, Alice and Bob have no further gains from trade between them. Hence, a second meeting immediately after the first would change nothing. In the second meeting, Bob's endowment is the allocation from the meeting with Alice. After this meeting, Bob holds a different quantity of the goods, so Bob and Alice can gain from trading again. I let the two meetings alternate indefinitely. I ask whether this process has a limit, and whether the limit depends on which pair meets first. I also ask how this limit compares with the outcome of the centralised market, in which all agents trade at the same time.

Figure 1 answers these questions for the case in which agents have quasilinear logarithmic utility over two goods. I write (AB, BC) for the sequence in which Alice and Bob meet first, (BC, AB) for the sequence in which Bob and Charlie meet first, and (ABC) for the centralised market. Each date in the figure is one meeting. The figure shows that the allocations of goods converge under both orders. More precisely, the final allocation of the non-numeraire good (x) coincides with that of the centralised market. However, since each group trades at its own price, the final allocation of the numeraire (y) differs, and it depends on the order of meetings. The case of Alice illustrates this. Alice values the good x most, so Alice buys it from Bob in every meeting. Under (BC, AB), Bob first buys it from Charlie. When Alice meets Bob for the first time, Bob holds more of the good x and values it less. Alice pays a lower price than under (AB, BC), and ends up

[1] In an over-the-counter market, participants trade directly with each other rather than through a central exchange. Prices are determined locally, so two transactions in the same asset may take place at different prices.

with more y than under that order. Moreover, under both orders, every price at which Alice buys is above the price of the centralised market, so Alice ends up with less y than in the centralised market. Charlie is in the opposite position. Charlie sells x in every meeting, always below the price of the centralised market, so Charlie also ends up with less y than in the centralised market. Bob ends up with more y under both orders, because Bob buys x from Charlie below the price of the centralised market and sells it to Alice above that price. However, Bob's position alone does not guarantee this gain.[2] These differences are relevant for policy design. A regulator could change the order of meetings or set up a more centralised market, in which larger groups meet. In this example, either decision changes who holds the numeraire.

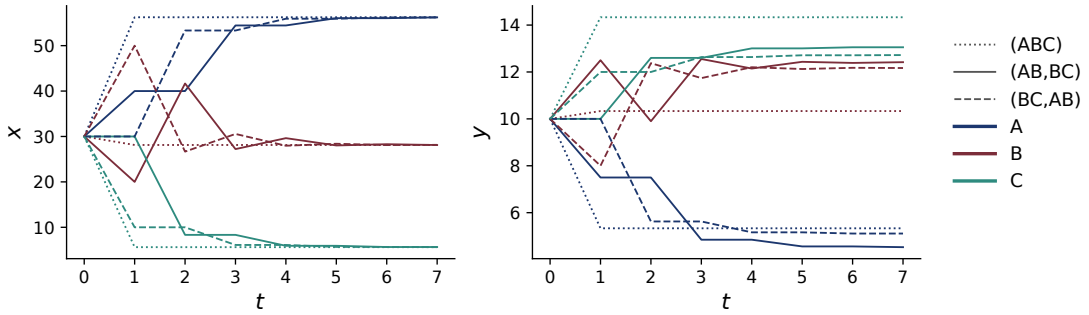


Figure 1: Allocations of the non-numeraire good (x, left) and of the numeraire (y, right) under the two orders of meetings and under the centralised market. Agents' utility is $u^i(x^i, y^i) = a^i \log(x^i) + y^i$ ($a^A = 10, a^B = 5$ and $a^C = 1$) and their endowment is $(\omega_x^i, \omega_y^i) = (30, 10)$.

This paper studies decentralised trade in networks through a sequence of Walrasian[3] exchanges. Each exchange takes place within a clique of the network, that is, a group of agents who are all connected to each other.[4] The allocation after each exchange becomes the endowment for the next one. Agents have quasilinear HARA preferences with common curvature, so the demand for the non-numeraire goods does not depend on wealth. I take the sequence of meetings as given, so no agent chooses when or with whom to trade. Moreover, agents are myopic, that is, they trade at each meeting without anticipating later meetings. Section 2 states the full set of assumptions.

I show that the allocation of every non-numeraire good converges to the

[2]With $a^A = 6$, $a^B = 1$ and $a^C = 10$ and the same endowments, Bob ends up with more y than in the centralised market under (AB, BC), and with less under (BC, AB).

[3]Agents take prices as given, and prices adjust until demand equals supply.

[4]The cliques of a network are one case. In general, the model takes as given a family of admissible coalitions, that is, a fixed collection of groups that are allowed to convene. The results do not depend on which family this is.

allocation of the centralised market. This result holds for any network and any sequence in which the relationships used infinitely often connect the economy, so that goods can move between any two agents. Therefore, every sequence ends at a Pareto efficient allocation. However, each clique trades at its own price, so the final allocation of the numeraire depends on the sequence. Hence, the sequence affects only the speed of convergence and the distribution of the numeraire. Finally, I show that larger groups need not accelerate convergence or reduce redistribution.

The sequence affects the numeraire because some agents belong to more than one clique. Without such agents, the economy is a collection of separate markets, and each market clears on its own. In the example, Bob belongs to two cliques. Under (AB, BC), Bob sells x to Alice first, so Bob holds less x at the meeting with Charlie. Therefore, the price in that meeting is higher than if Bob had met Charlie first. Moreover, until the economy converges, the price in a meeting can differ from the price of the centralised market. These price differences generate transfers of numeraire among the agents. Each exchange conserves the numeraire, so what some agents gain, the others lose.

The convergence of the non-numeraire goods allows me to describe the evolution of the numeraire. The centralised market gives each agent an allocation of non-numeraire goods and an amount of numeraire. Along every sequence, the non-numeraire goods converge to the first of these. Therefore, I measure each agent's holding of these goods at every date by its difference from this allocation, and I call this difference the *disequilibrium*. The *transfer* to an agent is the difference between the numeraire it holds in the limit and the amount that the centralised market would give it. This transfer equals the sum, over all meetings, of the quantity that the agent trades multiplied by the difference between the local price and the price of the centralised market.

The paper makes three contributions. *First, convergence under a weak hypothesis on the sequence of trades.* Under quasilinearity, all Pareto optimal allocations have the same allocation of non-numeraire goods. Therefore, convergence of these goods to the centralised allocation is equivalent to efficiency of the limit, and the efficiency of decentralised exchange is a classical question. Feldman (1973) shows that bilateral trade reaches efficiency when every pair of traders meets in each round of bounded length. Madden (1975) allows larger groups. That paper proves that a suitable order of meetings takes the economy arbitrarily close to efficiency, and that a fixed cycle of meetings does so under stronger conditions. The convergence result stated above requires only that the relationships used infinitely often connect the economy (Theorems 1 and 2). Therefore, the time between two meetings of a clique may grow without bound, and some cliques may never meet. In exchange, I restrict preferences to the HARA family and each meeting to Walrasian clearing. Under these restrictions, each exchange is a projection, so

the classical theory of alternating projections applies (Halperin, 1962; Amemiya and Ando, 1965). Moreover, every trade brings the economy weakly closer to the centralised allocation, whatever the order of trade.

Second, a closed form for the redistribution. The transfer is an infinite sum over all meetings, and each term depends on the full history before it. I obtain a closed-form approximation of the transfer in two cases, when the same finite sequence of cliques repeats indefinitely and when cliques are drawn at random. In the first case, I call the repeated sequence a *trading round*, and I require that it include every agent and connect the economy. Each term of the sum is a product of two factors that are approximately linear in the initial disequilibrium, so the approximation is quadratic in it. Its matrix is the unique solution of a system of linear equations, a discrete Lyapunov equation (Theorem 3). The size of this system depends only on the number of agents, and not on the length of the round. The approximation is exact for quadratic preferences, and for other HARA preferences its error is of third order in the initial disequilibrium.

In the second case, cliques are drawn independently from a fixed distribution, and I require that the cliques with positive probability connect the economy. The approximation of the expected transfer is again quadratic in the initial disequilibrium, and its matrix solves an equation of the same form as in the first case (Theorem 4 and Corollary 8). Moreover, this expectation is in general not zero. Therefore, randomising who trades does not restore, even in expectation, the numeraire allocation of the centralised market.

Along every sequence, most of the redistribution takes place in the early meetings, when local prices are still far apart. Near the limit, every clique trades close to the price of the centralised market, and little numeraire is redistributed.

Third, a comparison of trading structures. I ask whether a clique that meets as one group makes the economy converge faster, or redistribute less, than smaller cliques inside it. To answer, I replace one clique of a sequence by smaller cliques inside it, which together include all its members and connect them, and I call the resulting sequence a *refinement*. I compare a sequence with its refinement by measures of speed and of redistribution that depend only on the network, the order of meetings and preferences, so the comparison does not depend on endowments.

For speed, I find that, immediately after the whole clique has met, the economy is at least as close to the centralised allocation as under any refinement (Proposition 10). However, over longer horizons, when a trading round repeats, a refinement can converge at a strictly faster rate (Proposition 12). The reason is that the whole clique removes the part of the disequilibrium that would disappear quickly in any case, and the slow part remains. In contrast, when cliques are drawn at random, a refinement does not converge at a faster rate in expectation (Proposition 13). The reason is that a random sequence averages the exchanges,

whereas a round applies them one after another.

For redistribution, I measure how much a trading round redistributes by the sum of the absolute transfers across agents. The *redistribution capacity* of a round is the largest value of this sum over initial disequilibria of a given size. Speed does not determine redistribution. In the example above, the two orders converge at the same rate, and the transfers differ. Moreover, a refinement can raise capacity or lower it. The reason is that redistribution depends also on the cliques in which each agent meets others, and dividing a clique can change these in either direction.

These results have implications for regulation. A regulator concerned only with efficiency can be indifferent to the trading structure. Whether agents trade in one centralised market or through a connected network of smaller markets, the allocation of non-numeraire goods converges to the same one. A regulator concerned with speed faces a trade-off over time. Merging smaller markets into a larger one brings the economy weakly closer to the centralised allocation in the current exchange, but a more fragmented structure can converge faster when the same sequence of trades repeats. This trade-off disappears when the order of trade is drawn at random. Finally, a regulator concerned with distribution cannot be indifferent to the trading structure, because the order of trade affects who ends up holding the numeraire. Nevertheless, no simple rule, such as merging markets, controls this effect. Therefore, such a regulator must evaluate each trading structure directly, and the closed form of the transfer and the redistribution capacity make this evaluation possible.

The paper relates to several strands of literature, which Section 5 reviews in detail. The first strand studies the efficiency of sequential exchange (Feldman, 1973; Madden, 1975). As discussed above, I restrict preferences and obtain convergence under a weaker condition on the sequence, which may never repeat and may be random. The second strand studies trade in networks. In models of trading networks (Hatfield et al., 2013) and of decentralised exchanges (Malamud and Rostek, 2017), all trades take place at the same time. Finster et al. (2026) add dynamics to trading networks, in which agents adjust offers or prices until the outcome is stable. In all these models, goods change hands only once, at the final outcome. Here, each exchange is carried out, and the resulting holdings are the endowments of the next exchange. Models of strategic trade and intermediation find that trade in a network can be inefficient (Gofman, 2014; Manea, 2018). Here, preferences are quasilinear and every meeting is efficient for its members, so the limit allocation of goods is efficient on every network. The closest paper is Cassese and Pin (2025). They study a continuous-time Edgeworth process in which agents trade with their neighbours in a weighted network, and every trade divides the gain in utility equally. Given the endowments, each network leads to a different Pareto efficient allocation of all goods, and they study this relation by simulation.

In mine, local Walrasian exchange keeps the dynamics tractable, and quasilinearity confines the effect of the network to the numeraire. The third strand provides the tools. The disequilibrium follows a consensus process (DeGroot, 1974; Golub and Jackson, 2010), and the random case builds on randomised gossip (Boyd et al., 2006; Fagnani and Zampieri, 2008). In a consensus process, the network determines the limit. Here, preferences determine the limit, and the network determines the numeraire that agents transfer on the way.

The paper proceeds as follows. Section 2 presents the model and establishes the convergence of the non-numeraire goods. Section 3 studies the redistribution of the numeraire and obtains its closed-form approximation. Section 4 compares trading structures by speed and by redistribution. Section 5 reviews the related literature, and Section 6 concludes. All proofs are in the Appendix.

2 A Model of Sequential Exchange

Consider a pure exchange economy with $n \geq 2$ consumers and $L+1 \geq 2$ goods $(x_1, \dots x_L, y)$. Agent i holds an initial endowment $\boldsymbol{\omega}^i = (\omega_1^i, \dots, \omega_L^i, \omega_y^i) \in X := \mathbb{R}_+^L \times \mathbb{R}$ and has preferences over bundles $(\boldsymbol{x}^i, y^i) \in X$ represented by the additively separable, quasilinear utility

$$u^i(\boldsymbol{x}^i, y^i) = \sum_{l=1}^{L} a_l^i \frac{1-\gamma_l}{\gamma_l} \left(\frac{x_l^i}{1-\gamma_l} + b_l^i \right)^{\gamma_l} + y^i. \tag{1}$$

The utility that agents derive from each good l belongs to the Hyperbolic Absolute Risk Aversion (HARA) family. The parameter $a_l^i > 0$ scales agent i's marginal utility for good l relative to the other agents, and b_l^i sets a baseline requirement, measured in units of the good. The curvature γ_l is common across agents. It fixes the demand elasticity for good l, so that agents differ in the *intensity* a_l^i and the *baseline* b_l^i of their demand, and not in its *shape*. This common-γ_l restriction is the substantive assumption on preferences, since it renders the exchange affine in endowments (Lemma 1) and so makes the sequential dynamics explicit. It is satisfied by the standard specifications of trade (CES demand) and finance (CRRA utility), both special cases of HARA. Two limits of Equation (1) are also useful, although neither is a member of the family. The logarithmic case is the limit $\gamma_l \to 0$, and the CARA case is the limit $\gamma_l \to -\infty$, with their local equilibria in Remarks 4 and 5 of Appendix A. Every expression below applies to the logarithmic case with $\gamma_l = 0$.[5]

Because preferences are additively separable across goods, each good l admits

[5]Readers arriving from the CRRA literature should note that log utility corresponds to $\gamma_l = 0$ in this parametrisation, while $\gamma_l = 1$ is excluded from the family.

its own exchange operator and the dynamics run in parallel, coupled only through the conserved numeraire. Therefore, every result below holds good by good. I state the analysis for a representative good and suppress the subscript l where no ambiguity arises.[6] The numeraire enters utility linearly, and the remaining goods enter nonlinearly. The economy evolves in discrete time. At each date a single admissible coalition meets and clears among its members, the allocation it produces is the endowment its members carry into the next date, and everyone outside the coalition is untouched. The trading structure is fixed throughout, and what varies from one sequence to another is only which coalition meets when. A state is *stable* when no two agents who share an admissible coalition can gain from trading, that is, when their marginal rates of substitution agree within every admissible coalition. If the admissible coalitions connect the economy, equal rates propagate from one coalition to the next, so they agree across the economy. Since every exchange conserves the aggregate holding of each non-numeraire good, a stable state gives each of them the allocation of the centralised market.

Trading Structures. Which groups of agents may trade together is a primitive of the economy. I represent it as a family $\Omega \subseteq 2^N$ of *admissible coalitions*: a set $M \in \Omega$ is a group of agents who can convene and clear a local competitive equilibrium among themselves. I assume that the admissible coalitions connect N, in the sense that the agents cannot be split into two nonempty groups such that no admissible coalition contains members of both. The leading case, and the one that supplies the examples throughout the paper, is the connected *trading network* $G = (N, E)$, in which an undirected link $(i, j) \in E$ represents a bilateral trading relationship and the admissible coalitions are the *cliques* of G: the complete subgraphs of G on two or more agents,

$$\Omega(G) = \Big\{ M \subseteq N : |M| \geq 2,\ (i,j) \in E \text{ for all } i, j \in M \Big\}.$$

A group can clear jointly precisely when its members are already pairwise linked. OTC dealer networks, correspondent-banking relationships and trade blocs constrain exchange in this way, since agents in these markets trade within their relationships. Two features of $\Omega(G)$ matter for what follows. First, the clique family is a genuine restriction on the admissible family. A general Ω may contain a coalition $\{i, j, k\}$ whose members trade only together and never in pairs, for example a group that meets under a broker. No network represents this, because a clique containing i, j and k also contains the links (i, j), (j, k) and (i, k). Second, the clique family is closed under refinement: every clique decomposes into smaller

[6] Separability lets each good's market run on its own trading structure. A natural specialisation assigns each good its own network G_l, distinct dealer networks for distinct goods, coupled through a common vehicle numeraire.

cliques, in the limit into the edges of G, which are themselves admissible. This renders the refinement comparison of Section 4 unconditional under a network; on a general family it holds wherever the refining sub-cliques are admissible. My convergence and redistribution results require none of this, since each uses only the coalitions that appear in a given sequence. They therefore hold for an arbitrary admissible family Ω. A link records only that a relationship exists, and the terms of any single trade are fixed by market clearing within the coalition. The examples are all on networks, so I call their coalitions cliques, and every statement about cliques should be read as a statement about admissible coalitions.

Behaviour and Timing. Four assumptions are maintained throughout. First, agents are price takers, and each meeting clears at its Walrasian equilibrium. Second, the sequence of meetings is exogenous, so no agent chooses when or with whom to trade. Third, agents are myopic, in the sense that each coalition clears at the allocation that its members prefer at that date, without regard to the coalitions that meet later. Fourth, preferences are additively separable and quasilinear, with the common curvature γ_l of Equation (1). Two conditions on the domain complete the set-up. The bracket in Equation (1) is positive at the endowment, and every local equilibrium is interior. Lemma 15 in Appendix A shows that the first condition then holds at every date, and gives two cases in which the second holds automatically.[7] Finally, the analysis is sequential, with one coalition per date.

The remainder of this section proceeds as follows. I first solve for the competitive equilibrium within a single coalition and define the *exchange operator* that maps pre-trade to post-trade holdings. I then compose these operators along a finite sequence, define the trading round, and establish convergence of the non-numeraire goods to the centralised Walrasian allocation, whatever the round and whatever the order within it. Third, I replace periodicity by recurrence and extend the convergence result to arbitrary infinite sequences.

2.1 Local Exchange

To understand the competitive equilibrium of a socially restricted exchange economy, I first study what happens locally in the network. Trades within a clique satisfy the assumptions of a classical exchange economy, since there are no network restrictions among the connected agents. The central question is how the local competitive equilibria, within the cliques of the network, relate to the global competitive equilibrium of the socially unrestricted economy. I answer it in two steps. First, I

[7]The first condition is equivalent to $z_l^i/(1-\gamma_l) > 0$ for the shifted holding $z_l^i = x_l^i + (1-\gamma_l)b_l^i$ of Theorem 1. Hence shifted holdings and their clique aggregates carry the sign of $1-\gamma_l$.

characterise the local equilibrium within a single clique (Lemma 1). Second, I define the operator that maps endowments to local equilibrium allocations (Definition 1).

Lemma 1 (Equilibrium of a Local Economy)**.** *Let $M \subseteq N$ be a clique of the network G. The Walrasian equilibrium in M is given by:*

$$
\begin{aligned}
x^i(M;l) &= \frac{\alpha_l^i}{\alpha(M;l)} \sum_{j\in M} \omega_l^j + (1-\gamma_l)\left(\frac{\alpha_l^i}{\alpha(M;l)} \sum_{j\in M} b_l^j - b_l^i\right), \\
p(M;l) &= \left[\frac{\sum_{j\in M} \omega_l^j}{(1-\gamma_l)\,\alpha(M;l)} + \frac{\sum_{j\in M} b_l^j}{\alpha(M;l)}\right]^{\gamma_l - 1}, \\
y^i(M) &= \omega_y^i + \sum_{l\le L} p(M;l)\left(\omega_l^i - x^i(M;l)\right),
\end{aligned}
$$

where $\alpha_l^i := (a_l^i)^{1/(1-\gamma_l)}$, $\alpha(M;l) := \sum_{j\in M} \alpha_l^j$.

I assume that each clique settles at its Walrasian equilibrium. Appendix B justifies this assumption. Efficiency alone determines the allocation of the goods, so the assumption matters only for the numeraire. Moreover, $p(M)$ is the only price that clears the meeting, the resulting gains lie in the core of the meeting (Lemma 16), and the rule adds no bargaining primitive to the model.

Lemma 1 describes how the agents' endowments are transformed into their equilibrium allocations within a clique. If N itself were a clique of the network, the allocation $\left((\boldsymbol{x}^i(N), y^i(N))_{i\in N}, \boldsymbol{p}(N)\right)$ would be the competitive equilibrium of the unrestricted economy. This constitutes the benchmark against which I compare the dynamics of the socially constrained economy.

Two features of the local equilibrium matter for what follows. First, the equilibrium allocation of each non-numeraire good is an *affine* transformation of the aggregate local endowment, which I use to study the dynamics of the goods. Second, the equilibrium allocation of the numeraire is *not affine*, since it enters through the local price in the budget constraint. This interaction makes the numeraire path-dependent.

I now isolate the affine part. For a clique M and good l, let $\mathcal{A}(\cdot\,; M, l)$ denote the transformation that takes the endowments of good l and returns the Walrasian allocations after a local exchange among the agents in M. I call it the *exchange operator* of l in M (Definition 1), and its linear part the *exchange matrix* $A(M;l)$ of Equation (2). The exchange matrix is row-stochastic. It redistributes the endowment of good l among the clique members and leaves non-participants unchanged, through the unit entries on its diagonal. Example 1 illustrates it.

Definition 1 (Exchange Operator)*.* Let $M \subseteq N$ be a clique of G and let $\boldsymbol{\omega}_l$ be the vector of endowments of good l. The exchange operator of l in M is the affine

transformation

$$\mathcal{A}(\cdot\,; M, l) : \mathbb{R}^n \to \mathbb{R}^n$$
$$\boldsymbol{\omega}_l \mapsto \boldsymbol{\omega}_l\, A(M; l) + (1 - \gamma_l)\, \mathbf{b}_l\Big(A(M; l) - I\Big),$$

where I is the $n \times n$ identity, $\mathbf{b}_l = (b_l^1, \ldots, b_l^n)$, and $A(M; l)$ is the $n \times n$ matrix

$$A(M; l)_{i,j} = \begin{cases} \dfrac{\alpha_l^j}{\alpha(M; l)} & \text{if } i, j \in M, \\ 1 & \text{if } i = j \text{ and } i, j \notin M, \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Remark 1. Let $M \subseteq N$ be a clique of G and $\boldsymbol{\omega}_l$ the endowment vector of good l. Then $\mathcal{A}(\boldsymbol{\omega}_l; M, l) = x(M; l)$.

Example 1. Let $N = \{A, B, C\}$ and $u^i(x, y) = a^i \log(x) + y$. The exchange operator of l in the clique AB induces the equilibrium allocations

$$\boldsymbol{\omega}_x\, A(AB) = \begin{pmatrix} \omega_x^A & \omega_x^B & \omega_x^C \end{pmatrix} \begin{pmatrix} \frac{a^A}{a^A + a^B} & \frac{a^B}{a^A + a^B} & 0 \\ \frac{a^A}{a^A + a^B} & \frac{a^B}{a^A + a^B} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} x^A(AB) & x^B(AB) & \omega_x^C \end{pmatrix}. \qquad \diamond$$

2.2 Finite Sequences and Trading Rounds

With the local terms of trade defined, I now build global dynamics from them by tracking good l through a sequence of local exchanges. Consider a finite sequence of cliques $\mathcal{C} = (M_1, \ldots, M_K)$. The agents in M_1 trade, and the resulting allocation becomes the endowment that society carries into the exchange M_2. This repeats through the last clique of $\mathcal{C}$. The social allocation after the sequence is therefore the composition of the local operators as

$$\mathcal{A}(\cdot\,; \mathcal{C}; l) = \mathcal{A}(\cdot\,; M_K; l) \circ \cdots \circ \mathcal{A}(\cdot\,; M_1; l),$$

with linear part $A(\mathcal{C}; l) = A(M_1; l) \cdots A(M_K; l)$.

The outcome of an arbitrary sequence need not be an equilibrium of the global economy, since $\mathcal{C}$ may involve only some of the agents. Traders left out may still wish to exchange. To define a global notion, I restrict attention to sequences that reach every trader, the *trading rounds* of Definition 2. I call the sequence operator of a trading round a *trading round operator*, and Example 2 displays one.

Definition 2. A *trading round* is a finite sequence of cliques $\mathcal{R} = (M_1, \ldots, M_K)$ that spans every agent and whose cliques connect N, in the sense that the agents cannot be split into two nonempty groups sharing no clique of the round.[8]

[8]The condition is on the cliques of the round, not on G: the sequence (AB, CD) spans four agents and fails it, whatever links G happens to contain. For a general admissible family Ω the same definition applies with "clique" read as "admissible coalition".

Example 2. Consider three agents in a line B–A–C and the trading round $\mathcal{R} = (AB, AC)$. Let $u^i(x, y) = a^i \log(x) + y$ be the utility function of agent i. Then, the trading round operator is

$$\mathcal{A}(\boldsymbol{\omega}_x\,;\mathcal{R},l) = \begin{pmatrix} \omega_x^A & \omega_x^B & \omega_x^C \end{pmatrix} \begin{pmatrix} \frac{a^A}{a^A+a^B} & \frac{a^B}{a^A+a^B} & 0 \\ \frac{a^A}{a^A+a^B} & \frac{a^B}{a^A+a^B} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{a^A}{a^A+a^C} & 0 & \frac{a^C}{a^A+a^C} \\ 0 & 1 & 0 \\ \frac{a^A}{a^A+a^C} & 0 & \frac{a^C}{a^A+a^C} \end{pmatrix}$$

$$\boldsymbol{\omega}_x\, A(\mathcal{R};l) = \begin{pmatrix} \omega_x^A & \omega_x^B & \omega_x^C \end{pmatrix} \begin{pmatrix} \frac{a^A a^A}{(a^A+a^B)(a^A+a^C)} & \frac{a^B}{a^A+a^B} & \frac{a^A a^C}{(a^A+a^B)(a^A+a^C)} \\ \frac{a^A a^A}{(a^A+a^B)(a^A+a^C)} & \frac{a^B}{a^A+a^B} & \frac{a^A a^C}{(a^A+a^B)(a^A+a^C)} \\ \frac{a^A}{a^A+a^C} & 0 & \frac{a^C}{a^A+a^C} \end{pmatrix} \qquad \diamond$$

Under what conditions does a trading round drive society to a stable allocation of the non-numeraire goods, and hence to a global equilibrium? A single pass need not suffice. However, repeated passes always do, since iterating the round equalises marginal rates of substitution across the network. Theorem 1 establishes this, and the mechanism rests on the following four properties of the exchange matrices. (i) Each $A(M;l)$ is a *projection*, applying it twice equals applying it once, since agents who have optimised locally have no further incentive to trade, and its fixed points are the allocations in which the members' holdings are proportional to their weights. (ii) Each $A(M;l)$ is *row-stochastic*, so good l is conserved within every exchange. (iii) Products of row-stochastic matrices are row-stochastic, so the composed operator conserves good l, while each exchange in the sequence clears its own market. (iv) Because the round covers all agents and induces a connected subgraph, the composed operator is *irreducible and primitive*. By Perron–Frobenius, its iteration converges to a unique limit, which redistributes endowments according to the agents' relative preference weights.

Theorem 1 (Convergence of the Trading Round)**.** *Let $\mathcal{R} = (M_1, \ldots, M_K)$ be a trading round of G, and let $\mathcal{A}(\cdot\,;\mathcal{R};l)$ be the exchange operator for good l, with linear part $A(\mathcal{R};l)$. Then for each non-numeraire good l:*

(i) ***(Regularity)*** *$A(\mathcal{R};l)$ is row-stochastic, irreducible, and primitive, with spectral radius $\rho(A(\mathcal{R};l)) = 1$.*

(ii) ***(Unique Stationary Distribution)*** *$A(\mathcal{R};l)$ has a unique stationary distribution $\boldsymbol{\pi}_l \in \mathbb{R}^n$, with $\pi_l^i > 0$ and $\sum_i \pi_l^i = 1$, given by*

$$\pi_l^i = \frac{\alpha_l^i}{\alpha(N;l)}. \tag{3}$$

(iii) ***(Convergence)*** *For any $\boldsymbol{\omega}_l \in \mathbb{R}_+^n$, the allocations induced by the periodic repetition of $\mathcal{R}$ converge:*

$$\mathcal{A}^t(\boldsymbol{\omega}_l\,;\mathcal{R},l) \xrightarrow{t\to\infty} \boldsymbol{x}(N;l),$$

the competitive allocation of good l in the unrestricted economy (Lemma 1 at the clique N). Explicitly,

$$\boldsymbol{x}(N;l) = \boldsymbol{\omega}_l\,\Pi_l + \mathbf{d}_l(\Pi_l - I), \qquad \Pi_l = \mathbf{1}\boldsymbol{\pi}_l, \quad \mathbf{d}_l = (1-\gamma_l)\mathbf{b}_l. \tag{4}$$

(iv) ***(Geometric Rate)*** *Convergence is governed by the modulus of the second-largest eigenvalue of $A(\mathcal{R};l)$:*

$$\left\|\mathcal{A}^t(\boldsymbol{\omega}_l\,;\mathcal{R},l) - \boldsymbol{x}(N;l)\right\| = O\Big(t^{\,m-1}\,|\lambda_2(A(\mathcal{R};l))|^t\Big)\,,$$

where m is the index of λ_2, the size of the largest Jordan block attached to it. Since $A(\mathcal{R};l)$ has rank at most $n-\max_k|M_k|+1$ and 1 is a simple eigenvalue, $m \le n-\max_k|M_k|$; in particular $m=1$, and the bound is exactly geometric, whenever some clique of the round has at least $n-1$ members.

The stationary distribution (3) is the vector of the agents' preference weights in the *grand* coalition N, and it does not depend on the round. Therefore, every trading round redistributes good l according to the same shares $\pi_l^i = \alpha_l^i/\alpha(N;l)$, and its iteration reaches the centralised allocation (4), whatever the topology of G and whatever the order within the round. The *rate* does depend on the round, through $\lambda_2(A(\mathcal{R};l))$. Section 4 examines how the trading structure determines this rate.

Theorem 1 assumes that the round repeats in a fixed order. The classical sequential-exchange literature assumes the same kind of periodicity. Feldman (1973) obtains efficiency when every pair of traders meets in each round of bounded length. For larger groups, Madden (1975) shows that some sequence of meetings approaches efficiency, and that cyclical trade also does under stronger assumptions. Goldman and Starr (1982) ask instead when t-wise optimality implies Pareto optimality, which is a static question, and they leave the dynamics open. Relative to this literature, the analysis so far is a special case in two respects. I restrict preferences to the common-γ HARA family, and each meeting to Walrasian clearing rather than to an arbitrary Pareto-improving trade. In exchange, I obtain convergence to a unique limit, the same for every trading round, at a geometric rate.

Example 3 depicts the convergence of Theorem 1 across the HARA family, and Example 4 across two goods, whose dynamics run in parallel by additive separability.

Example 3. Consider four agents on a line A–B–C–D, under four preference scenarios: (i) logarithmic ($\gamma = 0, b = 0$); (ii) CRRA with ($\gamma = 0.5, b = 0$); (iii) CRRA with ($\gamma = -1, b = 0$) (more concave); (iv) quadratic with $\gamma = 2$ and $b^i = 35$ for every i.[9] Across scenarios $\boldsymbol{a} = (2,5,10,3)$ for agents A to D, and $\boldsymbol{\omega}^i = (30,10)$ for every i. Figure 2 shows the first eight stages of the repeated trading round (AB, BC, CD). ◇

[9]Scenario (iv) needs a positive baseline. With $b=0$, the bracket in Equation (1) is negative at the endowment. With $b^i = 35$, the bracket stays positive and every allocation stays interior along the path.

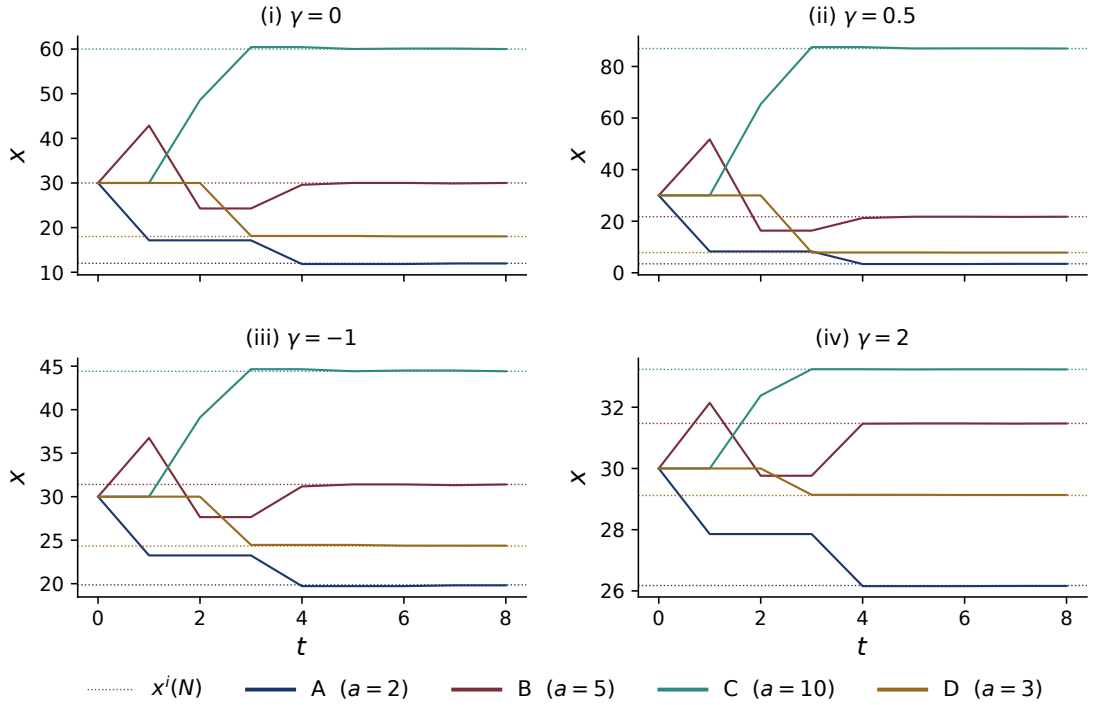


Figure 2: Dynamics of allocations for a family of HARA utility functions.

Example 4. Consider four agents on a star centred at A, with $L = 2$ non-numeraire goods and $u^i = a_1^i \log(x_1^i) + a_2^i \log(x_2^i) + y^i$, where $(a_1^i) = (2, 5, 10, 3)$ and $(a_2^i) = (8, 3, 1, 6)$ for agents A to D. Each agent i is endowed $\boldsymbol{\omega}^i = (30, 30, 10)$. The dynamics under $\mathcal{R} = (AB, AC, AD)$ appear in Figure 3. By separability, each non-numeraire good converges to its own centralised allocation $x_l^i(N)$. The numeraire also converges, and its limit differs from $y^i(N)$. ◇

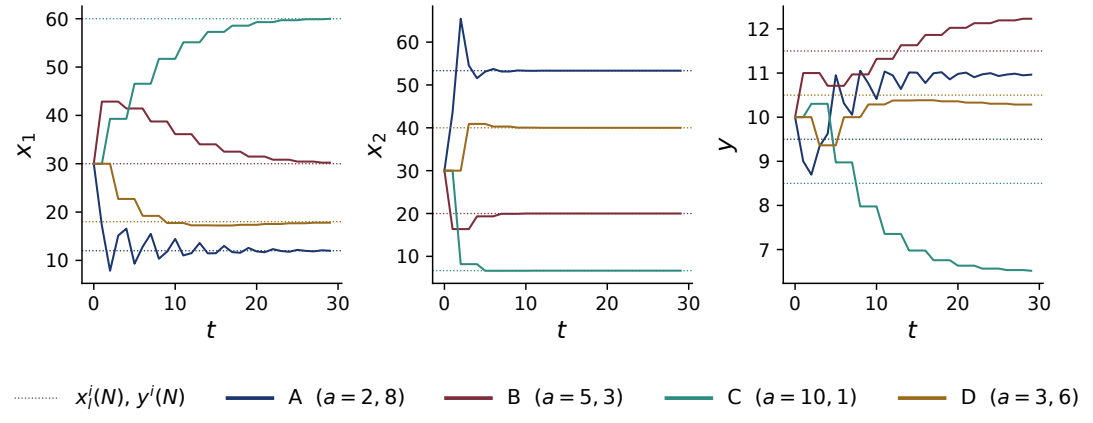


Figure 3: Dynamics of allocations with multiple goods. The dotted lines are the centralised benchmarks $x_l^i(N)$ and $y^i(N)$.

2.3 General Sequences of Trades

Periodicity is a strong and unrealistic assumption. A dealer calls counterparties in varying order, a consumer browses listings opportunistically, and bilateral negotiations follow no fixed calendar. On this dimension the analysis extends the classics rather than specialising them. Theorem 2 dispenses with periodicity, requiring only that the relationships which persist connect every agent, not that trade follow a fixed schedule. The condition is weaker than a bounded cycle, since the time between two uses of a clique may grow without bound, and not every admissible clique need ever be active.

Theorem 2. *Let $\mathcal{C} = (M_{(t)})_{t\in\mathbb{N}}$ be an infinite sequence of cliques in G such that the cliques appearing infinitely often cover N and connect it. Then for every non-numeraire good l,*

$$\boldsymbol{x}(M_{(t)};l) \xrightarrow{t\to\infty} \boldsymbol{x}(N;l), \qquad \textit{equivalently} \qquad \prod_{k\le t} A(M_{(k)};l) \xrightarrow{t\to\infty} \Pi_l = \mathbf{1}\boldsymbol{\pi}_l,$$

with $\boldsymbol{\pi}_l$ the stationary distribution of Theorem 1.

Remark 2 (Block Decomposition). Let $\mathcal{C}$ satisfy the hypothesis of Theorem 2, and let Ω_∞ be the set of cliques used infinitely often. After some date, $\mathcal{C}$ splits into consecutive blocks, each of which uses every clique of Ω_∞ and no other. Every block reduces the disequilibrium by at least a factor $q < 1$ that depends only on Ω_∞ (Lemma 21). Hence $\|E(t;l)\|_\pi \le q^{k(t)}$, where $k(t)$ is the number of blocks completed by date t. The rate is per block, not per trade. A block may be arbitrarily long, and it need not contract as much as a trading round it contains, since the trades interleaved with the round can weaken the contraction.

The economic content is that convergence needs neither coordination nor a schedule. The process self-corrects so long as the relationships that persist, those used infinitely often, keep the network connected. Individual links may fall dormant, or be abandoned entirely, without disrupting convergence. Each exchange carries every non-numeraire good no further from the centralised allocation, regardless of the order in which exchanges occur. This places the model in the tradition of non-tâtonnement, or Edgeworth barter, processes (Uzawa, 1962; Hahn and Negishi, 1962), from which it departs in one respect. There, trade proceeds at no market-clearing price, whereas here every exchange clears a competitive equilibrium of a sub-economy. Each clique trades at its own Walrasian price, and disequilibrium is a matter of these local prices differing across cliques, not of trade occurring out of equilibrium. Cassese and Pin (2025) study an Edgeworth process on a weighted network under general preferences. Since its dynamics has no analytical solution, they study the effect of the network on welfare by simulation.

The common-γ HARA restriction keeps the dynamics tractable and confines the influence of the network to the numeraire.

Theorem 1 came with an explicit rate, while Theorem 2 does not. Remark 2 gives a factor per block, but this factor has no closed form, and a block may take arbitrarily many dates. The question that survives is how far from equilibrium the economy stands at a given date. I therefore measure that distance in the norm induced by the inner product of Definition 3. Allocations act on the exchange matrices from the right, and weighting each agent by the reciprocal of its equilibrium share makes every exchange an orthogonal projection.

Definition 3 (Equilibrium-Weighted Inner Product). For row vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ set

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle_\pi := \sum_{i \in N} \frac{u^i v^i}{\pi^i}, \qquad \|\boldsymbol{u}\|_\pi^2 = \sum_{i \in N} \frac{(u^i)^2}{\pi^i}.$$

The weight $1/\pi^i$ makes a disequilibrium count for more when the agent's equilibrium share is smaller.

Write $A(t;l) = A(M_{(1)};l) \cdots A(M_{(t)};l)$ for the exchange accumulated up to date t, with $A(0;l) = I$, and $E(t;l) = A(t;l) - \Pi_l$ for its error. Since $\boldsymbol{\pi}_l A(M;l) = \boldsymbol{\pi}_l$ for every clique, $\Pi_l A(M;l) = \Pi_l$, and errors therefore compose along the sequence: $E(t;l) = E(t-1;l)\, A(M_{(t)};l)$. Each exchange matrix is idempotent and self-adjoint for $\langle \cdot, \cdot \rangle_\pi$, hence the orthogonal projection onto its range and non-expansive (Lemma 18 in the appendix). For a matrix E, write $\|E\|_\pi = \sup\Big\{ \|\boldsymbol{v}E\|_\pi : \|\boldsymbol{v}\|_\pi = 1 \Big\}$ for the operator norm induced by this inner product, with the operator acting on the right, as allocations do throughout. With $D = \operatorname{diag}(\boldsymbol{\pi})$, $\|E\|_\pi$ is the largest singular value of $D^{1/2} E D^{-1/2}$, which gives a direct way to compute it. The bound follows.

Proposition 2 (Monotone Convergence Bound)**.** *Along any sequence of cliques and for every non-numeraire good l,*

$$\|E(t;l)\|_\pi \;\le\; \|E(t-1;l)\|_\pi,$$

so the distance to the centralised allocation never increases, whichever cliques trade and in whatever order.

Two later results rest on this bound. Section 3 uses it to control the rate at which redistribution accumulates along a sequence that has no rate of its own. Section 4 uses it to compare trading structures, which the spectral gap cannot do, since the gap is defined only for a round repeated indefinitely, while $\|E(t;l)\|_\pi$ is defined along any sequence and at every date.

A natural next step randomises the sequence, drawing each clique from a distribution over the admissible family. This renders the effect of trading order on the numeraire (Equation (5)) tractable in expectation, and I take it up in Section 3.

Trades on separate parts of the network do not queue in practice, and two disjoint cliques may clear simultaneously without affecting any allocation. Disjoint exchange matrices commute, and each clique's price depends only on its own members. The model keeps sequential clearing for tractability. The distinction is immaterial for the allocations and their limits, although it changes the number of dates, and hence the per-trade statistics of Section 4.1.

3 The Redistribution of the Numeraire

Section 2 characterises the convergence of the non-numeraire goods. Every sequence in which the relationships used infinitely often connect the economy converges to $\boldsymbol{x}(N)$. Therefore the allocation of the non-numeraire goods is determined. These results do not determine the allocation of the numeraire. Each exchange settles at the price that clears the clique, and not at the price that would clear the whole economy, as a centralised market would. The difference between the two prices accumulates into a transfer of numeraire across agents, which survives in the limit.

To see this, fix an agent i and let $M_{(t)}$ be the clique active at period t. Since the period-$(t-1)$ allocations are the endowments at t, the budget constraint can be rewritten as

$$y^i(t) = y^i(t-1) - \sum_{l=1}^{L} p(M_{(t)};l)\Big[x^i(t;l) - x^i(t-1;l)\Big].$$

The change in the numeraire vanishes as $x^i(t)$ converges, so $y^i(t)$ converges. Write $y^i(\mathcal{C})$ for its limit along the sequence $\mathcal{C}$. Summing the budget constraint over dates and valuing each trade at the centralised price gives the decomposition below. The terms in $p(N;l)$ telescope to $p(N;l)\Big(\omega_l^i - x^i(N;l)\Big)$, which is what the centralised market would pay agent i, so they reconstitute $y^i(N)$ and leave the price gaps behind,

$$y^i(\mathcal{C}) = y^i(N) + \sum_{l=1}^{L}\sum_{t=1}^{\infty}\Big[p(N;l) - p(M_{(t)};l)\Big]\Big[x^i(t;l) - x^i(t-1;l)\Big]. \tag{5}$$

Equation (5) has two consequences. First, the numeraire has a well-defined limit allocation. Second, that limit depends on the order of exchange, and hence on the network, which determines the set of available sequences. Under quasilinear preferences, every Pareto optimal allocation gives the same allocation of the goods, so every sequence ends at an efficient allocation. The sequence determines only how the numeraire is divided, and the local prices along the path do not in general deliver $y^i(N)$, the numeraire allocation of the centralised market.[10]

[10]That decentralised group trade reaches *some* Pareto optimum is classical (Feldman, 1973; Madden, 1975). This section characterises which of the efficient allocations the sequence selects.

Each term of Equation (5) is a product of two factors, a price gap and a quantity traded. Both vanish as the economy approaches the centralised allocation, since prices converge to $p(N)$ and trades die out. Therefore the transfer accumulates while the economy is away from the centralised allocation, and it changes little near the limit. Section 3.1 makes this precise by writing both factors in terms of one object, the departure of the holdings from $\boldsymbol{x}(N)$.

Equation (5) decomposes the limit numeraire into the centralised allocation and a path-dependent transfer, and this section characterises that transfer. Each term of the sum over goods involves only that good's prices and quantities, so redistribution is additive across goods. I therefore set $L = 1$ for the remainder of the section, which costs no generality, since every identity below holds good by good and extends by summation.

Before developing the theory, an example shows what is at stake. Consider four agents on a diamond, each holding the same endowment, so that all disequilibrium comes from differences in preferences. By Theorem 1 every trading round on this network delivers the same allocation of the good. Table 1 reports what becomes of the numeraire.

Example 5. Consider four agents in a diamond, with logarithmic utility. Figure 4 shows the network, the utility functions and the endowments. The centralised market gives $\boldsymbol{x}(N) = (12, 30, 60, 18)$ and $\boldsymbol{y}(N) = (13, 10, 5, 12)$, at the price $p(N) = 1/6$. Table 1 reports the transfer $y^i(\mathcal{C}) - y^i(N)$ for three trading rounds, once the allocations have converged. ◇

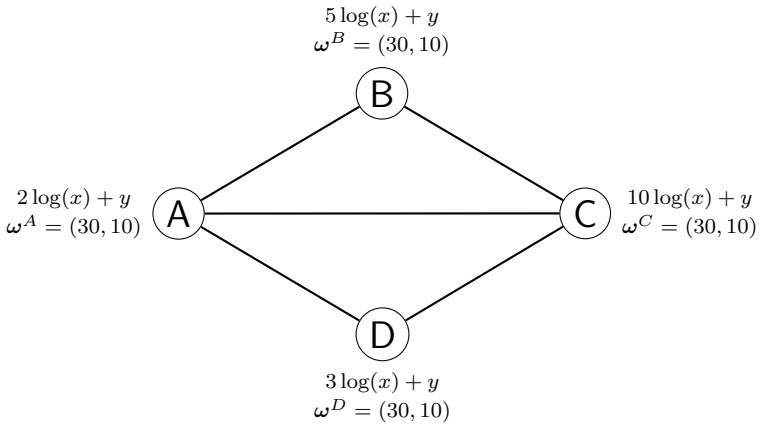


Figure 4: The diamond network of Example 5. The two triangles are ABC and ACD, and agents A and C belong to both.

The first two rounds convene the same two coalitions in opposite orders, and they leave the numeraire in different places. Under (ACD, ABC) every meeting clears at the centralised price $p(N) = 1/6$. Numeraire does change hands, and each

Round	Agent A	Agent B	Agent C	Agent D	Total moved
(ABC, ACD)	0.444	0.075	-0.447	-0.071	1.037
(ACD, ABC)	0	0	0	0	0
(AC, AB, BC, CD, AD)	0.678	0.052	-0.633	-0.097	1.460

Table 1: Transfers $y^i(\mathcal{C}) - y^i(N)$ on the diamond of Example 5. The last column is the sum of the absolute transfers.

agent ends with exactly what the centralised market would give it, so the transfer is zero. Agents A, C and D hold 90 together, which is their centralised total, so their meeting clears at $p(N)$ and leaves them at their centralised allocations. Agent B already holds $x^B(N)$, so the economy is at $\boldsymbol{x}(N)$ after one exchange. This is a feature of these endowments, since that coalition happens to hold its centralised total. In the opposite order the first meeting clears at 0.189 instead, and the round moves 1.037 of numeraire in total. The round of five bilateral meetings moves 1.460. Therefore, the order in which coalitions meet affects the transfer. In this example, the rounds of triangles move less numeraire than the round of links. However, Section 4.4 shows that the size of the coalitions does not rank the transfer in general.

The section proceeds in three steps. Section 3.1 expresses both factors of the transfer, the price gap and the quantity traded, in terms of the disequilibrium, the departure from the centralised allocation, which turns the accumulated transfer into a bilinear form in the initial disequilibrium. Section 3.2 sums that form along a trading round repeated indefinitely and obtains the closed form, a quadratic form whose kernel solves a discrete Lyapunov equation, exact under quadratic preferences and the leading term elsewhere. Section 3.3 replaces the fixed round by a distribution over cliques and obtains the expected transfer in the same shape, with the disequilibrium covariance in place of the initial disequilibrium.

3.1 Local Prices and Disequilibrium

I change variables from the allocation of the non-numeraire good at date t to its *disequilibrium*, the departure from the centralised allocation,

$$e^i(t) := x^i(t) - x^i(N), \qquad e^i(0) = \omega^i - x^i(N), \qquad \textstyle\sum_i e^i(t) = 0.$$

By the results of Section 2, this variable converges to zero along every sequence in which the relationships used infinitely often connect the economy.

The redistribution of agent i, the difference between the limit allocation of the numeraire and the centralised one, can be written as

$$\boldsymbol{y}(\mathcal{C}) - \boldsymbol{y}(N) = \sum_{t \geq 1} \Big(p(N) - p(M_{(t)}) \Big) \Big(\boldsymbol{x}(t) - \boldsymbol{x}(t-1) \Big), \tag{6}$$

which is Equation (5) with $L = 1$, and with the benchmark on the left. Equation (6) is a sum of products of two factors, a price gap and a quantity traded. Characterising the redistribution therefore amounts to expressing both in terms of a common variable, the disequilibrium. Lemma 3 does this for the quantity, since holdings and disequilibria differ by a constant, so their increments coincide and the disequilibrium is carried from one trade to the next by the exchange matrix alone. Lemma 4 does it for the price, which is determined by the gap between the active clique's aggregate holding and its centralised share. Together the two lemmas write the redistribution in terms of the initial disequilibrium and products of exchange matrices.

Lemma 3. *Along any sequence,* $\boldsymbol{e}(t) = \boldsymbol{e}(t-1)\,A(M_{(t)})$*, and hence* $\boldsymbol{e}(t) = \boldsymbol{e}(0)\prod_{s\le t} A(M_{(s)})$*. Moreover, let* $\mathbf{d} = (1-\gamma)\mathbf{b}$ *and* $\boldsymbol{z}(t) := \boldsymbol{x}(t) + \mathbf{d}$*, which absorbs the baseline into the origin. The shifted holdings then evolve by the linear part alone,* $\boldsymbol{z}(t) = \boldsymbol{z}(t-1)A(M_{(t)})$*, and the aggregate* $Z = \sum_i z^i(0)$ *is conserved.*

The shift absorbs the baseline entirely, since $\mathbf{b}$ reaches the problem only through $\boldsymbol{x}(N)$, and hence only through the initial disequilibrium. Once $\boldsymbol{e}(0)$ is fixed, the path is governed by the intensities and the curvature alone.

Recall from Lemma 1 the weights $\alpha^i = (a^i)^{1/(1-\gamma)}$ and $\alpha(M) = \sum_{j\in M}\alpha^j$, and write $\pi^i = \alpha^i/\alpha(N)$, $\pi(M) = \alpha(M)/\alpha(N)$ and $\mathbf{d} = (1-\gamma)\mathbf{b}$. Let $\mathbf{1}_M$ denote the indicator vector of a clique M.

Lemma 4 (Local Price)**.** *For every clique and every period,*

$$p(M_{(t)}) = p(N)\left(1 + \frac{\boldsymbol{e}(t-1)\,\mathbf{1}_{M_{(t)}}}{\pi\big(M_{(t)}\big)Z}\right)^{\gamma-1}, \tag{7}$$

where the bracket is positive by interiority of the local equilibrium. In particular $\boldsymbol{e}(t)\mathbf{1}_N = 0$*, so the grand clique always clears at* $p(N)$*. For quadratic preferences the price is exactly affine in the disequilibrium.*

Equation (7) relates the price of the clique active at date t to the centralised price. The denominator $\pi(M)Z = \sum_{i\in M} z^i(N)$ is the clique's aggregate holding at the centralised allocation, so the bracket measures the clique's departure from its own entitlement, as a fraction of that entitlement. A clique that holds more than its entitlement prices the good cheaply, and a clique that holds less prices it dearly.

The numerator of Equation (7) has a standard reading. Preferences are quasi-linear, so each agent demands $x^i(N)$ at the centralised price whatever it holds, and its excess demand there is $-e^i(t-1)$. Summing over the clique, the numerator is minus the clique's excess demand at $p(N)$. To first order, the local price departs from the centralised price in proportion to the clique's excess demand at that price.

Substituting Equation (7) into Equation (6), and using $\boldsymbol{x}(t) - \boldsymbol{x}(t-1) = \boldsymbol{e}(t-1)\Big(A(M_{(t)}) - I\Big)$, writes the redistribution in terms of the disequilibrium and the exchange matrices,

$$\boldsymbol{y}(\mathcal{C}) - \boldsymbol{y}(N) = p(N) \sum_{t \geq 1} \left[1 - \left(1 + \frac{\boldsymbol{e}(t-1)\mathbf{1}_{M_{(t)}}}{\pi\Big(M_{(t)}\Big) Z} \right)^{\gamma - 1} \right] \boldsymbol{e}(t-1)\Big(A(M_{(t)}) - I\Big). \quad (8)$$

Two properties are visible in Equation (8). First, each summand carries $\boldsymbol{e}(t-1)$ twice, once in the price factor and once in the quantity factor, and Proposition 5 makes this bilinearity precise. Second, $A(M_{(t)}) - I$ maps $\mathbf{1}$ to zero, so every summand sums to zero across agents, and redistribution is zero-sum by construction.

The change of variables does not affect the convergence of the redistribution, which Section 2 already established, and it determines the rate. Each summand is quadratic in the disequilibrium, one factor from the price gap and one from the quantity traded. Along a repeated trading round, the disequilibrium is $O\Big(t^{m-1}|\lambda_2(A(\mathcal{R}))|^t\Big)$ by Theorem 1, so the summands are $O\Big(t^{2(m-1)}|\lambda_2(A(\mathcal{R}))|^{2t}\Big)$, with twice the exponent and the polynomial factor squared, and both reduce to the geometric case when $m = 1$. Along an arbitrary sequence, the summands are bounded by the square of the π-norm of Proposition 2.

The redistribution still outstanding therefore converges at twice the rate of the goods, while the numeraire holding converges at the same rate as the goods. An agent goes on trading at prices close to $p(N)$, so its holding inherits the rate of the goods, and only after netting against the centralised price do both factors vanish. Redistribution is thus decided in the early rounds, while local prices are still far apart. Once the allocation is near $\boldsymbol{x}(N)$, every clique trades near $p(N)$, and little changes hands beyond $y^i(N)$. This governs the rate at which redistribution stops accumulating, and not its total size.

To first order, the departure of the local price from $p(N)$ is linear in the disequilibrium, and Proposition 5 turns this into a bound.

Proposition 5 (Second-Order Approximation)**.** *The t-th summand of Equation* (8) *equals*

$$-\,p(N)(\gamma - 1)\,\frac{\boldsymbol{e}(t-1)\,\mathbf{1}_{M_{(t)}}}{\pi\Big(M_{(t)}\Big) Z}\;\boldsymbol{e}(t-1)\Big(A(M_{(t)}) - I\Big) \;+\; O\Big(\|\boldsymbol{e}(0)\|^3\Big).$$

Since $\boldsymbol{e}(t-1) = \boldsymbol{e}(0)\prod_{s<t} A(M_{(s)})$ by Lemma 3, the leading term is a product of two linear functionals of $\boldsymbol{e}(0)$, and the error is cubic uniformly in t. When $\gamma = 2$ the error vanishes identically and the summand is exactly bilinear in $\boldsymbol{e}(0)$.

By Lemma 4, the local price is the power $\gamma - 1$ of a linear functional of the disequilibrium. Only $\gamma = 2$ makes that power equal to one, so the quadratic case is the only member of the HARA family with a closed form rather than an approximation.

3.2 A Closed Form

A trading round repeated indefinitely has a structure that an arbitrary sequence lacks. Every pass applies the same exchanges in the same order, so the disequilibrium left at the end of a pass is the image of the disequilibrium at its start under one and the same operator. The terms of Equation (8) therefore recur with the same structure from one pass to the next, shifted by that operator, and the infinite sum resums into a single quadratic form.

Consider a trading round $\mathcal{R} = (M_1, \dots, M_K)$ repeated indefinitely. The accumulated exchange $A(t)$ and its error $E(t) = A(t) - \Pi$ are those of Section 2, so that $A(\mathcal{R}) = A(K)$ is the round operator and $A(0), \dots, A(K-1)$ are its within-round prefixes. Write $E(\mathcal{R}) := A(\mathcal{R}) - \Pi$.

Disequilibria sum to zero at every date, and Π maps every such vector to zero. Therefore $A(\mathcal{R})$ and $E(\mathcal{R})$ have the same effect on disequilibria, and $E(\mathcal{R})$ contracts, with $\rho\Big(E(\mathcal{R})\Big) = |\lambda_2(A(\mathcal{R}))| < 1$ by Theorem 1. By Proposition 5, each summand of Equation (8) is bilinear in $\boldsymbol{e}(0)$, up to a cubic error and exactly when $\gamma = 2$.

Theorem 3 (Closed Form)**.** *Let $\mathcal{R} = (M_1, \dots, M_K)$ be a trading round repeated indefinitely, and set $E(\mathcal{R}) = A(\mathcal{R}) - \Pi$. For each agent there is a matrix $\mathcal{K}_i(\mathcal{R})$ with*

$$y^i(\mathcal{R}) - y^i(N) = \boldsymbol{e}(0)\, \mathcal{K}_i\, \boldsymbol{e}(0)^\top + O\Big(\|\boldsymbol{e}(0)\|^3\Big), \tag{9}$$

namely the unique solution of the discrete Lyapunov equation

$$\mathcal{K}_i - E(\mathcal{R})\, \mathcal{K}_i\, E(\mathcal{R})^\top = B_i, \tag{10}$$

where

$$B_i = \kappa \sum_{k=1}^{K} \frac{1}{\alpha(M_k)} \Big(A(k-1)\, \mathbf{1}_{M_k}\Big)\Big(A(k-1)\, (A(M_k) - I)\, \mathbf{1}_{\{i\}}\Big)^\top, \qquad \kappa := p(N)^{\frac{\gamma-2}{\gamma-1}}, \tag{11}$$

with $A(0) = I$. Uniqueness holds because $\rho\Big(E(\mathcal{R})\Big) < 1$ and Lemma 6 applies with $E = E(\mathcal{R})$.

Theorem 3 characterises $\mathcal{K}_i$ by an equation rather than by a formula, and the equation is finite. Equation (6) is an infinite sum over trading dates, and each term depends on the entire prefix of the sequence. Equation (10) replaces it by one linear system in the n^2 entries of $\mathcal{K}_i$, namely $\Big(\mathrm{Id}_{n^2} - E(\mathcal{R}) \otimes E(\mathcal{R})\Big)\, \mathrm{vec}(\mathcal{K}_i) = \mathrm{vec}(B_i)$, whatever the length of the round. The following lemma records why this works, and what it requires.

Lemma 6 (Lyapunov Equation)**.** *Let E be an $n \times n$ matrix with $\rho(E) < 1$, and let B be any $n \times n$ matrix. Then $\mathcal{K} - E\, \mathcal{K}\, E^\top = B$ has a unique solution, characterised equivalently by*

(i) the linear system

$$\Big(\mathrm{Id}_{n^2} - E \otimes E\Big)\,\mathrm{vec}(\mathcal{K}) = \mathrm{vec}(B),$$

whose matrix is invertible because every eigenvalue of $E \otimes E$ is a product $\lambda_\beta \lambda_{\beta'}$ of eigenvalues of E, hence of modulus at most $\rho(E)^2 < 1$;

(ii) the series $\mathcal{K} = \sum_{j \geq 0} E^j B (E^\top)^j$, whose terms are of order $\rho(E)^{2j}$.

If in addition E is diagonalisable, $E = \sum_\beta \lambda_\beta \mathbf{r}_\beta \boldsymbol{\ell}_\beta^\top$ with $\boldsymbol{\ell}_\beta^\top \mathbf{r}_{\beta'} = \delta_{\beta\beta'}$, then

$$\mathcal{K} = \sum_{\beta,\beta'} \frac{\boldsymbol{\ell}_\beta^\top B\, \boldsymbol{\ell}_{\beta'}}{1 - \lambda_\beta \lambda_{\beta'}}\, \mathbf{r}_\beta \mathbf{r}_{\beta'}^\top. \tag{12}$$

Part (i) is the computational content, since $\mathcal{K}_i$ comes from solving one linear system rather than from summing over repetitions. Part (ii) connects the two, and the proof of Theorem 3 constructs it. The decay $\rho(E)^{2j}$ that it records is the rate discussed above, regrouped by repetitions of the round rather than by trading dates. Written out, the series separates the transfer made within a single pass from the tail accumulated over repetitions,

$$\mathcal{K}_i = \underbrace{B_i}_{\text{within-round}} + \underbrace{\sum_{m \geq 1} E(\mathcal{R})^m\, B_i \left(E(\mathcal{R})^\top\right)^m}_{\text{across-round tail}}. \tag{13}$$

Equation (12) shows what that decay accumulates to, namely the damping factors $1/(1 - \lambda_\beta \lambda_{\beta'})$, the largest of which is $1/(1 - |\lambda_2|^2)$. However, it requires $E(\mathcal{R})$ to be diagonalisable, which is not automatic for a product of projections. When it fails, parts (i) and (ii) still give $\mathcal{K}_i$, without the spectral form.

Example 6 works the construction out on the smallest network whose round needs more than one clique, three agents in a line with two bilateral cliques. The kernels follow from nine equations in nine unknowns, and the damping factor appears in the arithmetic. Reversing the order of the two cliques leaves λ_2 unchanged, and yet changes the transfers in this example.

Example 6 (The closed form on a three-agent line). Consider three agents on the line *A*–*B*–*C* with quadratic preferences, $\gamma = 2$, so that Equation (1) reads $u^i(x^i, y^i) = -\frac{a^i}{2}\big(b^i - x^i\big)^2 + y^i$ with b^i the bliss point and $x^i = b^i - \alpha^i p$ for $\alpha^i = 1/a^i$. Take $a = \big(\frac{1}{9}, \frac{1}{16}, \frac{1}{4}\big)$, $b = (60, 48, 40)$ and $\boldsymbol{\omega}^i = (30, 30)$ for every i, and the trading round $\mathcal{R} = \big(\{A, B\}, \{B, C\}\big)$.

The economy. The weights are $\alpha = (9, 16, 4)$, so $\alpha(N) = 29$ and $\boldsymbol{\pi} = \frac{1}{29}(9, 16, 4)$. Since $z^i = x^i - b^i$ at $\gamma = 2$, the conserved aggregate is $Z = \sum_i (\omega^i - b^i) = -58$, and the competitive outcome is

$$p(N) = 2, \qquad \boldsymbol{x}(N) = (42, 16, 32), \qquad \boldsymbol{e}(0) = (-12, 14, -2).$$

The operators. Since $\alpha(M_1) = 25$ and $\alpha(M_2) = 20$,

$$A(M_1) = \tfrac{1}{25}\begin{pmatrix} 9 & 16 & 0 \\ 9 & 16 & 0 \\ 0 & 0 & 25 \end{pmatrix}, \qquad A(M_2) = \tfrac{1}{5}\begin{pmatrix} 5 & 0 & 0 \\ 0 & 4 & 1 \\ 0 & 4 & 1 \end{pmatrix}, \qquad A(\mathcal{R}) = \tfrac{1}{125}\begin{pmatrix} 45 & 64 & 16 \\ 45 & 64 & 16 \\ 0 & 100 & 25 \end{pmatrix},$$

whose spectrum is $\{1, \frac{9}{125}, 0\}$, so $\lambda_2 = \frac{9}{125}$ and

$$E = A(\mathcal{R}) - \Pi = \tfrac{1}{3625}\begin{pmatrix} 180 & -144 & -36 \\ 180 & -144 & -36 \\ -1125 & 900 & 225 \end{pmatrix}.$$

The within-round kernels. The prefixes are $A(0) = I$ and $A(1) = A(M_1)$, and since $\gamma = 2$ we have $\kappa = 1$. The price vectors $\mathbf{u}_k = A(k-1)\mathbf{1}_{M_k}$ are shared by all agents,

$$\mathbf{u}_1 = (1,1,0)^\top, \qquad \mathbf{u}_2 = A(M_1)\,(0,1,1)^\top = \left(\tfrac{16}{25}, \tfrac{16}{25}, 1\right)^\top,$$

while the quantity vectors $\mathbf{v}_{k,i} = A(k-1)\Big(A(M_k) - I\Big)\mathbf{1}_{\{i\}}$ are the columns of $A(M_k) - I$ carried through the prefix:

$$\begin{aligned} &\mathbf{v}_{1,A} = \left(-\tfrac{16}{25}, \tfrac{9}{25}, 0\right)^\top, \quad \mathbf{v}_{1,B} = \left(\tfrac{16}{25}, -\tfrac{9}{25}, 0\right)^\top, && \mathbf{v}_{1,C} = \mathbf{0}, \\ &\mathbf{v}_{2,A} = \mathbf{0}, && \mathbf{v}_{2,B} = \left(-\tfrac{16}{125}, -\tfrac{16}{125}, \tfrac{4}{5}\right)^\top, \quad \mathbf{v}_{2,C} = \left(\tfrac{16}{125}, \tfrac{16}{125}, -\tfrac{4}{5}\right)^\top. \end{aligned}$$

Agent A never trades in M_2 and agent C never in M_1, so each contributes a single term to Equation (11) while the intermediary contributes two:

$$B_A = \tfrac{1}{625}\begin{pmatrix} -16 & 9 & 0 \\ -16 & 9 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B_B = \tfrac{1}{15625}\begin{pmatrix} 336 & -289 & 400 \\ 336 & -289 & 400 \\ -100 & -100 & 625 \end{pmatrix}, \quad B_C = \tfrac{1}{15625}\begin{pmatrix} 64 & 64 & -400 \\ 64 & 64 & -400 \\ 100 & 100 & -625 \end{pmatrix}.$$

Solving for the kernels. By Lemma 6(i) each kernel solves a system of nine equations in nine unknowns,

$$\Big(\mathrm{Id}_9 - E \otimes E\Big)\,\mathrm{vec}(\mathcal{K}_i) = \mathrm{vec}(B_i), \qquad E \otimes E = \begin{pmatrix} E_{11}E & E_{12}E & E_{13}E \\ E_{21}E & E_{22}E & E_{23}E \\ E_{31}E & E_{32}E & E_{33}E \end{pmatrix},$$

where vec stacks columns and each block is 3×3. For agent A, $\mathrm{vec}(B_A) = \frac{1}{625}(-16, -16, 0, 9, 9, 0, 0, 0, 0)^\top$, and solving returns

$$\mathcal{K}_A = \tfrac{1}{2817350}\begin{pmatrix} -72176 & 40518 & 324 \\ -72176 & 40518 & 324 \\ 324 & 324 & -2025 \end{pmatrix},$$

whose first two rows coincide because those of E do. No summation over repetitions is needed, since the infinite sum over trading dates in Equation (6) is replaced by one finite solve. The same system with $\mathrm{vec}(B_B)$ and $\mathrm{vec}(B_C)$ returns the other two kernels.

A shortcut, particular to this round. Because $A(\mathcal{R})$ has spectrum $\{1, \frac{9}{125}, 0\}$, the operator E has a single nonzero eigenvalue. Moreover, once M_1 has cleared, agents A and B hold amounts proportional to their weights, so every zero-sum disequilibrium is then a multiple of one vector, and M_2 carries that line to another

line. Hence E has rank one. Writing $\mathbf{r} = (4, 4, -25)^\top$ and $\boldsymbol{\ell} = (-5, 4, 1)^\top$ for its right and left eigenvectors, with $c = \boldsymbol{\ell}^\top \mathbf{r} = -29$, every pair in Equation (12) involving a zero eigenvalue has denominator one, and those terms reassemble into B_i. Only the pair (λ_2, λ_2) is damped, leaving

$$\mathcal{K}_i = B_i + \frac{\lambda_2^2}{1-\lambda_2^2} \cdot \frac{\boldsymbol{\ell}^\top B_i \boldsymbol{\ell}}{c^2}\, \mathbf{r}\mathbf{r}^\top, \qquad \frac{\lambda_2^2}{1-\lambda_2^2} = \frac{81}{15544},$$

with $\boldsymbol{\ell}^\top B_i \boldsymbol{\ell} = \frac{1}{15625}(-2900,\ 3161,\ -261)$ for $i = A, B, C$, three scalars summing to zero.

The transfers. Evaluating the quadratic forms at $\boldsymbol{e}(0) = (-12, 14, -2)$,

$$\boldsymbol{y}(\mathcal{R}) - \boldsymbol{y}(N) = \left(\tfrac{1698}{1675},\ -\tfrac{3171}{3350},\ -\tfrac{9}{134}\right) = (1.014,\ -0.947,\ -0.067),$$

which sums to zero and reproduces the transfers obtained by simulating the round directly.

Order. Reversing the round to $\big(\{B, C\}, \{A, B\}\big)$ leaves the spectrum of $A(\mathcal{R})$ unchanged, so $\lambda_2 = \frac{9}{125}$ as before, and yet

$$\boldsymbol{y}(\mathcal{R}) - \boldsymbol{y}(N) = \left(-\tfrac{72}{67},\ -\tfrac{492}{335},\ \tfrac{852}{335}\right) = (-1.075,\ -1.469,\ 2.543).$$

The two rounds share λ_2, so redistribution accumulates at the same rate, while the size and the direction of the transfers differ. ◇

Example 6 is exact because the preferences are quadratic. How accurate is the closed form for other curvatures? To answer this, I change the curvature and keep everything else fixed. With logarithmic preferences $\alpha^i = a^i$, so choosing $a = (9, 16, 4)$ reproduces the weights of Example 6, and with them its exchange matrices, its round operator, its second eigenvalue and its kernels. The two economies differ only in the price, which is now the reciprocal of the clique's aggregate holding rather than an affine function of it. Example 7 measures the error, and the answer has two parts. At the scale of that example, the quadratic prediction is wrong by up to a quarter of the transfer, so the closed form does not replace the exact identity when the economy starts far from equilibrium. However, the error falls at exactly the rate of Proposition 5, so the same kernels describe the redistribution to leading order for every member of the family, and exactly for one.

Example 7 (Logarithmic preferences: the closed form as an approximation). Keep the network and round of Example 6, but replace preferences by logarithmic ones, $u^i(x^i, y^i) = a^i \log x^i + y^i$, with

$$a = (9, 16, 4), \qquad \boldsymbol{\omega}^A = (12, 10), \quad \boldsymbol{\omega}^B = (9, 10), \quad \boldsymbol{\omega}^C = (8, 10).$$

The logarithmic weights are $\alpha^i = a^i$, the shift vanishes and $z^i = x^i$, so $Z = 29$, $p(N) = \alpha(N)/Z = 1$ and

$$\boldsymbol{x}(N) = (9, 16, 4), \qquad \boldsymbol{e}(0) = (3, -7, 4), \qquad \kappa = p(N)^2 = 1.$$

Because $\boldsymbol{\alpha}$ is the same as in Example 6, so are $A(M_1)$, $A(M_2)$, $A(\mathcal{R})$, E, $\lambda_2 = \frac{9}{125}$ and every kernel B_i and $\mathcal{K}_i$. The two economies differ only in the curvature of the price.

Exact against approximate. Simulating the round gives the transfers in the first row below, while Theorem 3 predicts the second:

	A	B	C
exact $\boldsymbol{y}(\mathcal{R}) - \boldsymbol{y}(N)$	0.8301	−0.5794	−0.2507
quadratic prediction $\boldsymbol{e}(0)\mathcal{K}_i\boldsymbol{e}(0)^\top$	0.6949	−0.4263	−0.2687

The prediction is wrong by 16%, 26% and 7% respectively. Unlike the quadratic case, it is an approximation.

The order of the error. To isolate the size of the disequilibrium from everything else, scale the endowment along the ray towards the centralised allocation, $\boldsymbol{\omega}(\theta) = \boldsymbol{x}(N) + \theta\,\boldsymbol{e}(0)$, so that the economy starts θ times as far from equilibrium and the initial disequilibrium is exactly $\theta\,\boldsymbol{e}(0)$. Since $\sum_i e^i(0) = 0$ the aggregate Z is untouched, and with it $p(N)$, $\boldsymbol{x}(N)$, and every kernel of Theorem 3. The prediction is therefore θ^2 times its value at $\theta = 1$, by homogeneity of the quadratic form, so any departure of the realised transfer from a θ^2 law is the approximation error, and its order is the order at which the closed form first fails. Dividing that error by θ^3:

	$\big(\text{exact} - \text{predicted}\big)\big/\theta^3$		
θ	A	B	C
1	0.13513	−0.15309	0.01796
1/2	0.12337	−0.14195	0.01858
1/8	0.11581	−0.13488	0.01908
1/64	0.11377	−0.13300	0.01923

Each column settles at a finite limit, so the error is exactly cubic. Had it been of lower order, the ratios would diverge, and of higher order they would fall to zero. Since $\|\theta\,\boldsymbol{e}(0)\| = \theta\,\|\boldsymbol{e}(0)\|$, these are the error divided by the cubed norm of the initial disequilibrium, in units where the base economy has $\theta = 1$. Here $\|\boldsymbol{e}(0)\|^3 = 636.6$. The limits are finite and nonzero, so the rate of Proposition 5 is attained along this ray, and is not merely an upper bound. Each row sums to zero, because both the exact transfers and the prediction are zero-sum. The error therefore moves numeraire between agents, and does not change the total. ◇

3.3 Random Sequences

Theorem 2 dispenses with periodicity and asks only that trade be recurrent. Randomising the sequence is the probabilistic form of that relaxation, and it restores the structure that Section 3.2 exploits. The cliques at successive dates are drawn from the same distribution, so every date brings the same distribution over exchange operators, and the second moments of the disequilibrium follow a single linear recursion.

Let μ be a probability distribution on the admissible family, with μ_M the probability that clique M convenes at a given date, and let the cliques at successive dates be drawn independently. Throughout I maintain that the support of μ satisfies the hypothesis of Theorem 2, so the cliques carrying positive probability cover N and induce a connected subgraph. Every agent then trades infinitely often, and the allocation converges to $\boldsymbol{x}(N)$ almost surely.

The terminal numeraire $y^i(\mu)$ is a random variable, and the question is whether randomising who trades removes the redistribution in expectation. The answer has the same shape as Theorem 3. There the transfer accumulates the within-round kernel through powers of $E(\mathcal{R})$, and here it accumulates the disequilibrium covariance through the second-moment operator of the draw.

Theorem 4 (Expected Redistribution)**.** *Let μ satisfy the maintained hypothesis on its support. Then for every agent i the drift is the quadratic form*

$$\mathbb{E}\Big[y^i(\mu) - y^i(N)\Big] = \operatorname{tr}\Big(\bar{B}_i\,\mathcal{S}\Big) + O\Big(\|\boldsymbol{e}(0)\|^3\Big), \tag{14}$$

the remainder vanishing when $\gamma = 2$, where

$$\bar{B}_i = \kappa \sum_M \frac{\mu_M}{\alpha(M)}\,\mathbf{1}_M\Big((A(M) - I)\mathbf{1}_{\{i\}}\Big)^\top \tag{15}$$

collects i's trading opportunities under μ, weighted by clique, and $\mathcal{S}$ is the accumulated disequilibrium covariance, the total disequilibrium the economy passes through before settling, solving

$$\mathcal{S} - \sum_M \mu_M A(M)^\top \mathcal{S} A(M) = \boldsymbol{e}(0)^\top \boldsymbol{e}(0) \tag{16}$$

on $\{\Sigma : \Sigma\mathbf{1} = \mathbf{0}\}$, where it is unique by Lemma 7.

Remark 3. Since $(A(M) - I)\mathbf{1}_{\{i\}} = \mathbf{0}$ whenever $i \notin M$, only cliques containing i appear in Equation (15). If the grand coalition is admissible, it is among them, and it contributes nothing. Its term is proportional to $\mathbf{1}\Big((A(N) - I)\mathbf{1}_{\{i\}}\Big)^\top$, which pairs to zero with any Σ satisfying $\Sigma\mathbf{1} = \mathbf{0}$, and $\mathcal{S}$ is such. A centralised session settles

at $p(N)$, so it transfers nothing. Writing $s_M(\boldsymbol{e}) = \boldsymbol{e}\mathbf{1}_M$ for the disequilibrium a clique carries, the kernel acts on a single disequilibrium profile as

$$\operatorname{tr}\Big(\bar{B}_i\,\boldsymbol{e}^\top\boldsymbol{e}\Big) = \kappa \sum_{M\ni i} \frac{\mu_M}{\alpha(M)}\, s_M(\boldsymbol{e})\Big(\frac{\alpha^i}{\alpha(M)} s_M(\boldsymbol{e}) - e^i\Big). \tag{17}$$

The grand clique has $s_N(\boldsymbol{e}) = 0$, so its term vanishes.

Equation (16) defines $\mathcal{S}$ implicitly, and the restriction to $\{\Sigma : \Sigma\mathbf{1} = \mathbf{0}\}$ in Theorem 4 is not a technicality. On all of $\mathbb{R}^{n\times n}$ the operator $\mathcal{L}_\mu$ fixes $\boldsymbol{\pi}^\top\boldsymbol{\pi}$, because $\boldsymbol{\pi}A(M) = \boldsymbol{\pi}$ for every clique, so $\mathrm{Id} - \mathcal{L}_\mu$ is singular and the equation on its own does not pin down a solution. The restriction works because disequilibria are zero-sum. Since $\boldsymbol{e}(0)\mathbf{1} = 0$, the initial covariance lies in $\{\Sigma : \Sigma\mathbf{1} = \mathbf{0}\}$, every $A(M)$ keeps it there, and that subspace excludes $\boldsymbol{\pi}^\top\boldsymbol{\pi}$, so the operator contracts on it. The next lemma records this and gives two ways of reaching $\mathcal{S}$. The first is the one to compute, a single linear system of size n^2 solved once and reused for every agent. The second gives its meaning, the total disequilibrium the economy passes through before settling, rather than the disequilibrium it starts from.

Lemma 7. *Write $\mathcal{L}_\mu(\Sigma) = \sum_M \mu_M A(M)^\top\Sigma A(M)$, which preserves $\{\Sigma : \Sigma\mathbf{1} = \mathbf{0}\}$ and contracts on it. Then Equation* (16) *has a unique solution there, characterised equivalently by*

(i) the linear system $\Big(\mathrm{Id}_{n^2} - \sum_M \mu_M A(M)^\top \otimes A(M)^\top\Big)\operatorname{vec}(\mathcal{S}) = \operatorname{vec}\Big(\boldsymbol{e}(0)^\top\boldsymbol{e}(0)\Big)$, one solve of size n^2 shared by every agent;

(ii) the series $\mathcal{S} = \sum_{t\geq 0}\mathbb{E}\Big[\boldsymbol{e}(t)^\top\boldsymbol{e}(t)\Big]$, the total disequilibrium covariance the economy passes through before settling.

Theorems 3 and 4 are two forms of the same statement. Both make the transfer a pairing between an object built from the protocol and an object built from the initial disequilibrium, and they differ only in which of the two is propagated. The round form propagates the protocol, pushing agent i's one-period kernel B_i forward through $E(\mathcal{R})$ and pairing the result with the fixed outer product $\boldsymbol{e}(0)^\top\boldsymbol{e}(0)$. The random form propagates the disequilibrium, pushing that outer product forward through $\mathcal{L}_\mu$ and pairing the result with the fixed $\bar{B}_i$. The two accumulations are adjoint for the trace pairing, so either can be moved onto the other factor and each theorem rewritten in the other's form.

In covariance form, the transfer of Theorem 3 is

$$y^i(\mathcal{R}) - y^i(N) = \operatorname{tr}\Big(B_i\,\mathcal{S}_\mathcal{R}\Big) + O\Big(\|\boldsymbol{e}(0)\|^3\Big), \qquad \mathcal{S}_\mathcal{R} - E(\mathcal{R})^\top\mathcal{S}_\mathcal{R}E(\mathcal{R}) = \boldsymbol{e}(0)^\top\boldsymbol{e}(0),$$

with $\mathcal{S}_\mathcal{R}$ the accumulated disequilibrium covariance along the repeated round, uniquely determined because $\rho(E(\mathcal{R})) < 1$. Corollary 8 gives a random sequence in kernel form, which Section 4 uses to apply Definition 7 to random protocols.

Corollary 8 (Drift as a Kernel)**.** *Let μ satisfy the maintained hypothesis of Theorem 4, write $\mathcal{L}_\mu^*(\mathcal{K}) = \sum_M \mu_M A(M)\,\mathcal{K}\,A(M)^\top$ for the adjoint of $\mathcal{L}_\mu$, and set $\bar{\mathcal{K}}_i(\mu) = \sum_{t\geq 0}(\mathcal{L}_\mu^*)^t(\bar{B}_i)$. Then*

$$\mathbb{E}\big[y^i(\mu) - y^i(N)\big] = \boldsymbol{e}(0)\,\bar{\mathcal{K}}_i(\mu)\,\boldsymbol{e}(0)^\top + O\big(\|\boldsymbol{e}(0)\|^3\big), \qquad \bar{\mathcal{K}}_i(\mu) - \mathcal{L}_\mu^*\big(\bar{\mathcal{K}}_i(\mu)\big) = \bar{B}_i,$$

which is Equation (10) *with the round's error operator replaced by the second-moment operator of the draw.*

I use the kernel form for rounds and the covariance form for random sequences, because of what each one holds fixed. The kernel does not involve $\boldsymbol{e}(0)$ and is reusable across initial conditions, while the covariance does not involve the agent index and is one solve shared by all n agents.

Randomising the order does not restore the centralised outcome. The realised transfer varies with the draw, and its expectation does not vanish. The drift is a systematic transfer, determined by how $\bar{B}_i$, which records who trades with whom and at what weights, aligns with the accumulated disequilibrium covariance $\mathcal{S}$.

Two features of Equation (14) matter. First, by Remark 3 the kernel $\bar{B}_i$ is supported on the cliques containing i, so only its own counterparties enter directly. Agents who never trade with i reach its expected transfer through $\mathcal{S}$ alone, so the externality is real and indirect. Second, the drift vanishes when the grand clique convenes with probability one. A centralised session removes the outstanding disequilibrium and, by Remark 3, transfers nothing itself, so at $\mu = \delta_N$ the drift is exactly zero, and it vanishes continuously as $\mu(N) \to 1$.

Example 8 (Random matching and fixed orders on the complete network)*.* Four agents on the complete network with quadratic preferences, $u^i(x^i, y^i) = -\frac{a^i}{2}(b^i - x^i)^2 + y^i$, and

$$a^i = \tfrac{1}{16} \text{ for every } i, \qquad b = (60, 52, 40, 32) \text{ for agents } A \text{ to } D, \qquad \boldsymbol{\omega}^i = (30, 20).$$

Then $\alpha^i = 16$, so $\boldsymbol{\pi}$ is uniform, $\alpha(N) = 64$ and $Z = -64$, giving

$$p(N) = 1, \qquad \kappa = 1, \qquad \boldsymbol{x}(N) = (44, 36, 24, 16), \qquad \boldsymbol{e}(0) = (-14, -6, 6, 14).$$

Let μ be uniform on the six links, so $\alpha(M) = 32$ for each. Solving Equation (16) and evaluating Equation (15),

$$\mathcal{S} = \tfrac{1}{7}\begin{pmatrix} 3556 & 660 & -1356 & -2860 \\ 660 & 1316 & -620 & -1356 \\ -1356 & -620 & 1316 & 660 \\ -2860 & -1356 & 660 & 3556 \end{pmatrix}, \qquad \bar{B}_A = \tfrac{1}{384}\begin{pmatrix} -3 & 1 & 1 & 1 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix},$$

and the expected transfers are

$$\mathbb{E}\big[\boldsymbol{y}(\mu) - \boldsymbol{y}(N)\big] = \Big(-\tfrac{5}{3},\ \tfrac{5}{3},\ \tfrac{5}{3},\ -\tfrac{5}{3}\Big),$$

which a Monte Carlo over 40,000 draws reproduces as $(-1.669,\ 1.676,\ 1.663,\ -1.669)$, with standard errors below 0.01.

Against fixed bilateral orders. Keeping trade bilateral and changing only how the pairs are scheduled,

	A	B	C	D
$\mathcal{R}_2 = (AB, BC, CD, AD)$	-3.517	1.950	3.517	-1.950
$\mathcal{R}_3 =$ all six links in a fixed order	-2.993	1.550	3.481	-2.037
random matching	-1.667	1.667	1.667	-1.667

Randomising the order does not return the economy to the benchmark, and it does not reproduce the fixed orders either. Random matching produces a systematic transfer of its own, on the same cliques, differing from the deterministic ones in size and in who bears it. ◇

Example 8 also shows that a long sequence does not average the drift away. The realised transfer is a single quadratic functional of one realised sequence of cliques, and not an average of independent increments. By Lemma 7(ii) the terms of $\mathcal{S}$ decay at rate $\rho(\mathcal{L}_\mu)$, so the sum is dominated by its first $O\big(1/(1-\rho(\mathcal{L}_\mu))\big)$ trades, and the rest contribute geometrically little. Extending the horizon therefore averages nothing. In that economy $\rho(\mathcal{L}_\mu) = \frac{2}{3}$, and the ratio of across-path standard deviation to mean is $(0.91, 1.16, 1.16, 0.92)$ after twenty trades, against $(0.91, 1.16, 1.15, 0.91)$ after eighty. The dispersion is of the same order as the drift, and quadrupling the horizon leaves it unchanged.

The picture may differ across goods. With $L > 1$, the good-level terms share the clique draws but run on different operators, one for each good, so some averaging across goods is possible even though none occurs across time. Deciding this calls for the joint law of the good-level disequilibria rather than the good-by-good accounting used here, and I leave it to future work.

4 Comparing Trading Structures

Sections 2 and 3 show that every trading structure that connects the economy leads to the same allocation of the non-numeraire goods. However, the structure determines how fast the economy converges and how the numeraire is distributed. This section compares trading structures, that is, which coalitions meet and in what order, in these two dimensions. I ask three questions. First, how can one compare the speed of two structures when their rounds have different lengths, or when the sequence does not repeat? Second, does convening a coalition as one market, instead of as smaller coalitions inside it, make the economy converge faster?

Third, does it change how much numeraire the economy transfers? The second and third questions matter for a regulator who can merge or split markets.

Section 4.1 answers the first question and defines the statistics of speed. Sections 4.2 and 4.3 answer the second question. They compare a structure with its refinements, in which one coalition is replaced by smaller coalitions inside it, first at the end of a single exchange, and then over many repetitions of a round and under random draws. Section 4.4 answers the third question. It defines the redistribution capacity, which measures how much numeraire a structure transfers, and compares a structure with its refinements by this statistic. All these statistics depend only on the trading structure and on preferences, so the comparisons do not depend on endowments.

4.1 Measuring Speed

By Theorem 1, a trading round $\mathcal{R}$ converges at a geometric rate set by the second-largest eigenvalue modulus of the round operator, $|\lambda_2(A(\mathcal{R}))|$. The round applies the same operator in every repetition, so in the long run each repetition reduces the remaining disequilibrium by the same factor. However, Theorem 2 does not give a rate for general sequences. A general sequence need not repeat, so it has no such factor. I therefore use two measures of speed: (i) for a trading round, the spectral gap, which is defined from its asymptotic rate of convergence, and (ii) for any sequence, the distance between the exchange accumulated up to a given date and its limit.

The *spectral gap* of a trading round $\mathcal{R}$ is

$$\Lambda(\mathcal{R}) = 1 - \left|\lambda_2(A(\mathcal{R}))\right|. \tag{18}$$

It measures speed per repetition of the trading round. A larger gap means faster convergence, and a gap near 0 means that many repetitions are needed.

Rounds can differ in the number of coalitions they contain. A round with more coalitions may therefore have a larger gap only because each repetition contains more trades. To compare rounds of different lengths, I use two normalisations that measure speed per trade. Let $K(\mathcal{R})$ be the number of coalitions in $\mathcal{R}$. When the round is clear from the context, I write K for $K(\mathcal{R})$ and λ_2 for $\lambda_2(A(\mathcal{R}))$. The *effective gap* $\Lambda_{\text{eff}} = 1 - |\lambda_2|^{1/K}$ is the per-trade rate that gives the same long-run contraction over one repetition as the round. The *linear gap* $\Lambda_{\text{lin}} = \Lambda/K$ is a simpler rule, which spreads the gap evenly across the round's trades.

For any sequence, I measure speed by the distance between the accumulated exchange $A(t)$ and its limit Π. This distance is the norm $\|E(t)\|_\pi$ of the error $E(t) = A(t) - \Pi$, defined in Section 2. The norm has a simple meaning. Whatever the initial disequilibrium, at most a share $\|E(t)\|_\pi$ of it remains at date t, and

for some initial disequilibrium exactly this share remains.[11] By Proposition 2, this share never increases.

A general sequence has no spectral gap, but the norm can still be read through the rounds that the sequence contains. By Remark 2, a sequence that satisfies the hypothesis of Theorem 2 contains a succession of trading rounds $\mathcal{R}_1, \mathcal{R}_2, \ldots$, separated by single trades. The error of two consecutive rounds is the product of their errors,[12] and a single trade never increases the norm. Therefore, once m rounds are complete,

$$\|E(t)\|_\pi \;\leq\; \prod_{j\leq m} \|E(\mathcal{R}_j)\|_\pi.$$

Hence the rounds that a sequence contains bound its distance to the centralised allocation.

When the sequence repeats a single round $\mathcal{R}$, the error after k repetitions is $E(\mathcal{R})^k$, and the two measures are linked by Gelfand's formula,

$$\left|\lambda_2(A(\mathcal{R}))\right| = \lim_{k\to\infty} \left\|E(\mathcal{R})^k\right\|_\pi^{1/k}. \tag{19}$$

Hence $\Lambda(\mathcal{R})$ measures how fast the economy converges in the long run, and $\|E(t)\|_\pi$ measures how much disequilibrium can remain at date t. In particular, $|\lambda_2(A(\mathcal{R}))| \leq \|E(\mathcal{R})\|_\pi$, so a single repetition can remove less disequilibrium than the long-run rate suggests. Along a general sequence, $|\lambda_2(A(t))|$ is also well defined at each date. However, it is a counterfactual rate, namely the rate at which the economy would converge if the first t trades were repeated from then on. This counterfactual changes with t, so I describe a general sequence by the norm and not by the spectral gap.

I now return to trading rounds. A round is defined by its coalitions and by the order in which they meet. For given coalitions, how much does the order affect the spectral gap? Proposition 9 shows that some changes in the order do not affect it.

Proposition 9 (Invariance)**.** *Let $\mathcal{R} = (M_1, \ldots, M_K)$ be a trading round. Then $\Lambda(\mathcal{R})$ is unchanged by (i) cyclic permutation of the coalitions, (ii) reversal of their order, and (iii) the exchange of two consecutive coalitions with no agent in common. These operations reach every ordering when $K \leq 3$, so $\Lambda(\mathcal{R})$ is then independent of the order. For $K \geq 4$ they do not.*

Each invariance has a simple reading in terms of trade. First, a repeated round has no first meeting, so a rotation changes only the date from which one counts.

[11] Both disequilibria are measured in $\|\cdot\|_\pi$. The statement follows from $\boldsymbol{e}(t) = \boldsymbol{e}(0)A(t)$ (Lemma 3) and Lemma 20.

[12] Since $\Pi A(M) = \Pi$ and $A(M)\Pi = \Pi$ for every coalition, $\big(A(\mathcal{R}_1) - \Pi\big)\big(A(\mathcal{R}_2) - \Pi\big) = A(\mathcal{R}_1)A(\mathcal{R}_2) - \Pi$.

Second, the reversed round has the same meetings in the opposite order, and by Lemma 18 its operator is the adjoint of the original, with the same eigenvalue moduli. Third, two coalitions with no agent in common do not affect each other's trade. Therefore, the order of a round can change its gap only through the order of overlapping coalitions. Rotations and reversals give at most $2K$ of the $K!$ possible orders. These are all the orders when $K \leq 3$. However, they are only some of the orders when $K \geq 4$, and Example 9 shows that the gap can then depend on the order.

Example 9. Take four agents on the complete network with logarithmic preferences and intensities $\boldsymbol{a} = (2, 5, 10, 3)$ for agents A to D, and the round consisting of all six links. Traded in the order (AB, AC, AD, BC, BD, CD) the round has $\Lambda = 0.824$. Traded in the order (AB, CD, AC, BD, BC, AD) it has $\Lambda = 0.993$. The two rounds use the same cliques and differ only in when each pair meets. $\diamond$

In the long run, each repetition of a round multiplies the remaining disequilibrium by $1 - \Lambda = |\lambda_2|$. In Example 9, this factor is $1 - 0.824 = 0.176$ under the first order and $1 - 0.993 = 0.007$ under the second. Hence, in the long run, each repetition leaves about 18 per cent of the remaining disequilibrium under the first order, and less than 1 per cent under the second. Under both orders, the allocation of the non-numeraire goods converges to $\boldsymbol{x}(N)$.

4.2 Myopic Dominance

When a coalition meets as one market, all its members trade at one price, and the exchange leaves no gains from trade among them. When the same agents meet instead in smaller coalitions inside it, each smaller coalition clears at its own price. After these meetings, some members of the coalition may still gain from trading with each other. This subsection makes this intuition precise at the end of the coalition's turn.

To make the comparison, I first define when smaller coalitions refine a coalition (Definition 4), and then what it means to refine a sequence at one stage (Definition 5). The second definition applies to any sequence, because nothing in this subsection requires the sequence to repeat.

Definition 4 (Refinement of a Coalition). Let M be an admissible coalition. A family $M_1, \ldots, M_s$ of admissible coalitions *refines* M if $M_r \subseteq M$ for every r, every agent of M belongs to some M_r, and the M_r connect M, in the sense that its agents cannot be split into two nonempty groups sharing no M_r. The refinement is *non-trivial* if M does not itself appear among the M_r.

Definition 5 (Refinement of a Sequence). Let $\mathcal{C} = (M_{(t)})_{t \in \mathbb{N}}$ be a sequence, and let $M_1, \ldots, M_s$ refine $M_{(t)}$. A sequence $\mathcal{C}'$ *refines* $\mathcal{C}$ *at stage* t if $\mathcal{C}'$ replaces $M_{(t)}$ with

$M_1, \ldots, M_s$, in that order and consecutively, and leaves every other entry in place.

Two conditions of Definition 4 deserve comment. First, every agent of M must appear, so that a refinement breaks up a meeting without dropping a trader. Second, the smaller coalitions must connect M, so that goods can still move between any two of its members. However, they need not use every link of M. For example, BD and BE refine BDE. They meet through agent B and leave the link DE unused. The examples in this section use this refinement.

The order matters in Definition 5. For example, take $\mathcal{C} = (AB, BCD)$. The sequence (AB, BC, CD) refines $\mathcal{C}$. However, (BCD, AB) does not refine $\mathcal{C}$, because it only changes the order of the same coalitions.

Write $E(\mathcal{C}; t)$ for the error $E(t)$ along the sequence $\mathcal{C}$. Proposition 10 compares $\mathcal{C}$ with a refinement $\mathcal{C}'$ when the replaced stage has finished in each, that is, at date t along $\mathcal{C}$ and at date $t - 1 + s$ along $\mathcal{C}'$. The two sequences agree before this stage. Therefore, the comparison concerns only the stage that the refinement replaces.

Proposition 10. *Let $\mathcal{C}'$ refine $\mathcal{C}$ at stage t, replacing $M_{(t)}$ with s coalitions. Then*

$$\left\|E(\mathcal{C}; t)\right\|_\pi \;\leq\; \left\|E(\mathcal{C}'; t - 1 + s)\right\|_\pi. \tag{20}$$

The proof uses that the two sequences agree before stage t. After that stage, the two economies start each meeting from different holdings. A later coalition may remove more of the disequilibrium that the refinement leaves than of the disequilibrium that the single market leaves. Therefore, the comparison need not hold at later dates, and Section 4.3 shows that it can reverse over many repetitions of a round.

For a trading round, the comparison applies at the end of every pass that ends with the refined coalition. A repeated round has no first meeting, so it can be rotated to place the refined coalition last. Corollary 11 states the result for trading rounds.

Corollary 11 (Refinement Dominance)**.** *Let $\mathcal{R}$ be a trading round and let $\mathcal{R}'$ refine it at stage j. Write $\mathcal{R}_{\mathrm{rot}}$ and $\mathcal{R}'_{\mathrm{rot}}$ for their cyclic rotations placing M_j, respectively the coalitions replacing it, last. Then $\|E(\mathcal{R}_{\mathrm{rot}})\|_\pi \leq \|E(\mathcal{R}'_{\mathrm{rot}})\|_\pi$, whatever the stage j at which the refinement occurs.*

Proposition 10 and Corollary 11 make the same comparison. Whatever the coalitions that met before it, a coalition that meets as one market leaves the economy weakly closer to equilibrium at the end of its turn than any refinement of it. In this sense, the single market dominates its refinements myopically, that is, for the current turn only. Example 10 gives the numbers for every refinement of one triangle.

Example 10. Five agents trade on the bowtie of Figure 5, with weights

$$\boldsymbol{\alpha} = (4, 1, 20, 3, 8), \qquad \alpha(N) = 36, \qquad \boldsymbol{\pi} = \frac{\boldsymbol{\alpha}}{\alpha(N)} = \left(\tfrac{1}{9}, \tfrac{1}{36}, \tfrac{5}{9}, \tfrac{1}{12}, \tfrac{2}{9}\right).$$

The exchange matrices depend on preferences only through $\boldsymbol{\alpha}$, so any HARA parameters that give these weights give the same operators. Agent C holds the largest centralised share and agent B the smallest, and agent B is the only link between the two triangles. Take $\mathcal{R} = (ABC, BDE)$ and refine the triangle BDE in place. Since this triangle already ends the pass, Corollary 11 applies with no rotation: Every refinement leaves the economy further from equilibrium after one

Coalitions replacing BDE	K	$\|E\|_\pi$
— (no refinement)	2	0.9381
BD, BE	3	0.9571
DE, BD	3	0.9496
DE, BE	3	0.9400
BD, BE, DE	4	0.9383

Table 2: Distance from equilibrium after one pass, for every refinement of the triangle BDE.

pass, including the refinement that uses all three links. ◇

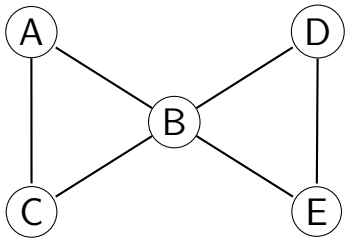


Figure 5: The bowtie network of Example 10. The two triangles are the largest coalitions available, and agent B is the only agent belonging to both.

4.3 Limits of Dominance

Section 4.2 compares a coalition with its refinements at the end of one turn. This subsection gives the asymptotic counterpart of that comparison, first over many repetitions of a trading round, and then when coalitions are drawn at random. Over many repetitions, the relevant measure is the spectral gap. A smaller norm after one pass does not imply a larger spectral gap. Proposition 12 shows that a refinement can have a larger spectral gap than the round it refines, and Example 11 gives such an economy. In that case, the refinement is not closer to equilibrium after one pass, but by Equation (19) it is closer after enough passes.

Proposition 12. *There exist a network, weights $\boldsymbol{\pi}$, a trading round $\mathcal{R}$ and a non-trivial refinement $\mathcal{R}'$ of it such that*

$$\Lambda(\mathcal{R}') > \Lambda(\mathcal{R}), \qquad \Lambda_{\text{lin}}(\mathcal{R}') > \Lambda_{\text{lin}}(\mathcal{R}), \qquad \Lambda_{\text{eff}}(\mathcal{R}') > \Lambda_{\text{eff}}(\mathcal{R}),$$

while $\|E(\mathcal{R})\|_\pi \leq \|E(\mathcal{R}')\|_\pi$, as Corollary 11 requires.

Example 11. Return to the bowtie of Figure 5 and the weights of Example 10, with parent round $\mathcal{R} = (ABC, BDE)$. Its refinement $\mathcal{R}'$ replaces the triangle BDE with BD and BE. Agent B meets each of its two neighbours separately, and agents D and E never meet. As in Example 10, the refined coalition ends the pass, so Corollary 11 applies with no rotation. Figure 6 draws the two rounds. Adding the three spectral gaps to the norms of Example 10 gives Under the

Coalitions replacing BDE	K	$\|E\|_\pi$	Λ	Λ_{lin}	Λ_{eff}
— (no refinement)	2	0.9381	0.1200	0.0600	0.0619
BD, BE	3	0.9571	**0.2000**	**0.0667**	**0.0717**
DE, BD	3	0.9496	0.0982	0.0327	0.0339
DE, BE	3	0.9400	0.1164	0.0388	0.0404
BD, BE, DE	4	0.9383	0.1273	0.0318	0.0335

Table 3: Speed statistics with the spectral gaps for the same refinements.

refinement BD, BE, the norm rises from $\|E(\mathcal{R})\|_\pi = 0.9381$ to $\|E(\mathcal{R}')\|_\pi = 0.9571$. Therefore, the refinement removes less disequilibrium over one pass, as Corollary 11 requires. However, the spectral gap also rises, from $\Lambda(\mathcal{R}) = 0.12 = 3/25$ to $\Lambda(\mathcal{R}') = 0.20 = 1/5$, so the refinement converges faster over many passes. Both normalised gaps rise as well, so the refinement is faster per trade, although each pass uses three trades instead of two. The last row shows why the normalised gaps are useful. The refinement that uses all three links has a larger gap than the triangle, 0.1273 against 0.1200, but it uses four trades per pass instead of two. Per trade, it is slower, and both normalised gaps show this. ◇

To see where the reversal comes from, I follow the trade in Example 11 from equal endowments. Every agent holds $\omega^i = 36$ of the good, and $\mathbf{b} = 0$. By Equation (4), the centralised market divides the aggregate in proportion to the weights,

$$\boldsymbol{x}(N) = \boldsymbol{\omega}\Pi = 180\,\boldsymbol{\pi} = (20, 5, 100, 15, 40).$$

Hence the initial disequilibrium is

$$\boldsymbol{e}(0) = \boldsymbol{\omega} - \boldsymbol{x}(N) = (16,\ 31,\ -64,\ 21,\ -4).$$

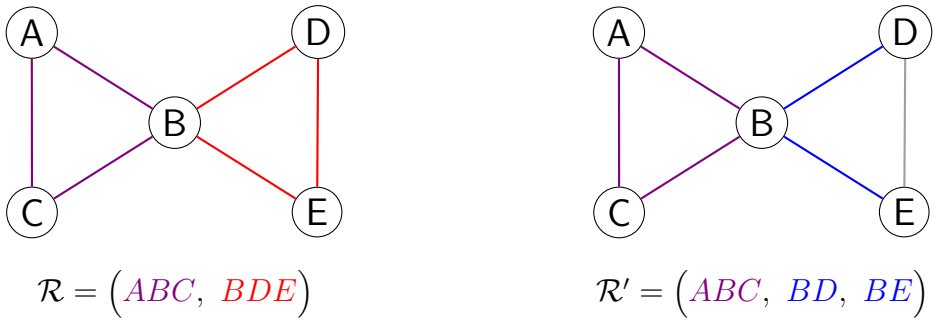


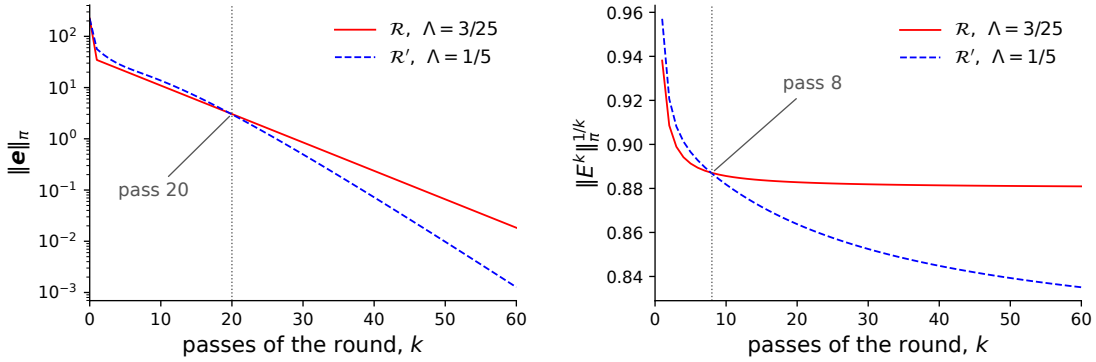
Figure 6: The two rounds of Example 11 on the bowtie. Red marks the coalition that only $\mathcal{R}$ convenes, blue the coalitions that only $\mathcal{R}'$ convenes, and purple the coalition ABC that both convene. The colours match Figure 7. The two rounds differ only in the link DE, which the refinement leaves unused.

Agents A, B and D hold more than their centralised allocation, and agents C and E hold less. In particular, agent C holds 36 of the 100 units that the centralised market gives it.

If BDE met on its own at these endowments, its members would bring 36 units each and divide the 108 units in the ratio $\pi^i/\pi(BDE) = (1, 3, 8)/12$. They would leave with 9, 27 and 72. Inside the round, agent B has already traded in ABC when the triangle meets, so the numbers are less round, but the rule is the same. At the end of the first pass, agents B, D and E hold 6.36, 19.08 and 50.88 under $\mathcal{R}$, and 5.12, 30.24 and 40.96 under $\mathcal{R}'$. Their centralised allocations are 5, 15 and 40, so their disequilibria are (1.36, 4.08, 10.88) under $\mathcal{R}$ and (0.12, 15.24, 0.96) under $\mathcal{R}'$. Over the whole economy, the parent round is well ahead after one pass, with $\|\boldsymbol{e}\|_\pi = 34.62$ against 56.49, as the proof of Proposition 10 shows.

Figure 7 follows the two economies pass by pass. A pass of $\mathcal{R}$ has two trades and a pass of $\mathcal{R}'$ has three, so the figure compares the two rounds after the same number of passes. Panel (a) starts from the equal-endowment profile, at $\|\boldsymbol{e}(0)\|_\pi = 222.8$. After 19 passes, the parent round is still ahead, 3.47 against 3.56. From pass 20 on, the refinement is ahead, and after 40 passes the two stand at 0.24 and 0.07. Compared after the same number of trades, the refinement leads only from date 360, in line with the small difference between the effective gaps.

Both panels count passes, but they start from different disequilibria. Corollary 11 compares the two curves of panel (b) at $k = 1$ only. Unless the round operators are normal, this comparison does not determine the comparison at later passes. Panel (b) shows, for each round, the largest share of disequilibrium that can remain after k passes, over all initial disequilibria, expressed per pass. Panel (a) starts from the equal-endowment profile, which is far from this worst case. After one pass, for instance, $\mathcal{R}$ leaves about 16 per cent of that profile's disequilibrium, while in the worst case it leaves 94 per cent. Therefore, the worst-case comparison changes



(a) From equal endowments

(b) Least favourable initial disequilibrium

Figure 7: Convergence of the two rounds of Example 11, against the number of passes k of each round. A pass of $\mathcal{R}$ has two trades and a pass of $\mathcal{R}'$ has three. Panel (a) shows $\|\boldsymbol{e}\|_\pi$ after k passes from the equal-endowment profile $\omega^i = 36$, on a logarithmic scale. Panel (b) shows $\|E^k\|_\pi^{1/k}$, the largest share of disequilibrium that can remain after k passes, expressed per pass.

at pass 8, and the comparison from the equal endowments of Example 11 changes at pass 20. By Equation (19), each curve in panel (b) converges to $|\lambda_2| = 1 - \Lambda$ of its round, so the refinement stays ahead from pass 8 on.

The reason lies in the pattern of the disequilibrium that each round leaves. When a coalition meets as one market, each member leaves with the share $\pi^i/\pi(M)$ of the coalition's total. Therefore, the members' disequilibria are proportional to their weights, and no trade among them remains.

Under $\mathcal{R}$, the disequilibria of agents B, D and E after the first pass, (1.36, 4.08, 10.88), stand exactly in the ratio $(1, 3, 8)$ of their weights. More generally, the ratio of disequilibrium to weight is then the same for agents A and C, and the same for agents B, D and E. This is the slowest pattern of $\mathcal{R}$. From the first pass on, each repetition of $\mathcal{R}$ reduces it by the factor $|\lambda_2| = 22/25$ exactly, and only agent B, the one agent in both triangles, can move it between the two groups.

Under $\mathcal{R}'$, the disequilibria (0.12, 15.24, 0.96) are not in the ratio $(1, 3, 8)$. Agent B carries 0.12, about an eleventh as much as under $\mathcal{R}$, and agent D holds almost all of the triangle's remaining disequilibrium. Agents D and E never meet, so agent B removes this disagreement a little at each visit. The slowest pattern of $\mathcal{R}'$ shrinks at rate $|\lambda_2| = 4/5$, which is faster than under $\mathcal{R}$. Hence the link DE, which allows BDE to meet as one market, offers an option that slows convergence over many passes.

Proposition 12 requires coalitions that overlap. If the clearings of both rounds commuted, both round operators would be self-adjoint, and the norm of each error

operator would equal its $|\lambda_2|$.[13] Then the one-pass comparison of Corollary 11 would determine the comparison over many passes, and no refinement could overtake the round it refines. Coalitions with no agent in common commute, which is why Proposition 9 allows their exchange. Hence a reversal requires agents who belong to more than one coalition of the round, such as agent B in the bowtie, who connects the two triangles. Overlap is necessary for a reversal, but it does not guarantee one.

The paper gives no simple test for when a refinement overtakes the round it refines. The condition is $\Lambda(\mathcal{R}') > \Lambda(\mathcal{R})$, and both gaps depend on the eigenvalues of products of exchange operators, which do not reduce to a condition on the network and the weights.

The reversal in Proposition 12, where a refinement is behind after one pass but ahead after many, depends on a fixed order of trade. When coalitions are drawn at random, the comparison of Section 4.2 extends to the asymptotic rate. As in Section 3.3, let μ be a distribution over admissible coalitions, and draw one coalition independently at each date.

Definition 6 (Refinement of a Distribution). Let $M_1, \ldots, M_s$ refine M. A distribution μ' on the admissible family *refines* μ *at* M if it replaces the mass μ_M with positive masses $w_1, \ldots, w_s$ on $M_1, \ldots, M_s$, which sum to μ_M, and leaves every other mass unchanged.

Unlike Definition 5, this definition involves no order. The refining coalitions may be drawn at dates that are not adjacent.

Proposition 13 (Randomised Order Restores Dominance)**.** *Let μ satisfy the maintained hypothesis of Theorem 4, write $\bar{A}_\mu = \sum_M \mu_M A(M)$ for the expected exchange operator, and set $\Lambda(\mu) = 1 - |\lambda_2(\bar{A}_\mu)|$. Let μ' refine μ at a coalition M, in the sense of Definition 6. Then*

$$\Lambda(\mu') \;\leq\; \Lambda(\mu).$$

Because the refining coalitions connect M, the support of μ' again connects the economy, so μ' also satisfies the hypothesis of Theorem 4. Each draw is one trade, so $\Lambda(\mu)$ is a rate per trade.

The difference from Proposition 12 lies in how the exchanges combine. A trading round applies its exchanges one after another, and a product of self-adjoint operators is in general not self-adjoint. A random draw averages the exchanges, and an average of self-adjoint operators is self-adjoint. Therefore, for the expected exchange operator, the norm equals $|\lambda_2|$, so the comparison after one trade and the asymptotic comparison coincide. The gap $\Lambda(\mu)$ is the rate of the expected

[13] A product of commuting self-adjoint operators is self-adjoint, and the norm of a self-adjoint operator equals its spectral radius.

disequilibrium, $\mathbb{E}[\boldsymbol{e}(t)] = \boldsymbol{e}(0)\bar{A}_{\mu}^{t}$. The rate along a realised sequence is a Lyapunov exponent, and it need not equal $\Lambda(\mu)$.

4.4 Redistribution Capacity

Sections 4.1 to 4.3 compare trading structures by speed. This subsection compares them by redistribution. Theorem 3 gives, for each round, the map from the initial disequilibrium to the transfer that each agent receives. The transfers are zero-sum, so I measure how much a round redistributes by the sum of the absolute transfers across agents. I use this sum to define a statistic that depends only on the trading structure, and I then ask whether refinement orders that statistic.

Definition 7 (Redistribution Capacity). The *redistribution capacity* of a trading round is

$$\operatorname{cap}(\mathcal{R}) := \sup_{\boldsymbol{e}\mathbf{1}=0,\ \|\boldsymbol{e}\|_{\pi}=1} \sum_{i} \left| \boldsymbol{e}\, \mathcal{K}_i(\mathcal{R})\, \boldsymbol{e}^{\top} \right|, \tag{21}$$

the largest sum of absolute transfers the round can induce over normalised disequilibria.

Three features of Equation (21) make it a comparator. First, the absolute values are taken agent by agent, because the transfers are zero-sum, so the signed sum is zero for every round and every endowment. Second, the disequilibrium is normalised, because the transfers are quadratic in it, so that a larger initial disequilibrium would otherwise raise the statistic without saying anything about the round. Third, the supremum is over initial disequilibria, as in the operator norm of Section 4.1, so capacity measures what a round can redistribute rather than what it redistributes in one economy. Therefore, capacity depends only on the network, the round and preferences, and two structures can be compared without fixing endowments. The form is homogeneous of degree two, so the sum of absolute transfers grows with the square of the size of the initial disequilibrium, and capacity is the largest factor of that growth.

Example 12. Take the bowtie of Figure 5 again, now with weights $\boldsymbol{\alpha} = (16, 9, 1, 1, 1)$, so that

$$\boldsymbol{\pi} = \left(\tfrac{4}{7}, \tfrac{9}{28}, \tfrac{1}{28}, \tfrac{1}{28}, \tfrac{1}{28}\right).$$

The network is the one of Example 11, and the weights differ because what follows turns on the size of the transfers rather than on the rate of convergence. Preferences are quadratic, so Theorem 3 is exact. Table 4 compares the round $\mathcal{R} = (ABC, BDE)$ with its reversal, and with the refinement that replaces BDE by BD and BE. ◇

The first two rows convene the same two coalitions in opposite orders, and every speed statistic agrees on them. The three gaps are invariant to where the

Round	$\|E\|_\pi$	Λ	Λ_{lin}	Λ_{eff}	cap
(ABC, BDE)	0.3448	0.8811	0.4406	0.6552	0.00962
(BDE, ABC)	0.3448	0.8811	0.4406	0.6552	**0.01281**
(ABC, BD, BE)	0.3508	0.9191	0.3064	0.5676	**0.00774**

Table 4: Speed statistics and capacity, with $\boldsymbol{\alpha} = (16, 9, 1, 1, 1)$.

cycle is cut (Proposition 9), and the norm is unchanged because reversing a round replaces its operator by the adjoint (Lemma 18), which has the same singular values. However, speed and redistribution are not aligned, as capacity differs by a third between the two rows, 0.01281 against 0.00962. Moreover, refining the second coalition lowers capacity, to 0.00774.

Example 13. Take the same network and the same three rounds, now with weights $\boldsymbol{\alpha} = (1, 16, 1, 16, 16)$, so that

$$\boldsymbol{\pi} = \left(\tfrac{1}{50}, \tfrac{8}{25}, \tfrac{1}{50}, \tfrac{8}{25}, \tfrac{8}{25}\right). \qquad \diamond$$

Round	$\|E\|_\pi$	Λ	Λ_{lin}	Λ_{eff}	cap
(ABC, BDE)	0.2722	0.9259	0.4630	0.7278	0.00566
(BDE, ABC)	0.2722	0.9259	0.4630	0.7278	**0.00369**
(ABC, BD, BE)	0.5172	0.8333	0.2778	0.4497	**0.01006**

Table 5: Speed statistics and capacity, with $\boldsymbol{\alpha} = (1, 16, 1, 16, 16)$.

In this economy the refinement raises capacity, from 0.00566 to 0.01006, and the reversal lowers it, to 0.00369. Both movements are the opposite of those in Example 12, although the network and the three rounds are the same. Therefore, refinement does not order capacity, and neither does the order in which two coalitions meet. The refinement is also slower here, with $\Lambda = 0.8333$ against 0.9259, so a round can redistribute more and converge more slowly at the same time.

A transfer comes from the difference between the price at which an agent trades and the centralised price, so what a round redistributes depends on which agents trade before prices have converged. The speed statistics do not record this. Two rounds that leave the economy equally close to the centralised allocation can have moved different agents at different prices, as the round and its reversal do in Examples 12 and 13. Capacity is therefore not redundant, and no statistic of Section 4.1 can replace it.

Only one inequality relates the two measures. Proposition 14 bounds capacity by the within-round kernels and the operator norm of the round. It follows from taking norms in the decomposition of Equation (13), which separates the transfer made during one pass from the transfer accumulated over later passes.

Proposition 14 (Capacity Bound)**.** *For every trading round,*

$$\operatorname{cap}(\mathcal{R}) \;\le\; \frac{\sum_i \|B_i\|_\pi}{1 - \|E(\mathcal{R})\|_\pi^2}, \tag{22}$$

where B_i is the within-round kernel of Equation (11)*.*[14]

The bound does not order rounds. Under a refinement the operator norm rises (Corollary 11), so the denominator of Equation (22) falls and the bound rises. However, the refinement also rebuilds the kernels in the numerator, and these can move either way. Table 6 reports both sides for the two economies. In Example 12 the bound rises from 0.0639 to 0.0643 while the capacity falls from 0.00962 to 0.00774, and in Example 13 the two rise together. The bound is also loose, since capacity lies roughly between a tenth and a fifth of it in all six rounds.

Weights	Round	cap	Bound
$(16, 9, 1, 1, 1)$	(ABC, BDE)	0.00962	0.0639
	(BDE, ABC)	0.01281	0.0697
	(ABC, BD, BE)	0.00774	0.0643
$(1, 16, 1, 16, 16)$	(ABC, BDE)	0.00566	0.0403
	(BDE, ABC)	0.00369	0.0374
	(ABC, BD, BE)	0.01006	0.0511

Table 6: Capacity and its bound for the rounds of the two examples.

A natural conjecture is that a refinement both slows the current exchange and redistributes more. Example 12 refutes it, since there the refinement redistributes less. The comparison does not depend on the horizon either. In that example the within-round kernels alone give 0.00927 against 0.00755, so the refinement redistributes less already within one pass.

Randomising the order does not restore an ordering. Proposition 13 holds because the expected exchange operator is an average of projections, and a smaller coalition projects onto a larger subspace, so the comparison holds term by term.

[14] Two norms appear in Equation (22). For the error operator, $\|E\|_\pi$ is the operator norm of Section 4.1. For the kernels it is the norm of the associated quadratic form, $\|B_i\|_\pi = \|D^{1/2} B_i D^{1/2}\|$ with $D = \operatorname{diag}(\boldsymbol{\pi})$, which is what the Rayleigh step below requires.

Expected capacity has no such structure. Its kernel is built from that operator and from $\bar{B}_i$, which records who attends each meeting, and a refinement rebuilds the second while raising the first. Drawing the two coalitions of Example 12 with equal probability, and splitting the mass of the triangle equally between BD and BE, expected capacity falls from 0.01116 to 0.00858 under the weights of Example 12, and rises from 0.00466 to 0.01030 under those of Example 13.

Capacity is a statistic of a repeated round or of a random protocol. For a protocol drawing its coalitions from μ, Corollary 8 supplies the kernel $\bar{\mathcal{K}}_i(\mu)$, and Equation (21) applies unchanged. The absolute value then falls outside the expectation, so capacity measures the systematic part of redistribution and not the variation around it. For an arbitrary deterministic sequence there is no kernel, since Theorem 3 builds $\mathcal{K}_i$ by iterating a single operator.

Two guarantees remain. Convening a coalition as one market leaves the economy weakly closer to equilibrium at the end of that coalition's turn than any refinement of it (Proposition 10). When the order of trade is drawn at random, a refinement converges no faster in expectation (Proposition 13). Neither concerns redistribution. How much numeraire a structure moves, and who receives it, depend on the order in which its coalitions meet.

5 Related Literature

Exchange economies are a foundational concept in microeconomic theory, representing an environment where rational agents interact solely through trade to reach mutually beneficial allocations. Standard general equilibrium theory, rooted in the work of Arrow and Debreu (1954) and Debreu (1959), relies on prices at which all markets clear simultaneously. This paper bridges classical out-of-equilibrium dynamics with modern network economics to study sequential, decentralised trade, contributing to four primary strands of literature.

Out-of-Equilibrium Dynamics and Sequential Exchange. Uzawa (1962) and Hahn and Negishi (1962) allow trade at prices common to the whole economy that are not yet equilibrium prices, and show that the economy approaches an equilibrium. In these processes, the final allocation depends on the path of trade, as Edgeworth already observed (Uzawa, 1962). Foley (1994) studies agents who meet and trade at random. That is, every mutually advantageous trade is equally probable, so agents trade at different price ratios. Therefore, two agents with the same preferences and endowments can reach different final bundles, and the final allocation only approximates Pareto efficiency. Werning and Lorenzoni (2026) return to the stability question with forward-looking agents who set prices

subject to rigidities. Away from equilibrium, rationing determines quantities, and each price adjusts to a discounted average of current and future excess demand. Here, in contrast, agents take prices as given within each clique, each exchange clears at its own local price, and I characterise how the path determines the final distribution of the numeraire.

Feldman (1973) studies bilateral trade and shows that it reaches Pareto optimality when every pair of traders meets in each round of bounded length. Madden (1975) allows larger groups, and each meeting reaches an allocation that is efficient for the group. That paper shows that some order of meetings, or under stronger conditions any cycle of bounded length, brings the economy close to Pareto optimality. Goldman and Starr (1982) study the static question of when t-wise optimality, the absence of improvements by groups of at most t traders, implies Pareto optimality, and they leave the dynamics open. Relative to this literature, I restrict preferences to the common-curvature HARA family and each meeting to Walrasian clearing. In exchange, I show convergence to a unique limit along every sequence in which the relationships used infinitely often connect the economy. Moreover, I characterise the distribution of the numeraire at that limit for repeated sequences and random sequences.

Trading Networks and Decentralised Exchange. A second strand studies trade in networks. Hatfield et al. (2013) study agents who choose which bilateral contracts to sign on a given network, so that the trades form a subnetwork. With quasilinear utility and fully substitutable preferences, stable outcomes exist, coincide with competitive equilibria, and are efficient (see also Hatfield et al., 2019). Candogan et al. (2021) obtain these results from a network flow problem. Finster et al. (2026) show that two decentralised processes, one in which agents adjust their offers and one in which a price clock adjusts, converge to stable outcomes in this framework. They also show that agents who are not essential receive zero utility in every core outcome. Kakade et al. (2004) study an exchange economy in which agents trade only with their neighbours in a network. In equilibrium, the price of a good differs across agents because of the network structure, and more connected networks equalise prices. Malamud and Rostek (2017) divide a market into exchanges, each formed by a group of traders, and all exchanges clear at the same time. With strategic traders and quasilinear-quadratic utility, a more decentralised market has a higher price impact, but it can achieve higher welfare. Their notion of a more decentralised market, in which every exchange is contained in an exchange of the original market, is close to the refinement I use in Section 4. In all these models, goods change hands only once, at the final outcome. Here, in contrast, each exchange is carried out, the resulting holdings become the endowments for the next exchange, and I study the dynamics that this

sequence generates. Prices differ across cliques along the way, but they converge to the price of the centralised market.

A large literature studies strategic trade in networks. Kranton and Minehart (2001) show that buyers and sellers who need a link to trade can form efficient networks. Corominas-Bosch (2004) studies buyers and sellers who make public offers to their neighbours, and characterises the networks in which the equilibrium coincides with the Walrasian outcome. Polanski (2007) and Abreu and Manea (2012) study bilateral bargaining in which players who agree leave the network. When a maximum matching bargains in each period, trade is efficient (Polanski, 2007). When a single link bargains in each period, Markov equilibria can be multiple and inefficient (Abreu and Manea, 2012). Other papers study intermediaries who earn rents from their position in the network (Blume et al., 2009; Condorelli et al., 2017; Manea, 2018; Siedlarek, 2025). Gale and Kariv (2007) show that sequential resale is efficient when each seller makes a take-it-or-leave-it offer to a buyer. Gofman (2014) shows that sequential trade is inefficient when intermediaries cannot extract the full surplus, and that adding trading relationships can reduce efficiency. Similarly, in Manea (2018), adding a link never lowers the seller's profit, but it can reduce welfare. For surveys, see Manea (2016) and Bramoullé et al. (2016). In over-the-counter markets, Babus and Kondor (2018) study dealers who trade through quantity-price schedules under private information, and Colliard and Demange (2021) study how dealers sell an asset along a chain of other dealers. In contrast, the agents here are price-takers, and each meeting is a local Walrasian exchange. Such an exchange is efficient for the members of the clique. Therefore, with quasilinear preferences, the allocation of goods is efficient on every network. The network affects only the speed of convergence and the distribution of the numeraire.

The closest paper is Cassese and Pin (2025). They model trade as a continuous-time Edgeworth process, in which agents exchange bilaterally with their neighbours in a weighted network, and every trade divides the gain in utility equally between the two agents. The process converges to a Pareto optimal allocation, and, given the endowments, the network determines which one. Since the dynamics has no analytical solution even with two agents and two goods, they study the effect of the network on welfare by simulation. Here, local Walrasian exchanges within cliques keep the dynamics tractable, and quasilinearity confines the effect of the network on the limit allocation to the numeraire.

Consensus and Randomised Gossip. The disequilibrium follows a consensus process, which economics and engineering have both studied. In DeGroot (1974), agents repeatedly average the opinions of others, and the consensus is a weighted average of initial opinions, with weights given by the stationary vector of the averaging matrix. Golub and Jackson (2010) study this process in networks,

where these weights measure each agent's influence, and the speed of convergence depends on the second eigenvalue of the averaging matrix. Boyd et al. (2006) study randomised gossip, in which a randomly chosen pair of agents averages at each date. Each averaging step is a symmetric projection, and the speed depends on the second eigenvalue of the expected averaging matrix. Fagnani and Zampieri (2008) show that the second moments of such processes follow a linear recursion. They also measure how far the limit moves from the initial average by a quadratic form in the initial state. I use these results through a change of variables. Each exchange matrix here is row-stochastic, and all of them have the same stationary vector, the equilibrium shares. Therefore, a diagonal change of basis makes every exchange matrix a symmetric projection, as in Boyd et al. (2006). The economic content differs in two respects. First, in a consensus process the network determines the limit through the influence weights. Here, preferences determine the weights, so every sequence of trades reaches the same limit for the goods. Second, I study the numeraire that agents transfer on the way to the limit. To leading order, this transfer is also a quadratic form in the initial state. When cliques are drawn at random, its expected value depends on the second moments of the disequilibrium summed over all dates, and these solve a linear equation of the same form as in Fagnani and Zampieri (2008). In Fagnani and Zampieri (2008), the limit moves only because the weights are random. Here, the weights never change, and the network moves the numeraire instead.

Equilibrium Computation. The process also relates to the computation of competitive equilibria. Scarf (1967) approximates equilibrium prices with a simplicial algorithm that uses the market excess demand. Manne et al. (1980) alternate between a master problem, which generates prices and balances the aggregate bundle, and a subproblem for each household. Both methods require a central procedure with information on the whole economy. As Werning and Lorenzoni (2026) point out, such methods are computational devices and do not describe decentralised markets. Here, each clique uses only the preferences and holdings of its members, and the sequence of local exchanges still reaches the allocation of the centralised market. Moreover, a computation method is judged by how accurately and how quickly it approximates the equilibrium. In this model, every admissible sequence reaches the same allocation of goods. Therefore, the sequences differ in speed and in the numeraire they transfer.

6 Conclusions and Future Research

This paper studies exchange in which agents trade sequentially with subsets of counterparties, and asks what the trading structure determines. The first result is a separation. Under additively separable, quasilinear HARA preferences with a common curvature, the allocation of every non-numeraire good converges to the allocation of the centralised market, along every sequence in which the relationships used infinitely often connect the economy. The allocation of the numeraire depends on the path. The hypothesis on the sequence is what is new here. Feldman (1973) requires every pair of traders to meet in each round of bounded length. In Madden (1975), some order of meetings brings the economy arbitrarily close to efficiency, and cyclical trade does the same under stronger conditions. Here the time between two meetings of a coalition may grow without bound, and some coalitions may never meet. The reason the numeraire does not follow the goods is that local prices differ from the centralised price until the economy converges. Each exchange therefore transfers numeraire, and these transfers accumulate to a finite limit that depends on the order of trade.

The second result computes that transfer. Along a trading round repeated indefinitely, the transfer to each agent is a quadratic form in the initial disequilibrium, and its kernel is the unique solution of a discrete Lyapunov equation (Theorem 3). The transfer is an infinite sum over trading dates, and each term depends on the whole history before it. Nevertheless, the sum reduces to one system of linear equations, whose size does not depend on the length of the round. When coalitions are drawn at random rather than scheduled, the expected transfer takes the same form, with the second-moment operator of the draw in place of the round's (Theorem 4). This expectation is in general not zero, and it does not vanish as the horizon grows. The form is exact under quadratic preferences, and elsewhere it is the leading term, with an error bounded by a cubic in the initial disequilibrium (Proposition 5).

The comparisons between trading structures rest on a common method. Each exchange is affine in the holdings and, in a preference-weighted inner product, an orthogonal projection. A sequence of exchanges is therefore a product of projections, which places the dynamics in the classical theory of alternating projections (Halperin, 1962; Amemiya and Ando, 1965) and yields both the rates of convergence and the comparisons of Section 4.

Those comparisons are mostly negative. Convening a coalition as one market leaves the economy weakly closer to the centralised allocation at the end of that coalition's turn than any refinement of it (Proposition 10). The guarantee ends with the turn, since a refinement can converge at a faster rate over many repetitions and overtake the round it refines (Proposition 12). When coalitions are drawn at random, a refinement converges no faster in expectation (Proposition 13). For

redistribution there is no ranking. No speed statistic orders it, and a refinement can raise the redistribution capacity or lower it, whether the order is fixed or drawn (Section 4.4). Coalition size therefore ranks trading structures in the current exchange, and over longer horizons only when the order is drawn at random.

These results bear on regulation. A regulator concerned only with efficiency can be indifferent to the trading structure, since the non-numeraire goods converge to the allocation of the centralised market whatever the structure. A regulator concerned with speed faces a trade-off over time. Merging coalitions into one market brings the economy closer to the centralised allocation in the current exchange. However, when the same sequence of trades repeats, a finer structure can converge faster over many exchanges. A regulator concerned with distribution cannot be indifferent, because the order in which coalitions meet determines who ends up holding the numeraire. However, no simple rule, such as merging markets, controls this effect. Such a regulator must therefore evaluate each trading structure directly, and the closed form of the transfer and the redistribution capacity make that evaluation possible.

The framework does not predict which agents gain. An agent's transfer is a quadratic form in the initial disequilibrium, and this form is in general indefinite, so the same agent in the same position can gain under some initial disequilibria and lose under others.

The framework suggests several extensions. The first is comparative statics of the classical kind, holding the trading structure fixed and varying preferences and endowments, which the closed form makes available and this paper does not pursue. A second is to average over initial disequilibria rather than over the order of trade. Both closed forms depend on the initial disequilibrium only through an outer product, so replacing that product by a covariance gives the expected transfer as a function of the trading structure alone, and a prediction about who gains would have to come from there. A third is to relax additive separability, which would require tracking the dependence between goods during exchange. A fourth is to endogenise the trading sequence, so that the choice of partner responds to the transfers it generates. Finally, the path-dependence of the numeraire poses a design question. Given an endowment and a target distribution of the numeraire, is that target in the range of the map from trading structures to transfers, and which structures attain it?

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A Domain, Interiority and Limit Cases

This appendix supports the footnote on the domain condition in Section 2. Lemma 15 shows that the model is well defined along every sequence, and it gives two cases in which every local equilibrium is interior. Remarks 4 and 5 then state the logarithmic and CARA cases, which are limits of Equation (1) and not members of it.

I state the results for a single non-numeraire good and suppress the subscript l. With L goods, they apply good by good. For agent i and holding x, write

$$B^i(x) = \frac{x}{1-\gamma} + b^i$$

for the bracket in Equation (1). Utility is well defined where $B^i(x) \geq 0$, and $B^i(x) = z^i/(1-\gamma)$ for the shifted holding $z^i = x + (1-\gamma)b^i$.

Lemma 15 (Well-Defined and Interior Local Equilibria)**.** *Suppose that $\omega^i \geq 0$ and $B^i(\omega^i) > 0$ for every agent i. Then, for every sequence of cliques and at every date t:*

(i) $B^i\Big(x^i(t)\Big) > 0$ for every agent i;

(ii) the local price $p\Big(M_{(t)}\Big)$ is positive and finite.

Moreover, $x^i(t) \geq 0$ for every agent i and every date t in each of the following cases:

(a) $\gamma < 1$ and $b^i \leq 0$ for every i;

(b) $b^i = \beta\,\alpha^i$ for every i, for some common β.

Proof. I argue by induction on t. At $t = 0$ the claims hold by assumption. Suppose they hold at date $t-1$, and let $M = M_{(t)}$. Agents outside M keep their holdings, so the claims hold for them at date t. For $i \in M$, Lemma 1 gives

$$x^i(t) = \frac{\alpha^i}{\alpha(M)} \sum_{j \in M} \Big(x^j(t-1) + (1-\gamma)b^j\Big) - (1-\gamma)b^i.$$

I divide by $1-\gamma$ and add b^i. This gives

$$B^i\Big(x^i(t)\Big) = \frac{\alpha^i}{\alpha(M)} \sum_{j \in M} B^j\Big(x^j(t-1)\Big), \tag{23}$$

which is positive because every term of the sum is positive. This proves (i). Similarly, Lemma 1 gives $p(M) = \Big(\sum_{j \in M} B^j(x^j(t-1))/\alpha(M)\Big)^{\gamma-1}$. The base is positive and finite, and therefore so is $p(M)$. This proves (ii).

For interiority, note that $x^i(t) = (1-\gamma)B^i\Big(x^i(t)\Big) - (1-\gamma)b^i$. In case (a), the first term is positive by (i), and the second term is nonnegative because $\gamma < 1$ and $b^i \leq 0$. In case (b), the terms in β cancel in the expression for $x^i(t)$, since $\sum_{j \in M} \beta\alpha^j = \beta\,\alpha(M)$. Hence $x^i(t) = \frac{\alpha^i}{\alpha(M)} \sum_{j \in M} x^j(t-1)$, which is nonnegative by the induction hypothesis. □

The positive bracket in (i) does not imply $x^i(t) \geq 0$ outside cases (a) and (b). For example, let $\gamma = 0$, $a = (1, 1)$, $b = (10, 0)$ and $\omega = (1, 1)$. Then $B^i(\omega^i) > 0$ for both agents, but Lemma 1 gives $x(M) = (-4, 6)$. Agent A values the good little, since $v^{A\prime}(0) = 1/10$, and efficiency would require it to sell more than it holds. In general, with $b^i > 0$, the condition $x^i(M) \geq 0$ holds if and only if $p(M) \leq a^i (b^i)^{\gamma-1}$, which is the marginal utility of agent i at zero. Outside cases (a) and (b), I therefore assume this condition at every meeting.

Remark 4 (Logarithmic Case). Equation (1) carries the factor $1/\gamma$, so the logarithmic case is its limit as $\gamma \to 0$ and not a member of the family. As $\gamma \to 0$,

$$a^i \frac{1-\gamma}{\gamma} \Big(B^i(x) \Big)^\gamma = a^i \frac{1-\gamma}{\gamma} + a^i (1-\gamma) \log B^i(x) + O(\gamma).$$

The first term is a constant and does not affect choices, because utility is quasilinear. Hence the limit utility is $u^i(x^i, y^i) = a^i \log\Big(x^i + b^i\Big) + y^i$, and the weights are $\alpha^i = a^i$. The local equilibrium of Lemma 1 becomes

$$x^i(M) = \frac{a^i}{a(M)} \sum_{j \in M} \Big(\omega^j + b^j \Big) - b^i, \qquad p(M) = \frac{a(M)}{\sum_{j \in M} \Big(\omega^j + b^j \Big)},$$

where $a(M) = \sum_{j \in M} a^j$. With $b^i = 0$, these reduce to $u^i = a^i \log x^i + y^i$, $x^i(M) = \frac{a^i}{a(M)} \sum_{j \in M} \omega^j$ and $p(M) = a(M) / \sum_{j \in M} \omega^j$. This case satisfies case (a) of Lemma 15, so every local equilibrium is interior.

Remark 5 (CARA Case). The CARA case is the limit $\gamma \to -\infty$, and it needs a normalisation of a^i. Fix $b^i > 0$ and $\kappa^i > 0$, and set $a^i = \kappa^i (b^i)^{1-\gamma}$. Then

$$v^{i\prime}(x) = a^i \Big(B^i(x) \Big)^{\gamma-1} = \kappa^i \bigg(1 + \frac{x}{b^i(1-\gamma)} \bigg)^{\gamma-1} \;\longrightarrow\; \kappa^i e^{-x/b^i}.$$

Hence the limit utility is $u^i(x^i, y^i) = -\kappa^i b^i e^{-x^i/b^i} + y^i$, with constant absolute risk aversion $1/b^i$. Without the normalisation, the factor $(b^i)^\gamma$ in Equation (1) tends to zero or to infinity, and the limit does not exist. The weights are $\alpha^i = (\kappa^i)^{1/(1-\gamma)}\, b^i \to b^i$. Write $R(M) = \sum_{j \in M} b^j$. The first-order conditions $\kappa^i e^{-x^i/b^i} = p$ and market clearing give

$$x^i(M) = \frac{b^i}{R(M)} \sum_{j \in M} \omega^j + b^i \bigg(\log \kappa^i - \frac{1}{R(M)} \sum_{j \in M} b^j \log \kappa^j \bigg),$$
$$p(M) = \exp\bigg(\frac{\sum_{j \in M} b^j \log \kappa^j - \sum_{j \in M} \omega^j}{R(M)} \bigg).$$

Therefore Definition 1 applies with $\alpha^i = b^i$ and with the offset $(1-\gamma)b^i$ replaced by $-b^i \log \kappa^i$. CARA utility is defined for every real x, so the domain condition holds automatically. However, CARA has no Inada condition at zero, and $x^i(M) \geq 0$ holds if and only if $p(M) \leq \kappa^i$. This condition holds at every meeting when κ^i is common to all agents, because then $x^i(M) = \frac{b^i}{R(M)} \sum_{j \in M} \omega^j$.

B Efficiency, Prices and the Core

Each meeting of a clique M determines two objects. The first is the allocation of the non-numeraire goods among the members of M. The second is the transfer of the numeraire among them. In this appendix I show that efficiency alone determines the first object. Therefore the Walrasian assumption matters only for the second. I then give three reasons to settle the numeraire at the Walrasian price. First, $p(M)$ is the only price that clears the meeting. Second, the Walrasian payoff lies in the core of the meeting. Third, the Walrasian rule adds no primitive to the model. Example 14 shows that two common alternatives, Nash bargaining and the Shapley value, fail the second property.

I state the argument for a single non-numeraire good and suppress the subscript l. With L goods, the same argument applies good by good, and gains add across goods. Write

$$v^i(x) = a^i \frac{1-\gamma}{\gamma} \left(\frac{x}{1-\gamma} + b^i \right)^{\gamma}$$

for the part of Equation (1) that depends on the good, so that $u^i(x^i, y^i) = v^i(x^i) + y^i$. Under the domain condition, v^i is strictly concave, with $v^{i\prime}(x) = a^i \left(\frac{x}{1-\gamma} + b^i \right)^{\gamma-1}$.

Efficiency determines the goods. Let $\boldsymbol{\omega}$ be the holdings of the members of M when they meet. The numeraire enters every utility linearly and is unbounded below, so utility is transferable within M. Hence an allocation of M is Pareto efficient if and only if its goods allocation solves

$$\max \left\{ \sum_{i \in M} v^i(x^i) \ : \ \sum_{i \in M} x^i = \sum_{i \in M} \omega^i \right\}. \tag{24}$$

Each v^i is strictly concave, so this problem has a unique solution. The solution is interior by the maintained assumption of Section 2 (see Lemma 15). Its first-order conditions equalise $v^{i\prime}(x^i)$ across the members of M. These are the equilibrium conditions of Lemma 1, and the common value is $p(M)$. Therefore the unique solution is $\boldsymbol{x}(M)$.

Consequently, every efficient settlement rule gives the goods allocation $\boldsymbol{x}(M)$ and the exchange operator $A(M)$. The rule can be Walrasian clearing, Nash bargaining, the Shapley value, or a broker who posts terms. Moreover, demand for the good does not depend on the numeraire, because preferences are quasilinear. Therefore the numeraire transfers of one meeting do not affect the goods allocation of the next. The convergence results of Section 2 hold for every efficient rule. The settlement rule matters only for the numeraire, and hence only for Section 3.

The settlement rule determines the numeraire. Given $\boldsymbol{x}(M)$, a settlement rule chooses transfers $(\tau^i)_{i \in M}$ with $\sum_{i \in M} \tau^i = 0$, and member i leaves the meeting

with $y^i = \omega^i_y + \tau^i$. Member i then gains

$$g^i = v^i\Big(x^i(M)\Big) - v^i(\omega^i) + \tau^i.$$

The Walrasian rule sets $\tau^i = p(M)\Big(\omega^i - x^i(M)\Big)$, as in Lemma 1. I write g^i_W for the resulting gain.

First reason: a unique clearing price. At a price q, member i demands the quantity $x^i(q)$ that solves $v^{i\prime}(x) = q$. This demand is strictly decreasing in q, because v^i is strictly concave. Hence the aggregate excess demand of M is strictly decreasing in q, and it vanishes only at $q = p(M)$. Therefore, if every member of M trades at one common price and buys what it demands at that price, the price is $p(M)$ and the transfers are the Walrasian ones. A rule that combines the efficient allocation with a common price $q \neq p(M)$ forces some members to trade quantities they would not choose at q.

Second reason: the core. Following Shapley and Shubik (1969), the meeting of M defines a market game. Its characteristic function is

$$V(S) = \max\Big\{\sum_{i\in S} v^i(x^i) \;:\; \sum_{i\in S} x^i = \sum_{i\in S}\omega^i\Big\} - \sum_{i\in S} v^i(\omega^i), \qquad S \subseteq M.$$

Every subset of a clique is itself a clique, so every $S \subseteq M$ with two or more members can meet on its own. A vector of gains $(g^i)_{i\in M}$ is in the core if $\sum_{i\in M} g^i = V(M)$ and $\sum_{i\in S} g^i \geq V(S)$ for every $S \subseteq M$. Shapley and Shubik (1969) show that every market game has a nonempty core and that competitive payoffs lie in it. Lemma 16 states this result for the present model and gives a direct proof.

Lemma 16 (Walrasian Payoff and the Core)**.** *For every clique M and all holdings $\boldsymbol{\omega}$, the Walrasian gains $(g^i_W)_{i\in M}$ lie in the core of the market game of M.*

Proof. Write $p = p(M)$. For each $i \in M$, the function $z \mapsto v^i(z) - p\,z$ is strictly concave, and its derivative vanishes at $z = x^i(M)$. Hence $v^i\Big(x^i(M)\Big) - p\,x^i(M) \geq v^i(z) - p\,z$ for every feasible z. Adding $p\,\omega^i - v^i(\omega^i)$ to both sides gives

$$g^i_W \;\geq\; v^i(z) - v^i(\omega^i) + p\,\Big(\omega^i - z\Big).$$

Fix $S \subseteq M$ and let $(z^i)_{i\in S}$ solve the maximisation that defines $V(S)$. I sum the inequality over $i \in S$. The price terms cancel because $\sum_{i\in S} z^i = \sum_{i\in S}\omega^i$. Therefore $\sum_{i\in S} g^i_W \geq V(S)$. For $S = M$ the solution is $\boldsymbol{x}(M)$, and equality holds because the Walrasian transfers sum to zero. □

Other efficient rules need not have this property. In Example 14, Nash bargaining and the Shapley value both give the pair $\{A, C\}$ less than the pair obtains when it meets alone. Under such a rule, two members of the clique prefer to meet without the third.

Third reason: no additional primitive. Nash bargaining requires a disagreement point and bargaining weights. A broker requires a rule for posting terms. Each of these adds a primitive to the model, and the redistribution of the numeraire then depends on it. Moreover, neither Nash bargaining nor the Shapley value corresponds to a single price for the meeting. Under the Walrasian rule, the transfers of a meeting depend only on the preferences and the holdings of its members. Therefore the final numeraire allocation $\boldsymbol{y}(\mathcal{C})$ is a function of the preferences, the endowments, the network G and the sequence $\mathcal{C}$ alone. Section 3 uses this property through Lemma 4.

Example 14. Consider three agents A, B and C who are all linked, with $u^i(x, y) = a^i \log x^i + y^i$, $(a^A, a^B, a^C) = (7, 5, 2)$ and $\omega_x^i = 30$ for every i. Let $M = \{A, B, C\}$ meet. The efficient allocation is $\boldsymbol{x}(M) = (45,\ 32.14,\ 12.86)$ and the price is $p(M) = 7/45 \approx 0.1556$. The pairs obtain $V(\{A, B\}) = 0.167$, $V(\{A, C\}) = 1.471$ and $V(\{B, C\}) = 0.664$, and the clique obtains $V(M) = 1.489$. Table 7 compares three efficient rules. Nash bargaining uses equal weights and no trade as the disagreement point. With transferable utility, it therefore splits $V(M)$ equally.

Rule	g^A	g^B	g^C	$g^A + g^C$
Walrasian	0.505	0.012	0.972	$1.477 \geq 1.471$
Nash bargaining	0.496	0.496	0.496	$0.992 < 1.471$
Shapley value	0.548	0.144	0.796	$1.344 < 1.471$

Table 7: Gains from the meeting of $M = \{A, B, C\}$ under three efficient rules. The last column compares the gains of $\{A, C\}$ with $V(\{A, C\}) = 1.471$.

All three rules give the same goods allocation $\boldsymbol{x}(M)$ and differ only in the numeraire. Only the Walrasian gains satisfy $g^A + g^C \geq V(\{A, C\})$. Moreover, no common price reproduces the Nash gains. The price that gives member i its Nash gain is $\Big(V(M)/3 - v^i(x^i(M)) + v^i(\omega^i)\Big)/\Big(\omega^i - x^i(M)\Big)$. This price equals 0.1561 for A, -0.0706 for B and 0.1278 for C. ◇

C Properties of the Exchange Operator

Lemma 17 (Properties of the Exchange Operator)**.** *Let M be a clique of the network G and let $\mathcal{A}(\cdot\,; M, l)$ be the associated exchange operator with linear part $A(M; l)$ and affine offset $\mathbf{d}_l = (1 - \gamma_l)\mathbf{b}_l$. Then:*

(i) ***(Idempotency)*** *$\mathcal{A}(\cdot\,; M, l) \circ \mathcal{A}(\cdot\,; M, l) = \mathcal{A}(\cdot\,; M, l)$. That is, applying the operator twice yields the same result as applying it once.*

(ii) ***(Row-stochasticity)*** *The matrix $A(M;l)$ is row-stochastic: $A(M;l)\mathbf{1} = \mathbf{1}$.*

(iii) ***(Spectral structure)*** *$A(M;l)$ has spectral radius $\rho(A(M;l)) = 1$. Write $V_M(l)$ for its 1-eigenspace, the set of row vectors $\boldsymbol{v}$ with $\boldsymbol{v}A(M;l) = \boldsymbol{v}$. A basis of $V_M(l)$ is $\{e_i : i \notin M\} \cup \left\{\sum_{i\in M} \alpha_l^i\, e_i\right\}$, where $\{e_i\}_{i\in N}$ is the standard basis of $\mathbb{R}^n$.*

Proof of Lemma 17. Let M be a clique of the Network G and $\mathcal{A}(\cdot\,; M, l)$ the associated exchange operator. Let $A = A(M;l)$ and $\mathbf{d} = \mathbf{d}_l = (1-\gamma_l)\mathbf{b}_l$. I will prove every part of the Lemma independently.

(i) **(Idempotency)**. We need to show $\mathcal{A}(\mathcal{A}(\boldsymbol{\omega}_l; M); M) = \mathcal{A}(\boldsymbol{\omega}_l; M)$ for all $\boldsymbol{\omega}_l$. Let $\boldsymbol{\omega}_l' = \mathcal{A}(\boldsymbol{\omega}_l; M) = \boldsymbol{\omega}_l A + \mathbf{d}(A - I)$. Then:

$$\begin{aligned}\mathcal{A}(\boldsymbol{\omega}_l'; M) &= \boldsymbol{\omega}_l' A + \mathbf{d}(A - I)\\ &= \Big[\boldsymbol{\omega}_l A + \mathbf{d}(A - I)\Big]A + \mathbf{d}(A - I)\\ &= \boldsymbol{\omega}_l A^2 + \mathbf{d}(A - I)A + \mathbf{d}(A - I).\end{aligned}$$

For agents $i \notin M$: $A_{ii} = 1$ and all other entries in row i are zero, so $A^2 = A$ on these rows trivially. For agents $i \in M$: the i-th row of A has entries $\alpha_l^j/\alpha(M;l)$ for $j \in M$ and 0 otherwise. Since all rows within the M-block are identical up to scaling, $A^2 = A$ on the M-block. Therefore $A^2 = A$, and:

$$\begin{aligned}\mathcal{A}(\boldsymbol{\omega}_l'; M) &= \boldsymbol{\omega}_l A + \mathbf{d}(A - I)A + \mathbf{d}(A - I)\\ &= \boldsymbol{\omega}_l A + \mathbf{d}\Big[(A - I)A + (A - I)\Big]\\ &= \boldsymbol{\omega}_l A + \mathbf{d}(A - I)\Big[A + I\Big].\end{aligned}$$

Since $A^2 = A$, we have $(A - I)A = A^2 - A = 0$, so $(A - I)(A + I) = (A - I)A + (A - I) = A - I$. Therefore:

$$\mathcal{A}(\boldsymbol{\omega}_l'; M) = \boldsymbol{\omega}_l A + \mathbf{d}(A - I) = \mathcal{A}(\boldsymbol{\omega}_l; M).$$

(ii) **(Row-stochasticity)** For $i \notin M$: $(A\mathbf{1})_i = 1$. For $i \in M$: $(A\mathbf{1})_i = \sum_{j\in M} \alpha_l^j/\alpha(M;l) = 1$. Therefore $A\mathbf{1} = \mathbf{1}$.

(iii) **(Spectral structure)** Since A is row-stochastic, $\rho(A) \le 1$. Since $A\mathbf{1} = \mathbf{1}$, the value 1 is an eigenvalue, so $\rho(A) = 1$. To find the 1-eigenspace, we seek all $\mathbf{v}$ with $\mathbf{v}A = \mathbf{v}$ (left eigenvectors, since dynamics are $\boldsymbol{\omega} \mapsto \boldsymbol{\omega}A$). Rewrite the M-block of A as a block-diagonal matrix $\left(\begin{smallmatrix} B & 0 \\ 0 & I_{n-|M|}\end{smallmatrix}\right)$ by reordering agents. The identity block contributes $\{e_i : i \notin M\}$ to the 1-eigenspace. For the B-block, $B_{ij} = \alpha_l^j/\alpha(M;l)$ for $i, j \in M$. The condition $\mathbf{v}B = \mathbf{v}$ yields $v_i = c \cdot \alpha_l^i$ for a constant c, giving the eigenvector $\sum_{i\in M} \alpha_l^i\, e_i$.

□

Lemma 18 (Self-Adjointness)**.** *Each exchange operator satisfies detailed balance with respect to* $\boldsymbol{\pi}$*,*

$$\pi^i A(M)_{ij} = \pi^j A(M)_{ji} \qquad \text{for all } i, j \in N,$$

and is self-adjoint for $\langle \cdot, \cdot \rangle_\pi$ *acting on the right:* $\langle \boldsymbol{u}A(M), \boldsymbol{v} \rangle_\pi = \langle \boldsymbol{u}, \boldsymbol{v}A(M) \rangle_\pi$ *for all* $\boldsymbol{u}, \boldsymbol{v}$*. Together with idempotence,* $A(M)$ *is therefore the orthogonal projection, in* $\langle \cdot, \cdot \rangle_\pi$*, onto its range* V_M *(Lemma 17(iii)), and in particular* $\|\boldsymbol{u}A(M)\|_\pi \le \|\boldsymbol{u}\|_\pi$ *with equality if and only if* $\boldsymbol{u} \in V_M$*.*

Proof of Lemma 18. Detailed balance is checked by cases. For $i, j \in M$, $\pi^i A(M)_{ij} = \frac{\alpha^i}{\alpha(N)} \cdot \frac{\alpha^j}{\alpha(M)}$, which is symmetric in i and j. For $i \in M$ and $j \notin M$ both sides vanish, since row i is supported on M and row j is a row of the identity; the case $i \notin M$, $j \in M$ is symmetric. For $i = j$ the identity is trivial, and for $i \neq j$ with both outside M both sides are zero.

Dividing $\pi^i A(M)_{ij} = \pi^j A(M)_{ji}$ by $\pi^i \pi^j$ gives $A(M)_{ij}/\pi^j = A(M)_{ji}/\pi^i$, and hence

$$\langle \boldsymbol{u}A(M), \boldsymbol{v} \rangle_\pi = \sum_{i,j} \frac{u^i A(M)_{ij} v^j}{\pi^j} = \sum_{i,j} \frac{u^i A(M)_{ji} v^j}{\pi^i} = \langle \boldsymbol{u}, \boldsymbol{v}A(M) \rangle_\pi.$$

By Lemma 17 the operator is idempotent, and a self-adjoint idempotent is an orthogonal projection. Orthogonal projections are non-expansive, with $\|\boldsymbol{u}\|_\pi^2 = \|\boldsymbol{u}A(M)\|_\pi^2 + \|\boldsymbol{u} - \boldsymbol{u}A(M)\|_\pi^2$, so equality holds exactly when $\boldsymbol{u} = \boldsymbol{u}A(M)$. □

Lemma 19 (Properties of the Trading Sequence Operator)**.** *Let* $\mathcal{C} = (M_k)_{k \in \{1,\dots,K\}}$ *be a non-empty sequence of cliques of the network* G*. Let* $\mathcal{A}(\cdot\,; \mathcal{C}; l)$ *denote the composed operator* $\mathcal{A}(\cdot\,; M_K; l) \circ \cdots \circ \mathcal{A}(\cdot\,; M_1; l)$ *with linear part* $A(\mathcal{C}; l) = A(M_1; l) \cdots A(M_K; l)$*. Let* $\mathcal{D}$ *be the set of maximal connected components of the graph on* $\cup_k M_k$ *in which two agents are linked if they belong to a common clique of* $\mathcal{C}$*. Then:*

(i) ***(Affine structure)*** *The trading round operator is an affine map of the same form as the exchange operator:*

$$\mathcal{A}(\boldsymbol{\omega}_l; \mathcal{C}) = \boldsymbol{\omega}_l\, A(\mathcal{C}; l) + \mathbf{d}_l \Big(A(\mathcal{C}; l) - I \Big). \tag{25}$$

(ii) ***(Row-stochasticity)*** $A(\mathcal{C}; l)$ *is row-stochastic.*

(iii) ***(Spectral structure)*** $A(\mathcal{C}; l)$ *has spectral radius* $\rho(A(\mathcal{C}; l)) = 1$*, its* 1*-eigenspace is* $\bigcap_k V_{M_k}(l)$*, and a basis of this eigenspace is* $\{e_i : i \notin \cup_k M_k\} \cup \Big\{ \sum_{i \in D} \alpha_l^i\, e_i : D \in \mathcal{D} \Big\}$*.*

Proof of Lemma 19. Let $\mathcal{C} = (M_k)_{k\in\{1,\dots,K\}}$ be a non-empty sequence of cliques of the network G. Let $\mathcal{A}(\cdot\,;\mathcal{C};l)$ denote the composed operator $\mathcal{A}(\cdot\,;M_K;l)\circ\cdots\circ\mathcal{A}(\cdot\,;M_1;l)$ with linear part $A(\mathcal{C};l) = A(M_1;l)\cdots A(M_K;l)$. We prove each part separately.

(i) **(Affine structure)** The proof is by induction on K.

- *Base case ($K = 1$):* A single clique M_1. By Definition 1, $\mathcal{A}(\boldsymbol{\omega}_l; M_1) = \boldsymbol{\omega}_l A(M_1;l) + \mathbf{d}_l(A(M_1;l) - I)$, which has the same functional form as Equation (25) with $A(\mathcal{C};l) = A(M_1;l)$.
- *Inductive step:* Suppose the result holds for a sequence of length K, with composed matrix $A_K = A(M_1;l)\cdots A(M_K;l)$ and operator $\mathcal{A}_K(\boldsymbol{\omega}_l) = \boldsymbol{\omega}_l A_K + \mathbf{d}_l(A_K - I)$. Appending clique M_{K+1}:

$$\begin{aligned}
\mathcal{A}_{K+1}(\boldsymbol{\omega}_l) &= \mathcal{A}\Big(\mathcal{A}_K(\boldsymbol{\omega}_l); M_{K+1}\Big) \\
&= \Big[\boldsymbol{\omega}_l A_K + \mathbf{d}_l(A_K - I)\Big]A(M_{K+1};l) + \mathbf{d}_l\Big(A(M_{K+1};l) - I\Big) \\
&= \boldsymbol{\omega}_l A_K A(M_{K+1};l) + \mathbf{d}_l(A_K - I)A(M_{K+1};l) + \mathbf{d}_l(A(M_{K+1};l) - I) \\
&= \boldsymbol{\omega}_l A_{K+1} + \mathbf{d}_l\Big[A_K A(M_{K+1};l) - A(M_{K+1};l) + A(M_{K+1};l) - I\Big] \\
&= \boldsymbol{\omega}_l A_{K+1} + \mathbf{d}_l(A_{K+1} - I),
\end{aligned}$$

  where $A_{K+1} = A_K A(M_{K+1};l) = A(M_1;l)\cdots A(M_{K+1};l)$.

(ii) **(Row-stochasticity)** The product of stochastic matrices is stochastic:

$$A(\mathcal{C};l)\mathbf{1} = A(M_1;l)\cdots A(M_K;l)\mathbf{1} = \mathbf{1}$$

(iii) **(Spectral structure)** By part (ii), $A(\mathcal{C};l)$ is row-stochastic, so $\rho(A(\mathcal{C};l)) \le 1$. Since $A(\mathcal{C};l)\mathbf{1} = \mathbf{1}$, the value 1 is an eigenvalue. Therefore $\rho(A(\mathcal{C};l)) = 1$.

We first show that the 1-eigenspace of $A(\mathcal{C};l)$ is $\bigcap_k V_{M_k}(l)$, that is, a vector is fixed by $A(\mathcal{C};l)$ if and only if it is fixed by every $A(M_k;l)$. A vector fixed by every $A(M_k;l)$ is clearly fixed by their product. Conversely, suppose $\boldsymbol{v}A(\mathcal{C};l) = \boldsymbol{v}$. Define $\boldsymbol{v}_0 = \boldsymbol{v}$ and $\boldsymbol{v}_k = \boldsymbol{v}_{k-1}A(M_k;l)$ for $k = 1,\dots,K$, so that $\boldsymbol{v}_K = \boldsymbol{v}$. By Lemma 18, each $A(M_k;l)$ is an orthogonal projection in $\langle\cdot,\cdot\rangle_\pi$. Hence

$$\|\boldsymbol{v}\|_\pi = \|\boldsymbol{v}_0\|_\pi \ge \|\boldsymbol{v}_1\|_\pi \ge \cdots \ge \|\boldsymbol{v}_K\|_\pi = \|\boldsymbol{v}\|_\pi.$$

The first and last terms are equal, so every inequality is an equality. Suppose $\boldsymbol{v}_{k-1} = \boldsymbol{v}$ for some k. Then $\|\boldsymbol{v}A(M_k;l)\|_\pi = \|\boldsymbol{v}\|_\pi$, and by Lemma 18 this equality holds only if $\boldsymbol{v}A(M_k;l) = \boldsymbol{v}$. Therefore $\boldsymbol{v}_k = \boldsymbol{v}$. Since $\boldsymbol{v}_0 = \boldsymbol{v}$,

induction on k gives $\boldsymbol{v}A(M_k;l) = \boldsymbol{v}$ for every k. We use this below, within each component.

We group the cliques by component. Each clique of $\mathcal{C}$ lies in exactly one component $D \in \mathcal{D}$, since its members are linked to each other. For each $D \in \mathcal{D}$, let $\mathcal{C}^D$ be the subsequence of cliques of $\mathcal{C}$ contained in D, in the same order. Two cliques in different components share no agent. Moreover, $A(M;l)$ differs from the identity only in the rows and columns of M. Therefore the matrices of two cliques in different components commute, and we can reorder the product $A(M_1;l)\cdots A(M_K;l)$ as $A(\mathcal{C};l) = \prod_{D\in\mathcal{D}} A(\mathcal{C}^D;l)$, keeping the order within each component. Each factor differs from the identity only in the rows and columns of its component, and the components are disjoint. Therefore, after reordering the agents, $A(\mathcal{C};l)$ is block diagonal. It has one block for each $D \in \mathcal{D}$ and an identity block on the agents outside $\cup_k M_k$. The identity block contributes the vectors e_i for $i \notin \cup_k M_k$. It remains to find, for each D, the vectors supported on D that are fixed by $A(\mathcal{C}^D;l)$.

Fix $D \in \mathcal{D}$. By the argument above, a vector $\boldsymbol{v}$ supported on D is fixed by $A(\mathcal{C}^D;l)$ if and only if it is fixed by every clique of $\mathcal{C}^D$. By Lemma 17(iii), each clique M of $\mathcal{C}^D$ requires $v_i = c_M\,\alpha_l^i$ for all $i \in M$, for some constant c_M. If two cliques M and M' share an agent j, then $c_M\,\alpha_l^j = v_j = c_{M'}\,\alpha_l^j$, and since $\alpha_l^j > 0$, we get $c_M = c_{M'}$. Since D is connected, any two cliques of $\mathcal{C}^D$ are joined by a chain of cliques in which consecutive cliques share an agent. Therefore all the constants are equal, and the fixed vectors supported on D are the multiples of $\sum_{i\in D} \alpha_l^i\, e_i$.

Therefore the 1-eigenspace of $A(\mathcal{C};l)$ is spanned by $\{e_i : i \notin \cup_k M_k\} \cup \left\{\sum_{i\in D} \alpha_l^i\, e_i : D \in \mathcal{D}\right\}$. These vectors are non-zero and have disjoint supports, so they are linearly independent and form a basis.

□

D Proofs for Section 2

Proof of Lemma 1. Let M be a clique of G. The utility maximisation problem that consumer i faces in the local economy given by M is:

$$\begin{aligned} \max_{x_l^i, y^i} \quad & \sum_{l=1}^{L} a_l^i \frac{1-\gamma_l}{\gamma_l}\left(\frac{x_l^i}{1-\gamma_l} + b_l^i\right)^{\gamma_l} + y^i \\ \text{s.t} \quad & \forall l \le L,\ \sum_{j\in M} x_l^j = \sum_{j\in M} \omega_l^j \\ & y^i + \sum_{l\le L} p_l x_l^i = \omega_y^i + \sum_{l\le L} p_l \omega_l^i \end{aligned}$$

To solve this optimisation problem, let me consider the following Lagrangian.

$$\mathcal{L} = \sum_{l=1}^{L} a_l^i \frac{1-\gamma_l}{\gamma_l} \left(\frac{x_l^i}{1-\gamma_l} + b_l^i \right)^{\gamma_l} + y^i - \lambda \left(y^i + \sum_{l \leq L} p_l x_l^i - \omega_y^i - \sum_{l \leq L} p_l \omega_l^i \right)$$

We obtain the following FOC:

$$\frac{\partial L}{\partial x_l^i} = a_l^i \left(\frac{x_l^i}{1-\gamma_l} + b_l^i \right)^{\gamma_l - 1} - \lambda p_l = 0, \ \forall l \leq L \tag{26}$$

$$\frac{\partial L}{\partial y^i} = 1 - \lambda = 0 \tag{27}$$

$$\frac{\partial L}{\partial \lambda} = y^i + \sum_{l \leq L} p_l x_l^i - \omega_y^i - \sum_{l \leq L} p_l \omega_l^i = 0 \tag{28}$$

Let me note that given the additive separable property of u^i, the price of each good is determined independently from the other goods (Equation (26)). Moreover, given the quasilinear property of u^i, the consumption of the numeraire good is determined by the budget constraint (Equation (28)). Therefore, we only need to focus on finding the optimal conditions of a generic good l.

We start by combining Equations (26) and (27) and then solving for x_l^i. This gives us the following demand function of agent i for good l:

$$x_l^i = (1-\gamma_l) \left(\frac{p_l}{a_l^i} \right)^{\frac{1}{\gamma_l - 1}} - (1-\gamma_l)\, b_l^i = (1-\gamma_l)\, \alpha_l^i\, p_l^{\frac{1}{\gamma_l - 1}} - (1-\gamma_l)\, b_l^i, \tag{29}$$

where the second equality uses $(a_l^i)^{-\frac{1}{\gamma_l - 1}} = (a_l^i)^{\frac{1}{1-\gamma_l}} = \alpha_l^i$. Each agent i has a similar equation so we combine them using the market clearing conditions to obtain the equilibrium prices $p(M; l)$.

$$\begin{aligned}
\sum_{j \in M} x_l^j &= \sum_{j \in M} \omega_l^j \\
(1-\gamma_l)\, p_l^{\frac{1}{\gamma_l - 1}} \sum_{j \in M} \alpha_l^j &= \sum_{j \in M} \omega_l^j + (1-\gamma_l) \sum_{j \in M} b_l^j \\
\Big(p(M;l) \Big)^{\frac{1}{\gamma_l - 1}} &= \frac{\sum\limits_{j \in M} \omega_l^j}{(1-\gamma_l)\, \alpha(M;l)} + \frac{\sum\limits_{j \in M} b_l^j}{\alpha(M;l)}
\end{aligned}$$

Raising both sides to the power $\gamma_l - 1$ gives the price in Lemma 1.

Once we have computed the equilibrium price, we can compute the equilibrium allocations of the consumption good l by replacing the equilibrium prices into the demand function of agent i (Equation (29)) to obtain the equilibrium allocation $x^i(M; l)$:

$$x^i(M;l) = \frac{\alpha_l^i}{\alpha(M;l)} \sum_{j \in M} \omega_l^j + (1-\gamma_l) \left(\frac{\alpha_l^i}{\alpha(M;l)} \sum_{j \in M} b_l^j - b_l^i \right)$$

Now, from the budget constraint we can compute the equilibrium allocations of the *numeraire* good

$$y^i(M) = \omega_y^i + \sum_{l \le L} p(M;l)\left(\omega_l^i - x^i(M;l)\right)$$

Finally, the argument M notes that these equilibrium allocations correspond to exchanges in the local economy given by clique M. □

Proof of Remark 1. Let M be a clique and $A(M;l)$ be the matrix representation of the operator $\mathcal{A}(\cdot\,; M, l)$. Note that for every consumer $j \notin M$ the column and the row of $A(M;l)$ coincide with those of the identity matrix $I_{n\times n}$. Therefore $\left(\boldsymbol{\omega}_l A(M;l)\right)_j = \omega_l^j$ and $\left(\mathbf{b}_l(A(M;l) - I)\right)_j = 0$, and the operator does not change the endowments of these consumers.

We can focus on the consumers in M. Fix the consumer $i \in M$. Since endowments are row vectors, the i-th coordinate of $\boldsymbol{\omega}_l A(M;l)$ is the product of $\boldsymbol{\omega}_l$ with the i-th *column* of $A(M;l)$: $\left(\boldsymbol{\omega}_l A(M;l)\right)_i = \sum_j \omega_l^j A(M;l)_{ji}$. This column has entry $\alpha_l^i/\alpha(M;l)$ in every row $j \in M$ and zero in the other rows. Therefore

$$\left(\boldsymbol{\omega}_l A(M;l)\right)_i = \frac{\alpha_l^i}{\alpha(M;l)} \sum_{j\in M} \omega_l^j,$$

$$\left((1-\gamma_l)\,\mathbf{b}_l(A(M;l) - I)\right)_i = (1-\gamma_l)\left(\frac{\alpha_l^i}{\alpha(M;l)} \sum_{j\in M} b_l^j - b_l^i\right).$$

The sum of the two terms is $x^i(M;l)$ in Lemma 1. Since the choice of i was generic, this applies to every consumer i in the clique M. □

Lemma 20 (Disequilibrium Subspace)**.** *Let* $V_0 = \{\boldsymbol{v} \in \mathbb{R}^n : \boldsymbol{v}\mathbf{1} = 0\}$. *Then*

(i) V_0 *is invariant under* $A(M)$ *for every clique* M*, and* $\mathbb{R}^n = V_0 \oplus \operatorname{span}(\boldsymbol{\pi})$ *is orthogonal for* $\langle\cdot,\cdot\rangle_\pi$*;*

(ii) $E(t)$ *vanishes on* $\operatorname{span}(\boldsymbol{\pi})$ *and agrees with* $A(t)$ *on* V_0*, so that*

$$\left\|E(t)\right\|_\pi = \sup\Big\{\|\boldsymbol{v}A(t)\|_\pi : \boldsymbol{v} \in V_0,\ \|\boldsymbol{v}\|_\pi = 1\Big\}. \tag{30}$$

Proof of Lemma 20. For *(i)*, row-stochasticity gives $\boldsymbol{v}A(M)\mathbf{1} = \boldsymbol{v}\mathbf{1}$, so V_0 is invariant. For $\boldsymbol{v} \in V_0$ we have $\langle \boldsymbol{\pi}, \boldsymbol{v}\rangle_\pi = \sum_i \pi^i v^i/\pi^i = \boldsymbol{v}\mathbf{1} = 0$, while $\boldsymbol{\pi}\mathbf{1} = 1$ places $\boldsymbol{\pi}$ outside V_0; as $\dim V_0 = n-1$ the sum is direct, and orthogonal by the displayed identity.

For *(ii)*, $\boldsymbol{\pi}A(t) = \boldsymbol{\pi}$ and $\boldsymbol{\pi}\Pi = \boldsymbol{\pi}$ give $\boldsymbol{\pi}E(t) = \mathbf{0}$, and for $\boldsymbol{v} \in V_0$, $\boldsymbol{v}\Pi = (\boldsymbol{v}\mathbf{1})\boldsymbol{\pi} = \mathbf{0}$, so $\boldsymbol{v}E(t) = \boldsymbol{v}A(t)$. Writing $\boldsymbol{w} = c\boldsymbol{\pi} + \boldsymbol{v}$ orthogonally, $\boldsymbol{w}E(t) = \boldsymbol{v}E(t)$ while $\|\boldsymbol{w}\|_\pi \ge \|\boldsymbol{v}\|_\pi$, so the supremum defining the operator norm is attained on V_0. □

Proof of Theorem 1. Let $\mathcal{R} = (M_k)_{k\in\{1,\dots,K\}}$ be a trading round of the network $G = (N, E)$. Fix a non-numeraire good l and suppress the index l. Write $A = A(\mathcal{R}; l)$, $\mathbf{d} = \mathbf{d}_l$, and $\mathcal{A} = \mathcal{A}(\cdot\,; \mathcal{R}; l)$.

(i) **(Regularity)** First, by Lemma 19(ii), A is row-stochastic with $\rho(A) = 1$. Second, every factor has a positive diagonal. For any agent $i \in N$ and any clique M_k: if $i \in M_k$, then $A(M_k; l)_{ii} = \alpha_l^i/\alpha(M_k; l) \in (0, 1)$; if $i \notin M_k$, then $A(M_k; l)_{ii} = 1$. In both cases $A(M_k; l)_{ii} > 0$. Third, we prove *irreducibility*. Let i and j be two agents that belong to a common clique M_k of the round. Since $A(M_k; l)_{ij} = \alpha_l^j/\alpha(M_k; l) > 0$,

$$A_{ij} \geq \Big(\prod_{k'<k} A(M_{k'}; l)_{ii}\Big)\, A(M_k; l)_{ij} \Big(\prod_{k'>k} A(M_{k'}; l)_{jj}\Big) > 0.$$

Hence $A_{ij} > 0$ and $A_{ji} > 0$ whenever i and j share a clique of the round. Since the cliques of a trading round connect N, the graph of A is strongly connected, and A is irreducible. Fourth, we prove that A is *primitive*. Since the diagonal of a product of non-negative matrices satisfies $A_{ii} \geq \prod_{k=1}^{K} A(M_k; l)_{ii} > 0$, the matrix A has a positive diagonal entry. An irreducible non-negative matrix with a positive diagonal entry is aperiodic, hence primitive.

(ii) **(Unique Stationary Distribution)** Since A is primitive and row-stochastic, the Perron–Frobenius theorem gives $A^t \to \Pi_l = \mathbf{1}\boldsymbol{\pi}_l$ as $t \to \infty$, where $\boldsymbol{\pi}_l$ is the unique stationary distribution. By Lemma 19(iii) applied to the trading round, whose cliques connect N so that $\mathcal{D} = \{N\}$, $\pi_l^i = \alpha_l^i/\alpha(N; l)$.

(iii) **(Convergence)** By Lemma 19(i), a single application of the trading round gives $\mathcal{A}(\boldsymbol{\omega}) = \boldsymbol{\omega}A + \mathbf{d}(A - I)$. Applying $\mathcal{A}$ again, using $\mathcal{A}(\boldsymbol{\omega})$ as the new endowment:

$$\mathcal{A}^2(\boldsymbol{\omega}) = \Big[\boldsymbol{\omega}A + \mathbf{d}(A - I)\Big]A + \mathbf{d}(A - I) = \boldsymbol{\omega}A^2 + \mathbf{d}(A^2 - I).$$

By induction, the allocation after t repetitions of the trading round is:

$$\mathcal{A}^t(\boldsymbol{\omega}) = \boldsymbol{\omega}\, A^t + \mathbf{d}(A^t - I).$$

By the property of unique stationary distribution we have that:

$$\lim_{t\to\infty} \mathcal{A}^t(\boldsymbol{\omega}) = \boldsymbol{\omega}\, \Pi_l + \mathbf{d}(\Pi_l - I).$$

This is the exchange operator $\mathcal{A}(\cdot\,; N; l)$ applied to $\boldsymbol{\omega}$, with $A(N; l) = \Pi_l$. By Lemma 1, this equals the competitive equilibrium allocation $\boldsymbol{x}(N; l)$ of the socially unconstrained economy.

(iv) **(Geometric Rate)** The error at iteration t is

$$\mathcal{A}^t(\boldsymbol{\omega}) - \boldsymbol{x}(N; l) = \boldsymbol{\omega}(A^t - \Pi_l) + \mathbf{d}(A^t - \Pi_l) = (\boldsymbol{\omega} + \mathbf{d})(A^t - \Pi_l),$$

so it suffices to bound $A^t - \Pi_l$. On the disequilibrium subspace V_0 of Lemma 20 the operator A acts with spectrum contained in the disc of radius $|\lambda_2(A)| < 1$, and $A^t - \Pi_l$ agrees with A^t there. Writing $A|_{V_0}$ in Jordan form, a block of size m_β attached to an eigenvalue λ_β contributes entries of order $t^{m_\beta - 1}|\lambda_\beta|^t$ to the t-th power, so

$$\left\| A^t - \Pi_l \right\| = O\Big(t^{\,m-1} \, |\lambda_2(A)|^t \Big),$$

with m the size of the largest Jordan block attached to an eigenvalue of modulus $|\lambda_2(A)|$. Primitivity alone would not give this: it delivers convergence and $|\lambda_2| < 1$, but the clean geometric form needs λ_2 semisimple, and A is a product of exchange matrices that do not commute.

For the bound on m, each $A(M_k; l)$ has rank $n - |M_k| + 1$, so the product has rank at most $n - \max_k |M_k| + 1$. The eigenvalue 1 is simple, so the eigenvalues of modulus $|\lambda_2|$ occupy at most $n - \max_k |M_k|$ dimensions and $m \le n - \max_k |M_k|$. In particular $m = 1$, and the bound is exactly geometric, whenever some clique of the round has at least $n - 1$ members.

□

Proof of Remark 2. This is shown in the proof of Theorem 2. □

Proof of Proposition 2. The proof relies on two observations. First, $\boldsymbol{\pi} A(M) = \boldsymbol{\pi}$ for every clique M: the global equilibrium distribution is a fixed point of every local exchange. This follows from the detailed balance condition (Lemma 18): $\sum_i \pi^i A(M)_{ij} = \pi^j$ for all j. Consequently, $\Pi \cdot A(M) = \Pi$, and the error operator satisfies $E(t+1) = A(t) \cdot A(M_{(t+1)}) - \Pi = (A(t) - \Pi) \cdot A(M_{(t+1)}) = E(t) \cdot A(M_{(t+1)})$.

Second, since $A(M_{(t+1)})$ is an orthogonal projection in $\langle \cdot, \cdot \rangle_\pi$, it is non-expansive: $\|A(M)\|_\pi \le 1$. Taking operator norms: $\|E(t+1)\|_\pi = \|E(t) \cdot A(M_{(t+1)})\|_\pi \le \|E(t)\|_\pi \cdot \|A(M_{(t+1)})\|_\pi \le \|E(t)\|_\pi$. □

Lemma 21 (Uniform Contraction)**.** *Let l be a non-numeraire good. We drop the index l from the notation. For a non-empty set $\mathcal{F}$ of cliques, let $V_\mathcal{F} = \bigcap_{M \in \mathcal{F}} V_M$, and write $d_\pi(\boldsymbol{x}, V) = \min_{\boldsymbol{y} \in V} \|\boldsymbol{x} - \boldsymbol{y}\|_\pi$. There is a constant $c_\mathcal{F} \ge 1$ such that, for every finite sequence $\mathcal{S}$ whose cliques are exactly those of $\mathcal{F}$ and every $\boldsymbol{x} \in \mathbb{R}^n$,*

$$d_\pi(\boldsymbol{x}, V_\mathcal{F})^2 \;\le\; c_\mathcal{F}^2 \Big(\|\boldsymbol{x}\|_\pi^2 - \|\boldsymbol{x} A(\mathcal{S})\|_\pi^2 \Big).$$

The constant does not depend on the length or the order of $\mathcal{S}$.

Proof of Lemma 21. The proof adapts an argument of Amemiya and Ando (1965, pp. 242–243) to the finite-dimensional case. By Lemma 18, $A(M)$ is the orthogonal projection onto V_M, so

$$d_\pi(\boldsymbol{y}, V_M)^2 = \|\boldsymbol{y}\|_\pi^2 - \|\boldsymbol{y}A(M)\|_\pi^2 \qquad \text{for every } \boldsymbol{y}. \tag{31}$$

Moreover, there is $\kappa_{\mathcal{F}} \geq 1$ with $d_\pi(\boldsymbol{x}, V_{\mathcal{F}}) \leq \kappa_{\mathcal{F}} \max_{M \in \mathcal{F}} d_\pi(\boldsymbol{x}, V_M)$ for every $\boldsymbol{x}$. Indeed, both sides depend only on the component of $\boldsymbol{x}$ orthogonal to $V_{\mathcal{F}}$ and are positively homogeneous. On the unit sphere of $V_{\mathcal{F}}^\perp$, the right-hand side is continuous. It is also positive. Indeed, a vector at distance zero from every V_M lies in $V_{\mathcal{F}}$, and the only such vector in $V_{\mathcal{F}}^\perp$ is zero. The sphere is compact, so the right-hand side attains a positive minimum there.

We argue by induction on the number of cliques in $\mathcal{F}$. Write $\Delta = \|\boldsymbol{x}\|_\pi^2 - \|\boldsymbol{x}A(\mathcal{S})\|_\pi^2$. By Lemma 18, each step of $\mathcal{S}$ decreases the squared norm by a non-negative amount, and these amounts add up to Δ. If $\mathcal{F} = \{M\}$, then $A(\mathcal{S}) = A(M)$ by Lemma 17(i), and Equation (31) gives the claim with $c_{\mathcal{F}} = 1$. Otherwise, let M^* be the clique of $\mathcal{F}$ whose first appearance in $\mathcal{S}$ comes last, and let $\mathcal{U}$ be the initial segment of $\mathcal{S}$ before it. The cliques of $\mathcal{U}$ are exactly those of $\mathcal{F}' = \mathcal{F} \setminus \{M^*\}$. By the induction hypothesis, $\boldsymbol{x} = \boldsymbol{a} + \boldsymbol{r}$ with $\boldsymbol{a} \in V_{\mathcal{F}'}$ and $\|\boldsymbol{r}\|_\pi \leq c_{\mathcal{F}'}\sqrt{\Delta}$. Every clique of $\mathcal{U}$ fixes $\boldsymbol{a}$, so $\|\boldsymbol{x} - \boldsymbol{x}A(\mathcal{U})\|_\pi \leq 2\|\boldsymbol{r}\|_\pi$. The next step applies M^*, so Equation (31) gives $d_\pi(\boldsymbol{x}A(\mathcal{U}), V_{M^*}) \leq \sqrt{\Delta}$. Hence $d_\pi(\boldsymbol{x}, V_M) \leq (1 + 2c_{\mathcal{F}'})\sqrt{\Delta}$ for every $M \in \mathcal{F}$. The inequality with $\kappa_{\mathcal{F}}$ gives the claim with $c_{\mathcal{F}} = \kappa_{\mathcal{F}}\Big(1 + 2\max_{\mathcal{F}'} c_{\mathcal{F}'}\Big)$, where the maximum runs over the finitely many sets with one clique fewer than $\mathcal{F}$. □

Proof of Theorem 2. The argument is the same for every non-numeraire good. We fix one and drop the index l. Let Ω_∞ be the set of cliques that appear infinitely often in $\mathcal{C}$. There are finitely many cliques, so after some date T_0 only cliques of Ω_∞ appear. By hypothesis, the cliques of Ω_∞ cover N and connect it. We can therefore choose dates $T_0 = t_0 < t_1 < \cdots$ such that each block $\mathcal{S}_k = (M_{(t_{k-1}+1)}, \ldots, M_{(t_k)})$ contains every clique of Ω_∞ and no other.

The cliques of $\mathcal{S}_1$ are those of Ω_∞, and they connect N. Therefore Lemma 19(iii), applied to $\mathcal{S}_1$ with $\mathcal{D} = \{N\}$, gives $V_{\Omega_\infty} = \operatorname{span}(\boldsymbol{\pi})$, whose orthogonal complement is V_0 by Lemma 20(i). Hence $d_\pi(\boldsymbol{x}, V_{\Omega_\infty}) = \|\boldsymbol{x}\|_\pi$ for $\boldsymbol{x} \in V_0$, and Lemma 21 gives

$$\|\boldsymbol{x}A(\mathcal{S}_k)\|_\pi \;\leq\; q\,\|\boldsymbol{x}\|_\pi \quad \text{for every } k \text{ and } \boldsymbol{x} \in V_0, \qquad q = \Big(1 - c_{\Omega_\infty}^{-2}\Big)^{1/2} < 1.$$

Let $\boldsymbol{v} \in V_0$ with $\|\boldsymbol{v}\|_\pi = 1$. By Lemma 20(i), $\boldsymbol{v}A(t_{k-1}) \in V_0$, so $\|\boldsymbol{v}A(t_k)\|_\pi \leq q\,\|\boldsymbol{v}A(t_{k-1})\|_\pi$. Taking the supremum over $\boldsymbol{v}$ and using Equation (30), $\|E(t_k)\|_\pi \leq q\,\|E(t_{k-1})\|_\pi$. Since $\|E(0)\|_\pi = 1$, Proposition 2 gives $\|E(t)\|_\pi \leq q^k$ for every $t \geq t_k$. Therefore $A(t) \to \Pi$.

Finally, by Lemma 19(i), the allocation at date t is $\boldsymbol{\omega}A(t) + \mathbf{d}\Big(A(t) - I\Big)$, which converges to $\boldsymbol{\omega}\Pi + \mathbf{d}(\Pi - I)$. Since $\Pi = A(N)$, this limit is the exchange operator at N applied to $\boldsymbol{\omega}$, which is $\boldsymbol{x}(N)$ by Lemma 1. □

E Proofs for Section 3

Proof of Lemma 3. By Definition 1, $\boldsymbol{x}(t) = \boldsymbol{x}(t-1)A + \mathbf{d}(A-I)$ with $A = A(M_{(t)})$, so $\boldsymbol{x}(t) + \mathbf{d} = \big(\boldsymbol{x}(t-1) + \mathbf{d}\big)A$, which is the shifted statement; Z is conserved because A is row-stochastic. By Equation (4), $\boldsymbol{x}(N) + \mathbf{d} = \boldsymbol{z}(0)\Pi = \boldsymbol{\pi}Z$, and $\boldsymbol{\pi}Z\,A = \boldsymbol{\pi}Z$ since $\boldsymbol{\pi}A = \boldsymbol{\pi}$ by Theorem 1. Subtracting, $\boldsymbol{e}(t) = \boldsymbol{z}(t) - \boldsymbol{\pi}Z = \big(\boldsymbol{z}(t-1) - \boldsymbol{\pi}Z\big)A = \boldsymbol{e}(t-1)A$. □

Proof of Lemma 4. By Lemma 1 and the definition of $\boldsymbol{z}$,

$$p(M) = \Big[\frac{\sum_{j\in M}\omega^j}{(1-\gamma)\alpha(M)} + \frac{\sum_{j\in M} b^j}{\alpha(M)}\Big]^{\gamma-1} = \Big[\frac{Z_M}{(1-\gamma)\,\alpha(M)}\Big]^{\gamma-1}, \qquad Z_M := \sum_{j\in M} z^j.$$

Applying this at M and at N and taking the ratio,

$$p(M)/p(N) = \Big[Z_M\alpha(N)/(Z\alpha(M))\Big]^{\gamma-1} = \Big[Z_M/(\pi(M)Z)\Big]^{\gamma-1}.$$

By Lemma 3, $Z_M = \pi(M)Z + \boldsymbol{e}(t)\mathbf{1}_M$, which gives Equation (7); interiority of the local equilibrium gives $Z_M/(1-\gamma) > 0$, so the bracket, a ratio of two quantities of equal sign, is positive. Finally $\boldsymbol{e}(t)\mathbf{1}_N = \sum_i e^i(t) = 0$. □

Proof of Proposition 5. Write $\epsilon_t := \boldsymbol{e}(t-1)\mathbf{1}_{M_{(t)}}\big/\big(\pi(M_{(t)})Z\big)$ for the relative disequilibrium of the clique active at t, so that by Lemma 4 the t-th summand of Equation (8) is

$$p(N)\Big[1-(1+\epsilon_t)^{\gamma-1}\Big]\,\boldsymbol{e}(t-1)\Big(A(M_{(t)}) - I\Big).$$

By Lemma 3, $\boldsymbol{e}(t-1) = \boldsymbol{e}(0)\prod_{s<t}A(M_{(s)})$, so both ϵ_t and the quantity factor $\boldsymbol{e}(t-1)\big(A(M_{(t)}) - I\big)$ are exactly linear in $\boldsymbol{e}(0)$; the quantity factor involves no approximation at all. The only expansion is in the price factor. Writing $f(\epsilon) = (1+\epsilon)^{\gamma-1}$, a second-order Taylor expansion about $\epsilon = 0$ gives

$$1 - f(\epsilon_t) = -(\gamma-1)\,\epsilon_t + R_t, \qquad R_t = -\tfrac{1}{2}f''(\xi_t)\,\epsilon_t^2 \ \text{ for some } \xi_t \text{ between } 0 \text{ and } \epsilon_t,$$

where $f''(\epsilon) = (\gamma-1)(\gamma-2)(1+\epsilon)^{\gamma-3}$ is bounded because interiority keeps $1+\epsilon_t$ away from zero. Substituting the linear term and writing ϵ_t out gives the stated leading expression, and the error is $p(N)R_t\,\boldsymbol{e}(t-1)\big(A(M_{(t)}) - I\big)$: quadratic from R_t, one further power from the quantity factor, hence cubic. It is cubic in $\boldsymbol{e}(0)$ rather than merely in $\boldsymbol{e}(t-1)$, uniformly in t, because Proposition 2 gives $\|\boldsymbol{e}(t-1)\|_\pi \le \|\boldsymbol{e}(0)\|_\pi$ at every date. Finally $f'' \equiv 0$ exactly when $(\gamma-1)(\gamma-2) = 0$; since $\gamma = 1$ is excluded from the family, $\gamma = 2$ is the only curvature at which R_t vanishes identically. □

Proof of Theorem 3. Write $t = mK + k$ with $1 \le k \le K$, so the date falls in step k of repetition m, and set $\boldsymbol{f} = \boldsymbol{e}(0)E^m$. Then $\boldsymbol{e}(t-1) = \boldsymbol{f}A(k-1)$: the first m repetitions contribute $A(\mathcal{R})^m$, which acts as E^m on disequilibria, and the remaining $k-1$ steps contribute the within-round prefix. By Lemma 4 the price factor is $\kappa\,\boldsymbol{f}A(k-1)\mathbf{1}_{M_k}/\alpha(M_k)$ to leading order, and the i-th component of the quantity factor is $\boldsymbol{f}A(k-1)\big(A(M_k)-I\big)\mathbf{1}_{\{i\}}$. Their product is $\boldsymbol{f}C_{k,i}\boldsymbol{f}^\top$, where $C_{k,i}$ is the outer product of the two column vectors, so summing over k gives $\boldsymbol{f}B_i\boldsymbol{f}^\top$. Summing over $m \ge 0$ gives $\mathcal{K}_i = \sum_{m\ge 0} E^m B_i (E^\top)^m$, the unique solution of Equation (10), the series converging because $\rho(E) < 1$. The per-summand errors of Proposition 5 aggregate to $O(\|\boldsymbol{e}(0)\|^3)$ by the geometric contraction of Theorem 1(iv), with constant proportional to $1/(1-\rho^3)$ for $\rho = |\lambda_2(A(\mathcal{R}))|$, and vanish identically when $\gamma = 2$. □

Proof of Lemma 6. Substituting the equation into itself m times gives

$$\mathcal{K} = \sum_{j<m} E^j B (E^\top)^j + E^m \mathcal{K} (E^\top)^m,$$

and $\rho(E) < 1$ sends the remainder to zero, which is (ii); the terms are $O(\rho(E)^{2j})$, so the series converges absolutely. For (i), apply $\operatorname{vec}(XYZ) = (Z^\top \otimes X)\operatorname{vec}(Y)$ with $X = E$ and $Z = E^\top$ to obtain $\operatorname{vec}(E\mathcal{K}E^\top) = (E \otimes E)\operatorname{vec}(\mathcal{K})$; the spectrum of $E \otimes E$ consists of the products $\lambda_\beta \lambda_{\beta'}$, so $\mathrm{Id} - E \otimes E$ is invertible and the solution is unique. Under diagonalisability, $E^j = \sum_\beta \lambda_\beta^j \mathbf{r}_\beta \boldsymbol{\ell}_\beta^\top$; substituting into (ii) and summing each geometric series $\sum_j (\lambda_\beta \lambda_{\beta'})^j$ separately gives Equation (12). □

Proof of Theorem 4. Telescoping the budget recursion against the Walrasian benchmark,

$$y^i(\mu) - y^i(N) = \sum_{t\ge 1} \Big(p(N) - p(M_{(t)})\Big)\Big[x^i(t) - x^i(t-1)\Big].$$

By Proposition 5 the t-th summand equals

$$\frac{\kappa}{\alpha(M_{(t)})}\Big(\boldsymbol{e}(t-1)\mathbf{1}_{M_{(t)}}\Big)\Big(\boldsymbol{e}(t-1)(A(M_{(t)})-I)\mathbf{1}_{\{i\}}\Big) + O\Big(\|\boldsymbol{e}(t-1)\|^3\Big),$$

the remainder being absent when $\gamma = 2$, where the expansion is exact. Write $\mathcal{F}_{t-1} = \sigma(M_{(1)}, \dots, M_{(t-1)})$. Since $\boldsymbol{e}(t-1)$ is $\mathcal{F}_{t-1}$-measurable and $M_{(t)} \sim \mu$ is independent of $\mathcal{F}_{t-1}$, averaging the leading term over the draw gives

$$\mathbb{E}\Big[\,\cdot \mid \mathcal{F}_{t-1}\Big] = \boldsymbol{e}(t-1)\,\bar{B}_i\,\boldsymbol{e}(t-1)^\top = \operatorname{tr}\Big(\bar{B}_i\,\boldsymbol{e}(t-1)^\top \boldsymbol{e}(t-1)\Big),$$

with $\bar{B}_i$ as in Equation (15). Taking expectations and writing $\Sigma_t = \mathbb{E}\Big[\boldsymbol{e}(t)^\top \boldsymbol{e}(t)\Big]$, the t-th summand contributes $\operatorname{tr}(\bar{B}_i \Sigma_{t-1})$. From $\boldsymbol{e}(t) = \boldsymbol{e}(t-1)A(M_{(t)})$ we have $\boldsymbol{e}(t)^\top \boldsymbol{e}(t) = A(M_{(t)})^\top \boldsymbol{e}(t-1)^\top \boldsymbol{e}(t-1) A(M_{(t)})$, and conditioning as above gives $\Sigma_t = \mathcal{L}_\mu(\Sigma_{t-1})$, hence $\Sigma_t = \mathcal{L}_\mu^t(\Sigma_0)$ with $\Sigma_0 = \boldsymbol{e}(0)^\top \boldsymbol{e}(0)$. Since $\boldsymbol{e}(0)\mathbf{1} = 0$ we have

$\Sigma_0\mathbf{1} = \mathbf{0}$, so by Lemma 7 the series $\mathcal{S} = \sum_{t\geq 0}\Sigma_t$ converges and solves Equation (16). Summing over t gives $\operatorname{tr}(\bar{B}_i\mathcal{S})$.

For the remainder, each $A(M)$ is non-expansive in $\|\cdot\|_\pi$ by Lemma 18, so $\|\boldsymbol{e}(t)\| \leq C\|\boldsymbol{e}(0)\|$ almost surely for a constant C depending only on $\boldsymbol{\pi}$. Hence

$$\mathbb{E}\|\boldsymbol{e}(t)\|^3 \leq C\|\boldsymbol{e}(0)\|\,\mathbb{E}\|\boldsymbol{e}(t)\|^2 = C\|\boldsymbol{e}(0)\|\operatorname{tr}(\Sigma_t) = O\Big(\rho(\mathcal{L}_\mu)^t\|\boldsymbol{e}(0)\|^3\Big),$$

using the contraction of Lemma 7. These are summable, and the total remainder is $O\Big(\|\boldsymbol{e}(0)\|^3/(1-\rho(\mathcal{L}_\mu))\Big)$. □

Proof of Remark 3. If $i \notin M$ then column i of $A(M)$ is $\mathbf{1}_{\{i\}}$: rows indexed by M are supported on M and so vanish there, and rows outside M are rows of the identity. Hence $(A(M)-I)\mathbf{1}_{\{i\}} = \mathbf{0}$ and only cliques containing i survive in Equation (15). The $M = N$ term is $\frac{\kappa\mu_N}{\alpha(N)}\mathbf{1}\boldsymbol{v}^\top$ with $\boldsymbol{v} = (A(N)-I)\mathbf{1}_{\{i\}}$, and for any Σ with $\Sigma\mathbf{1} = \mathbf{0}$, $\operatorname{tr}(\mathbf{1}\boldsymbol{v}^\top\Sigma) = \boldsymbol{v}^\top\Sigma\mathbf{1} = 0$. For Equation (17), note first that $\operatorname{tr}(\bar{B}_i\boldsymbol{e}^\top\boldsymbol{e}) = \boldsymbol{e}\bar{B}_i\boldsymbol{e}^\top$, so by Equation (15)

$$\operatorname{tr}\Big(\bar{B}_i\,\boldsymbol{e}^\top\boldsymbol{e}\Big) = \kappa\sum_M \frac{\mu_M}{\alpha(M)}\Big(\boldsymbol{e}\mathbf{1}_M\Big)\Big(\boldsymbol{e}(A(M)-I)\mathbf{1}_{\{i\}}\Big).$$

For $i \in M$ the i-th coordinate of $\boldsymbol{e}A(M)$ is $\sum_{j\in M} e^j\alpha^i/\alpha(M) = \alpha^i s_M(\boldsymbol{e})/\alpha(M)$, so $\boldsymbol{e}(A(M)-I)\mathbf{1}_{\{i\}} = \alpha^i s_M(\boldsymbol{e})/\alpha(M) - e^i$, which gives Equation (17). Finally $s_N(\boldsymbol{e}) = \boldsymbol{e}\mathbf{1} = 0$ for every disequilibrium profile, since $\boldsymbol{e}$ is a difference of allocations of the same aggregate endowment. □

Proof of Lemma 7. Write $V = \{\Sigma : \Sigma\mathbf{1} = \mathbf{0}\}$. Each $A(M)$ is row-stochastic, so $A(M)\mathbf{1} = \mathbf{1}$ and $\mathcal{L}_\mu(\Sigma)\mathbf{1} = \sum_M \mu_M A(M)^\top\Sigma A(M)\mathbf{1} = \sum_M \mu_M A(M)^\top\Sigma\mathbf{1} = \mathbf{0}$, so V is invariant.

For the contraction, let $A_{(t)} = A(M_{(1)})\cdots A(M_{(t)})$. Because the draws are independent, conditioning successively gives $\mathcal{L}_\mu^t(\Sigma) = \mathbb{E}\Big[A_{(t)}^\top\Sigma A_{(t)}\Big]$. Under the maintained hypothesis Theorem 2 gives $A_{(t)} \to \Pi$ almost surely, so $A_{(t)}^\top\Sigma A_{(t)} \to \Pi^\top\Sigma\Pi$ almost surely; and for $\Sigma \in V$ we have $\Sigma\Pi = \Sigma\mathbf{1}\boldsymbol{\pi} = 0$, so the limit is 0. Each $A(M)$ is non-expansive in $\|\cdot\|_\pi$ by Lemma 18, whence $\|A_{(t)}^\top\Sigma A_{(t)}\| \leq C\|\Sigma\|$ uniformly in t, and dominated convergence gives $\mathcal{L}_\mu^t(\Sigma) \to 0$ for every $\Sigma \in V$. A linear operator whose powers vanish pointwise has spectral radius below one, so $\rho(\mathcal{L}_\mu|_V) < 1$ and $\mathrm{Id} - \mathcal{L}_\mu$ is invertible on V. Since $\boldsymbol{e}(0)\mathbf{1} = 0$ puts $\boldsymbol{e}(0)^\top\boldsymbol{e}(0)$ in V, Equation (16) has exactly one solution there.

For (i), $\operatorname{vec}(A^\top\Sigma A) = (A^\top\otimes A^\top)\operatorname{vec}(\Sigma)$, so Equation (16) is the stated $n^2\times n^2$ system; it does not involve i, so one solve serves every agent. For (ii), invertibility gives the Neumann series $\mathcal{S} = \sum_{t\geq 0}\mathcal{L}_\mu^t(\Sigma_0)$, and $\mathcal{L}_\mu^t(\Sigma_0) = \mathbb{E}[\boldsymbol{e}(t)^\top\boldsymbol{e}(t)]$ as established in the proof of Theorem 4. □

Proof of Corollary 8. By Theorem 4, $\mathbb{E}[y^i(\mu) - y^i(N)] = \operatorname{tr}(\bar{B}_i\,\mathcal{S}) + O(\|\boldsymbol{e}(0)\|^3)$ with $\mathcal{S} = \sum_{t\geq 0} \mathcal{L}_\mu^t(\Sigma_0)$ and $\Sigma_0 = \boldsymbol{e}(0)^\top \boldsymbol{e}(0)$, the series converging by Lemma 7(ii). For any matrices K and Σ, cyclicity of the trace gives

$$\begin{aligned}\operatorname{tr}\Big(K\,\mathcal{L}_\mu(\Sigma)\Big) &= \sum_M \mu_M \operatorname{tr}\Big(KA(M)^\top \Sigma A(M)\Big)\\ &= \sum_M \mu_M \operatorname{tr}\Big(A(M)\,K\,A(M)^\top \Sigma\Big)\\ &= \operatorname{tr}\Big(\mathcal{L}_\mu^*(K)\,\Sigma\Big),\end{aligned}$$

so $\mathcal{L}_\mu^*$ is the adjoint of $\mathcal{L}_\mu$ for the trace pairing.

We now show that the series defining $\bar{\mathcal{K}}_i$ converges in $\mathbb{R}^{n\times n}$. Lemma 7 does not give this directly, because $\mathcal{L}_\mu^*$ fixes $\mathbf{1}\mathbf{1}^\top$. Write $\Pi = \mathbf{1}\boldsymbol{\pi}$. Since $\boldsymbol{\pi}\Big(A(M) - I\Big) = \mathbf{0}$, every term of Equation (15) vanishes when multiplied on the right by $\boldsymbol{\pi}^\top$. Hence $\bar{B}_i\boldsymbol{\pi}^\top = \mathbf{0}$, and

$$\bar{B}_i = (I - \Pi)\,\bar{B}_i\,(I - \Pi)^\top + \Pi\bar{B}_i.$$

Consider the first part. Since $A(M)\Pi = \Pi A(M) = \Pi$, every $A(M)$ commutes with $I - \Pi$. Therefore $\mathcal{L}_\mu^*$ preserves the matrices of the form $(I-\Pi)K(I-\Pi)^\top$, and $\mathcal{L}_\mu$ preserves those of the form $(I-\Pi)^\top\Sigma(I-\Pi)$. The two maps $K \mapsto (I-\Pi)K(I-\Pi)^\top$ and $\Sigma \mapsto (I-\Pi)^\top\Sigma(I-\Pi)$ are adjoint for the trace pairing. Hence the restriction of $\mathcal{L}_\mu^*$ to the first set is the adjoint of the restriction of $\mathcal{L}_\mu$ to the second. The second set lies in $\{\Sigma : \Sigma\mathbf{1} = \mathbf{0}\}$, where $\mathcal{L}_\mu$ contracts by Lemma 7. Therefore $(\mathcal{L}_\mu^*)^t$ applied to the first part decays geometrically. Consider the second part. The draws are independent, so $(\mathcal{L}_\mu^*)^t(K) = \mathbb{E}\Big[A(t)KA(t)^\top\Big]$ and $\mathbb{E}[A(t)] = \bar{A}_\mu^t$, with $\bar{A}_\mu = \sum_M \mu_M A(M)$. Since $A(t)\Pi = \Pi$ and $\bar{B}_i\Pi^\top = 0$,

$$(\mathcal{L}_\mu^*)^t(\Pi\bar{B}_i) = \Pi\bar{B}_i\Big(\bar{A}_\mu^t\Big)^\top = \Pi\bar{B}_i\Big(\bar{A}_\mu^t - \Pi\Big)^\top.$$

The matrix $\bar{A}_\mu$ is a convex combination of orthogonal projections in $\langle\cdot,\cdot\rangle_\pi$, so it is self-adjoint, with eigenvalues in $[0,1]$, and it maps V_0 into itself. If $\boldsymbol{v} \in V_0$ satisfies $\boldsymbol{v}\bar{A}_\mu = \boldsymbol{v}$, then $\|\boldsymbol{v}\|_\pi \leq \sum_M \mu_M \|\boldsymbol{v}A(M)\|_\pi \leq \|\boldsymbol{v}\|_\pi$, so every clique in the support of μ fixes $\boldsymbol{v}$, and $\boldsymbol{v} = \mathbf{0}$ by Lemma 19(iii). Hence the largest eigenvalue of $\bar{A}_\mu$ on V_0, which is its norm there, is below one. Since $\boldsymbol{\pi}\bar{A}_\mu = \boldsymbol{\pi}$, it follows that $\bar{A}_\mu^t - \Pi \to 0$ geometrically. Both parts decay geometrically, so the series converges.

Iterating and summing,

$$\operatorname{tr}\Big(\bar{B}_i\,\mathcal{S}\Big) = \sum_{t\geq 0} \operatorname{tr}\Big(\bar{B}_i\,\mathcal{L}_\mu^t(\Sigma_0)\Big) = \sum_{t\geq 0} \operatorname{tr}\Big((\mathcal{L}_\mu^*)^t(\bar{B}_i)\,\Sigma_0\Big) = \operatorname{tr}\Big(\bar{\mathcal{K}}_i(\mu)\,\Sigma_0\Big),$$

and $\operatorname{tr}(\bar{\mathcal{K}}_i\Sigma_0) = \boldsymbol{e}(0)\bar{\mathcal{K}}_i\boldsymbol{e}(0)^\top$ because Σ_0 is the outer product of $\boldsymbol{e}(0)$ with itself. Applying $\mathcal{L}_\mu^*$ to the series and telescoping gives $\bar{\mathcal{K}}_i - \mathcal{L}_\mu^*(\bar{\mathcal{K}}_i) = \bar{B}_i$. Since $\mathrm{Id} - \mathcal{L}_\mu^*$ is singular on all of $\mathbb{R}^{n\times n}$, that equation determines $\bar{\mathcal{K}}_i$ only up to matrices vanishing as quadratic forms on $\{\boldsymbol{v} : \boldsymbol{v}\mathbf{1} = 0\}$, where the disequilibria live; the series is what selects one. □

F Proofs for Section 4

Proof of Proposition 9. For (i), the nonzero spectrum of a product is invariant under cyclic permutation: if $AB\,\boldsymbol{v} = \lambda \boldsymbol{v}$ with $\lambda \neq 0$ then $BA\,(B\boldsymbol{v}) = \lambda(B\boldsymbol{v})$, so AB and BA share nonzero eigenvalues. Writing $A(\mathcal{R}) = A(M_1)\big(A(M_2)\cdots A(M_K)\big)$ and cycling gives the claim for $|\lambda_2|$, hence for Λ.

For (ii), each $A(M_k)$ is self-adjoint for $\langle\cdot,\cdot\rangle_\pi$ by Lemma 18, so the adjoint of $A(M_1)\cdots A(M_K)$ is $A(M_K)\cdots A(M_1)$, the operator of the reversed round. An operator and its adjoint have complex-conjugate spectra and hence equal eigenvalue moduli.

For (iii), two cliques with no agent in common act on disjoint blocks and therefore commute, so exchanging them leaves the product $A(\mathcal{R})$ itself unchanged.

For the last claim, the orderings reachable by (i) and (ii) form the orbit of $\mathcal{R}$ under the group generated by the cyclic shift and the reversal, of order at most $2K$. For $K \leq 3$ this is at least $K!$, so the orbit is every ordering; for $K \geq 4$ it is a proper subgroup, and Example 9 exhibits two orderings of the same six cliques with different spectral gaps. □

Lemma 22. *Let M be a clique and let $M_1, \ldots, M_s$ satisfy $M_r \subseteq M$ for every r. Then for every $\boldsymbol{u} \in \mathbb{R}^n$,*

$$\|\boldsymbol{u}A(M)\|_\pi \;\leq\; \|\boldsymbol{u}A(M_1)\cdots A(M_s)\|_\pi.$$

Proof of Lemma 22. Write $\mathcal{S} = (M_1, \ldots, M_s)$ and $A(\mathcal{S}) = A(M_1)\cdots A(M_s)$. By Lemma 18 each exchange matrix $A(M)$ is the orthogonal projection onto V_M. By Lemma 17(iii), $V_M = \{\boldsymbol{x} : x^i/\pi^i = x^j/\pi^j \text{ for all } i, j \in M\}$, since π^i is proportional to α^i. Since $M_r \subseteq M$, the condition defining V_M implies that defining V_{M_r}, so $V_M \subseteq V_{M_r}$ and $A(M_r)$ fixes V_M pointwise; hence so does $A(\mathcal{S})$.

Each $A(M_r)$ therefore maps $V_M^\perp$ into itself: for $\boldsymbol{w} \in V_M$ and $\boldsymbol{z} \in V_M^\perp$, self-adjointness gives $\langle \boldsymbol{z}A(M_r), \boldsymbol{w}\rangle_\pi = \langle \boldsymbol{z}, \boldsymbol{w}A(M_r)\rangle_\pi = \langle \boldsymbol{z}, \boldsymbol{w}\rangle_\pi = 0$, and the same then holds for $A(\mathcal{S})$.

Decompose $\boldsymbol{u} = \boldsymbol{u}A(M) + \boldsymbol{z}$ with $\boldsymbol{z} = \boldsymbol{u}\big(I - A(M)\big) \in V_M^\perp$. Then $\boldsymbol{u}A(\mathcal{S}) = \boldsymbol{u}A(M)A(\mathcal{S}) + \boldsymbol{z}A(\mathcal{S}) = \boldsymbol{u}A(M) + \boldsymbol{z}A(\mathcal{S})$, with the two summands in V_M and $V_M^\perp$ respectively. By Pythagoras,

$$\|\boldsymbol{u}A(\mathcal{S})\|_\pi^2 = \|\boldsymbol{u}A(M)\|_\pi^2 + \|\boldsymbol{z}A(\mathcal{S})\|_\pi^2 \;\geq\; \|\boldsymbol{u}A(M)\|_\pi^2. \qquad \square$$

Proof of Proposition 10. The two sequences share the exchange accumulated before the replaced stage; write $A(t-1)$ for it. Fix $\boldsymbol{v} \in V_0$ with $\|\boldsymbol{v}\|_\pi = 1$, where V_0 is the disequilibrium subspace of Lemma 20. By Lemma 20*(i)* the vector $\boldsymbol{u} := \boldsymbol{v}A(t-1)$

again lies in V_0, so Lemma 22 applies to it and gives

$$\Big\|\boldsymbol{v}A(t-1)A\big(M_{(t)}\big)\Big\|_\pi = \Big\|\boldsymbol{u}A\big(M_{(t)}\big)\Big\|_\pi \leq \Big\|\boldsymbol{u}A(M_1)\cdots A(M_s)\Big\|_\pi = \Big\|\boldsymbol{v}A(t-1)A(M_1)\cdots A(M_s)\Big\|_\pi.$$

Taking suprema over such $\boldsymbol{v}$ and applying Equation (30) to each side gives Equation (20). □

Proof of Corollary 11. Rotating places the refined block last, so the two rotated rounds share the prefix $X = A(M_{j+1})\cdots A(M_{j-1})$ and differ only in what follows it. Proposition 10 applied with $A(t-1) = X$ gives the claim, since $A(\mathcal{R}_{\mathrm{rot}}) = XA(M_j)$ and $A(\mathcal{R}'_{\mathrm{rot}}) = XA(M_1)\cdots A(M_s)$. □

Proof of Proposition 12. Example 11. □

Proof of Proposition 13. Each $A(M)$ is idempotent and self-adjoint for the inner product of Definition 3. Hence $A(M)$ is the orthogonal projection onto V_M (Lemma 18), and $\langle \boldsymbol{v}A(M), \boldsymbol{v}\rangle_\pi = \|\boldsymbol{v}A(M)\|_\pi^2$ for every $\boldsymbol{v}$. Therefore

$$\langle \boldsymbol{v}\bar{A}_\mu, \boldsymbol{v}\rangle_\pi = \sum_M \mu_M \Big\|\boldsymbol{v}A(M)\Big\|_\pi^2 \;\geq\; 0.$$

So $\bar{A}_\mu$ is self-adjoint and its eigenvalues are nonnegative. By Lemma 20*(i)*, $\bar{A}_\mu$ maps V_0 into itself, and V_0 is orthogonal to $\boldsymbol{\pi}$, which $\bar{A}_\mu$ fixes. Consequently $|\lambda_2(\bar{A}_\mu)|$ is the largest eigenvalue of $\bar{A}_\mu$ on V_0, and

$$\Big|\lambda_2(\bar{A}_\mu)\Big| = \max\Big\{\sum_M \mu_M \Big\|\boldsymbol{v}A(M)\Big\|_\pi^2 :\; \boldsymbol{v}\in V_0,\; \|\boldsymbol{v}\|_\pi = 1\Big\}. \tag{32}$$

This holds even if the eigenvalue one has multiplicity greater than one, since then both sides equal one.

Now I compare the two sums term by term. For $M_r \subseteq M$, the condition defining V_M implies the condition defining V_{M_r}, so $V_M \subseteq V_{M_r}$. The projection onto the larger subspace keeps at least as much of each vector. Hence $\|\boldsymbol{v}A(M)\|_\pi \leq \|\boldsymbol{v}A(M_r)\|_\pi$ for every r. Since the weights w_r sum to μ_M,

$$\mu_M \Big\|\boldsymbol{v}A(M)\Big\|_\pi^2 \;\leq\; \sum_r w_r \Big\|\boldsymbol{v}A(M_r)\Big\|_\pi^2.$$

The other terms are the same under μ and μ'. Therefore the sum in Equation (32) is weakly larger under μ' at every $\boldsymbol{v}$, and so is its maximum. This gives $|\lambda_2(\bar{A}_{\mu'})| \geq |\lambda_2(\bar{A}_\mu)|$, that is, $\Lambda(\mu') \leq \Lambda(\mu)$. The argument uses only $M_r \subseteq M$. Moreover, it does not require the maintained hypothesis for μ'. □

Proof of Proposition 14. Write $E = E(\mathcal{R})$ and $F = D^{1/2} E D^{-1/2}$, so that $\|F\| = \|E\|_\pi$. Conjugating Equation (13) gives

$$D^{1/2} \mathcal{K}_i D^{1/2} = \sum_{m \geq 0} F^m \Big(D^{1/2} B_i D^{1/2} \Big) \Big(F^m \Big)^\top,$$

whose norm is at most $\|B_i\|_\pi \sum_{m \geq 0} \|E\|_\pi^{2m} = \|B_i\|_\pi / \Big(1 - \|E\|_\pi^2 \Big)$, the geometric series converging because $\|E\|_\pi < 1$: equality would require a unit vector with $\boldsymbol{v}\mathbf{1} = 0$ whose π-norm survives every projection in the round, and the ranges V_{M_k} of a round covering N connectedly meet only in $\operatorname{span}(\boldsymbol{\pi})$. For $\boldsymbol{e}$ with $\boldsymbol{e}\mathbf{1} = 0$ and $\|\boldsymbol{e}\|_\pi = 1$ the vector $\boldsymbol{w} = \boldsymbol{e} D^{-1/2}$ is a Euclidean unit vector and $\boldsymbol{e} \mathcal{K}_i \boldsymbol{e}^\top = \boldsymbol{w} \Big(D^{1/2} \mathcal{K}_i D^{1/2} \Big) \boldsymbol{w}^\top$, so each term of Equation (21) is bounded by that quantity. Summing over agents gives the claim. $\square$